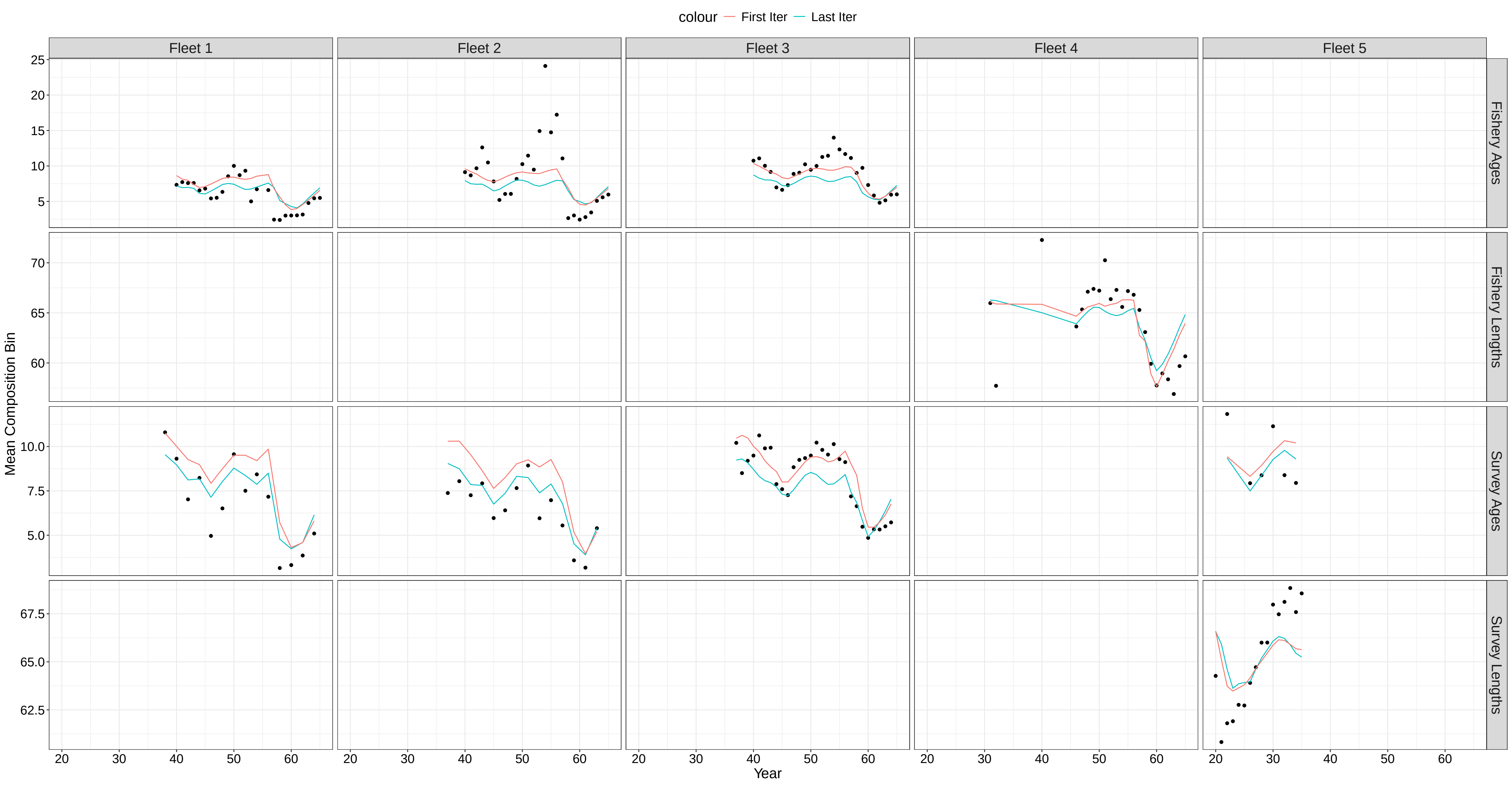
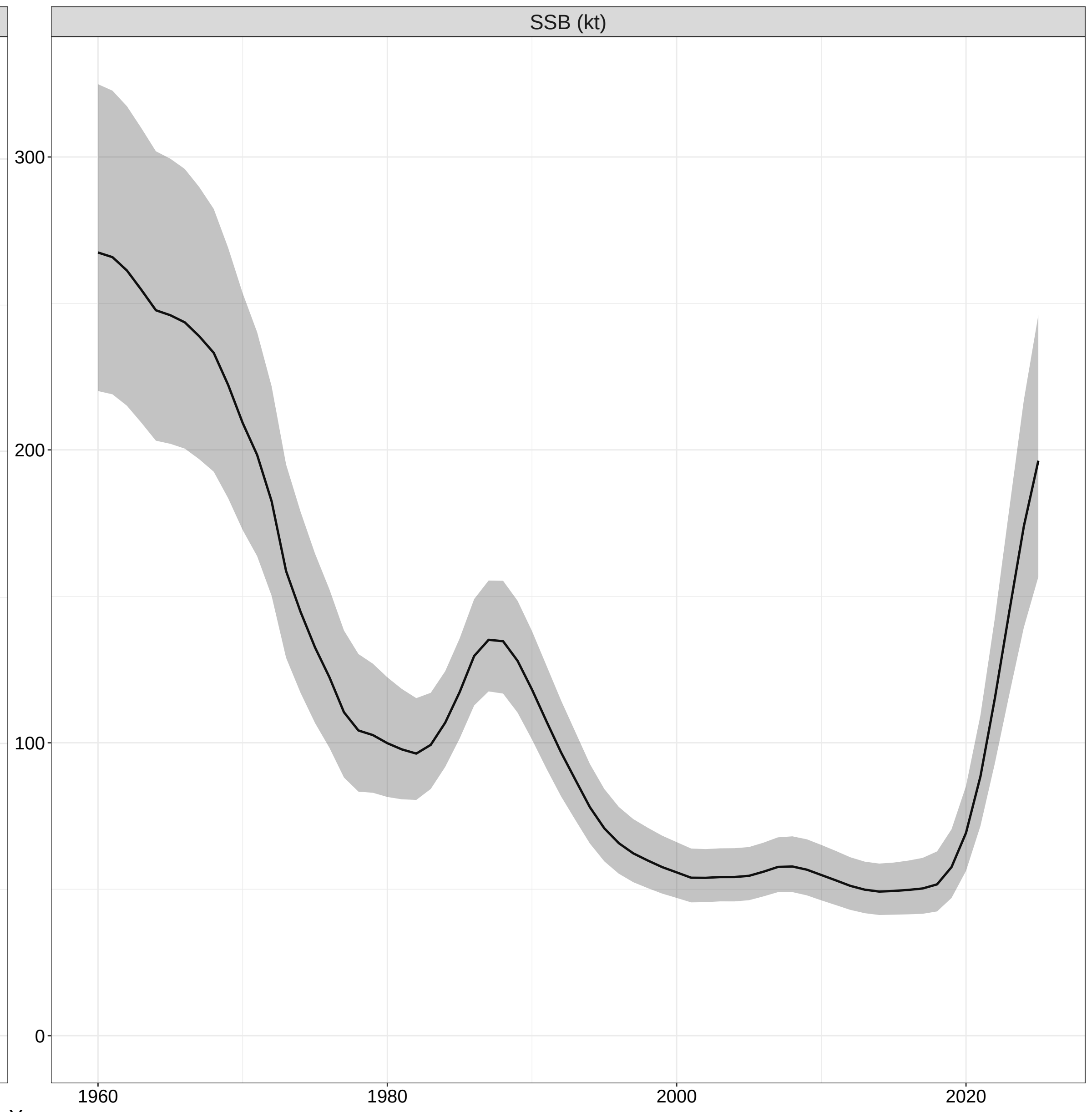
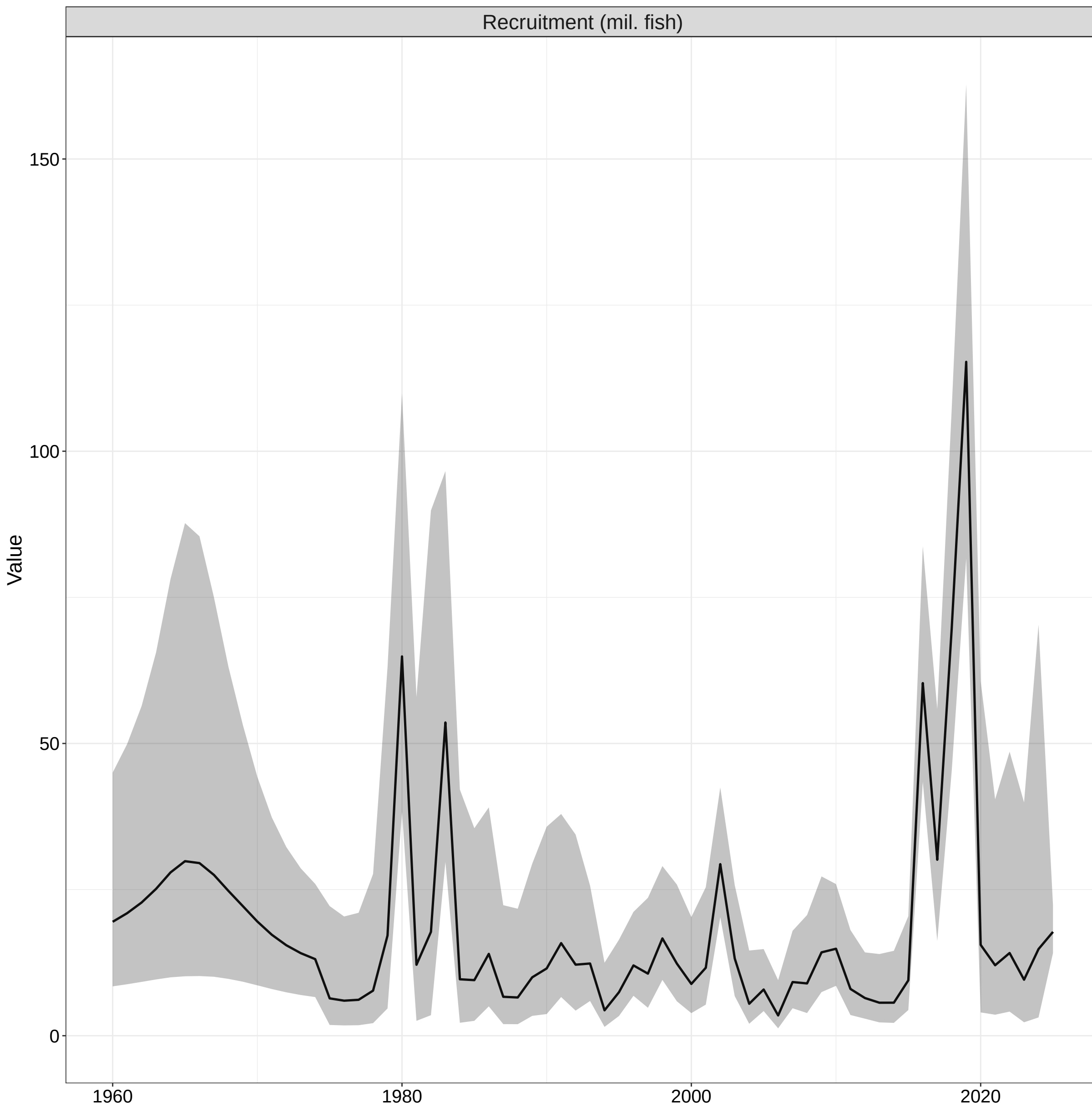


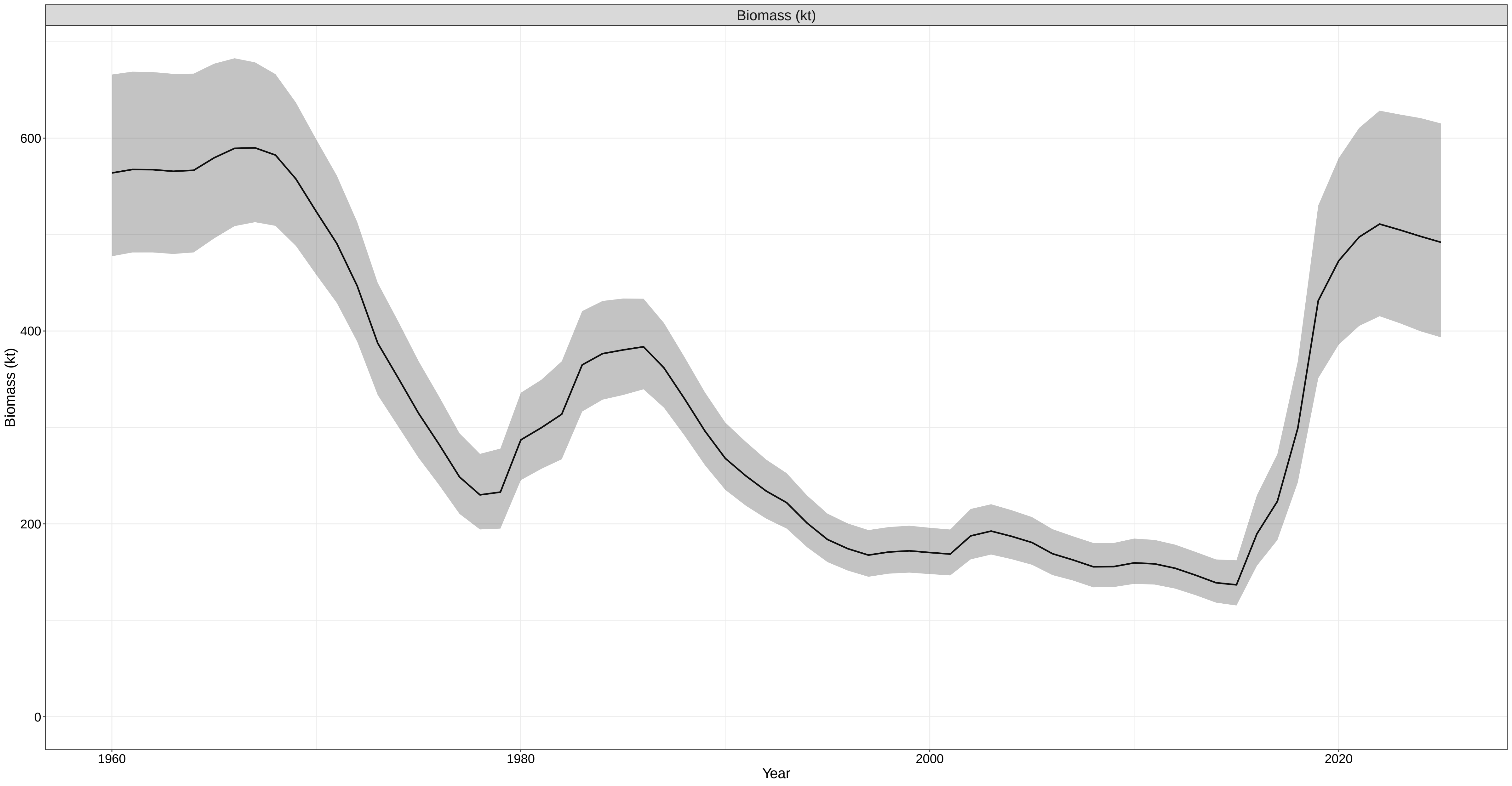
Source	SE	Eff_SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	0.20	1.00	NA	NA
AI LL Survey Index	0.24	0.24	1.00	NA	NA
GOA LL Survey Index	0.08	0.08	1.00	NA	NA
JPN LL Survey Index	0.07	0.07	1.00	NA	NA
BS Fixed Gear Fish Ages	NA	NA	0.33	26.92	8.81
AI Fixed Gear Fish Ages	NA	NA	0.04	29.65	1.14
GOA Fixed Gear Fish Ages	NA	NA	0.11	64.15	6.96
BS LL Survey Ages	NA	NA	0.56	37.43	20.87
AI LL Survey Ages	NA	NA	1.81	39.14	70.68
GOA LL Survey Ages	NA	NA	0.33	77.14	25.84
JPN LL Survey Ages	NA	NA	0.17	83.33	14.56
Trawl Fish Lengths	NA	NA	0.15	45.39	6.73
JPN LL Survey Lengths	NA	NA	0.34	54.62	18.61

G3;Successfully passed post-optim-sanity checks



	Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
1	1	1	196.30591	353.5244	0.0586490038616642	33.14317	120.41233	0.08484	1.63028	0.55528	0.69129

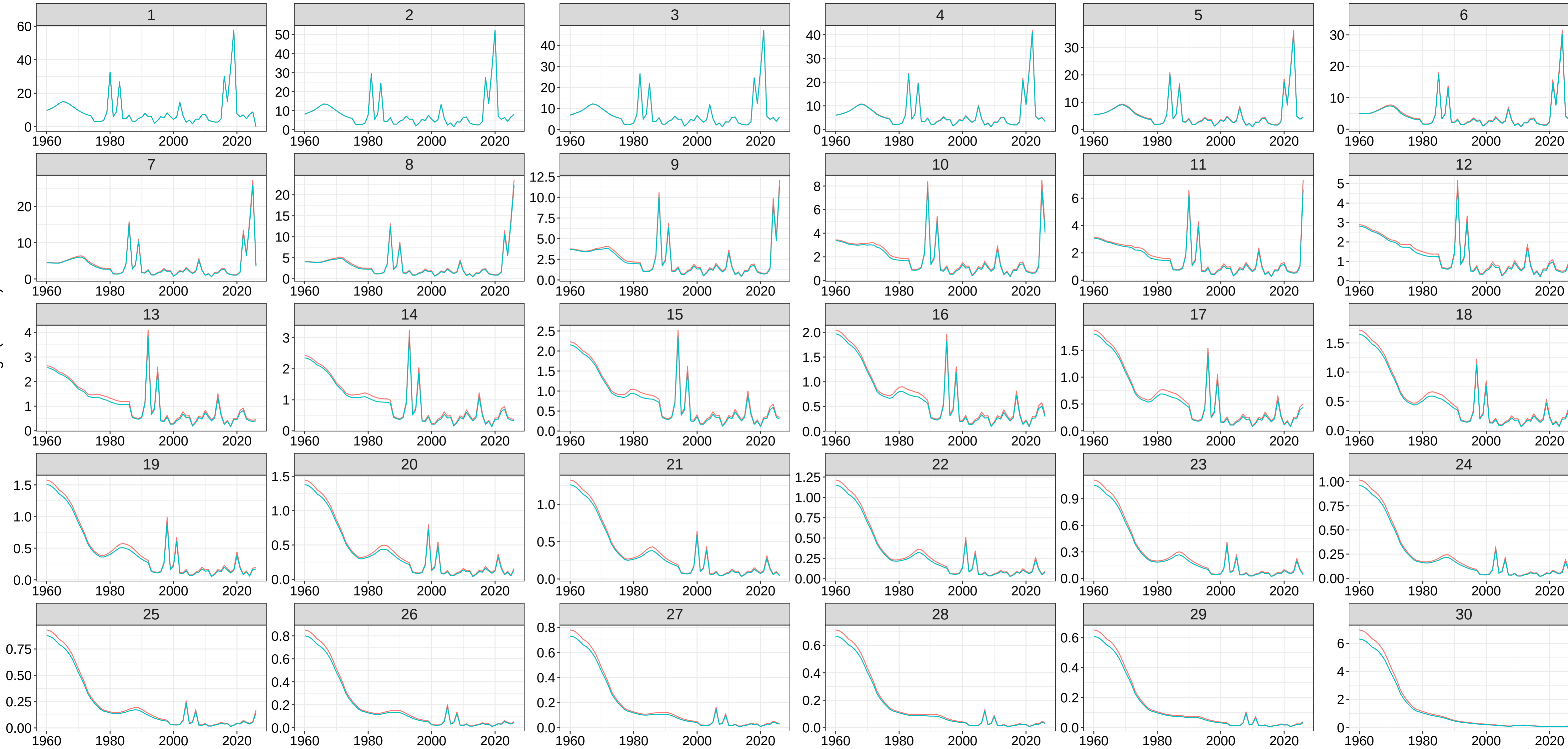




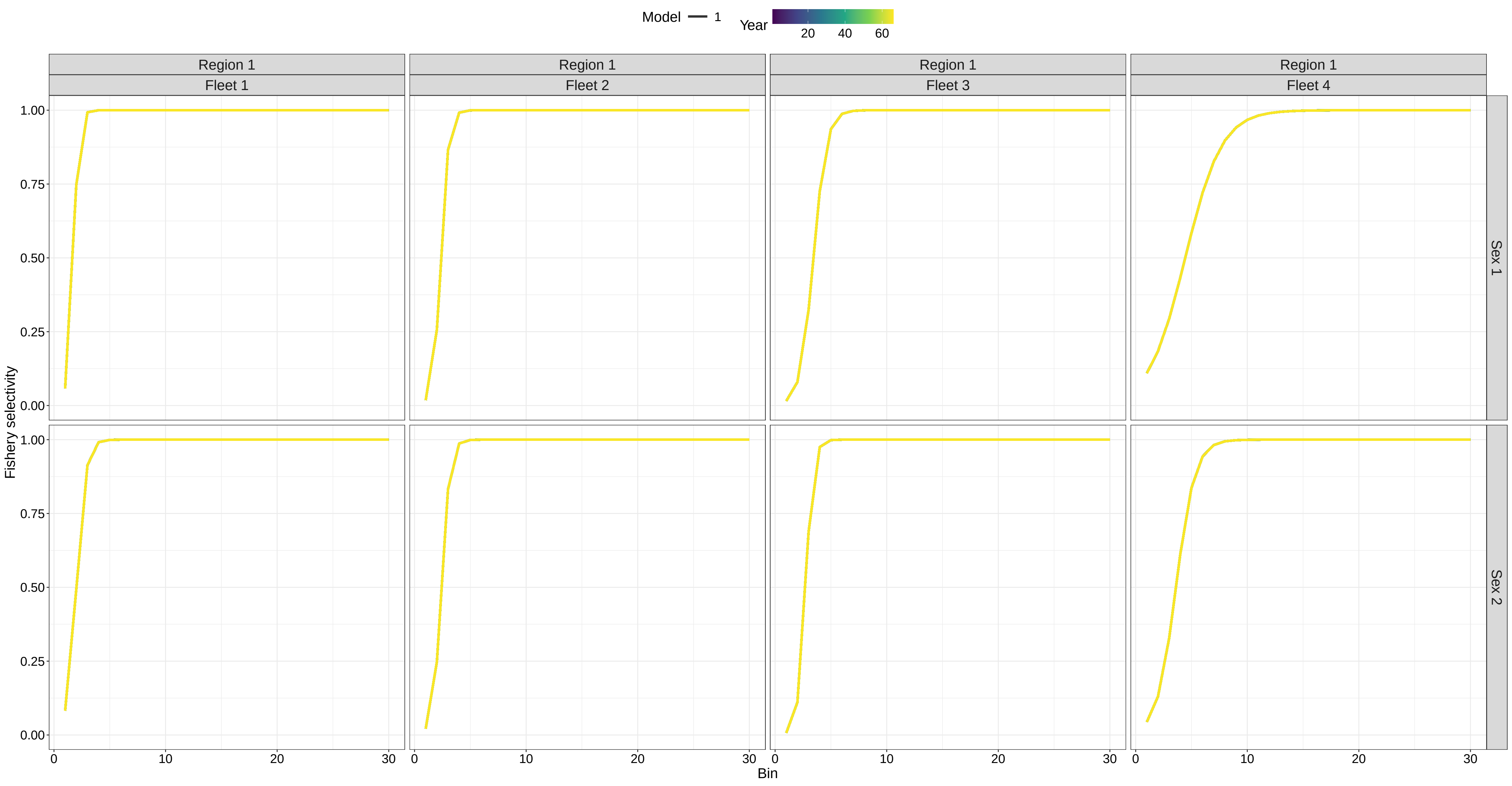
Numbers-at-Age

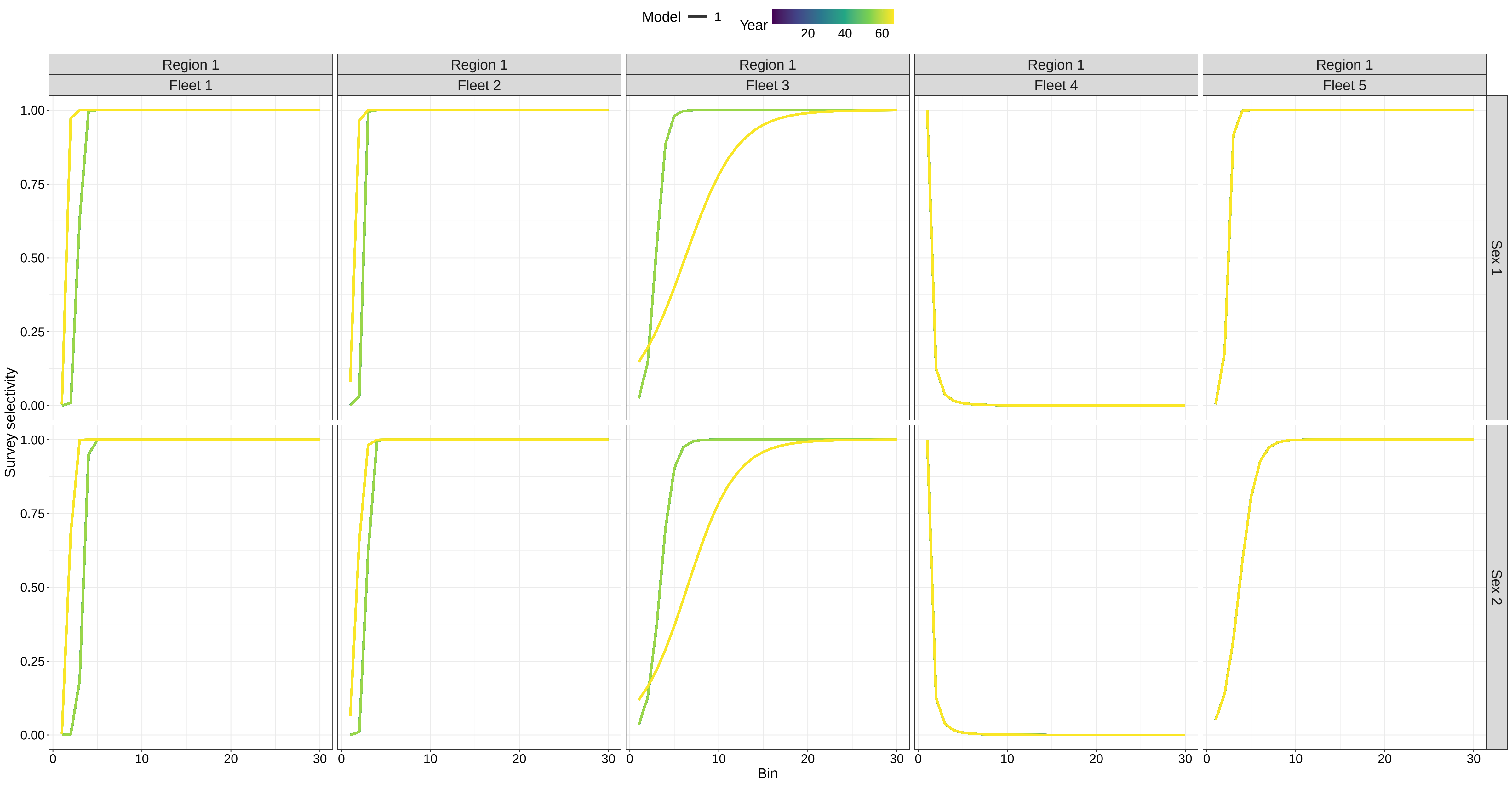
Sex — Female — Male

Numbers-at-age (millions)



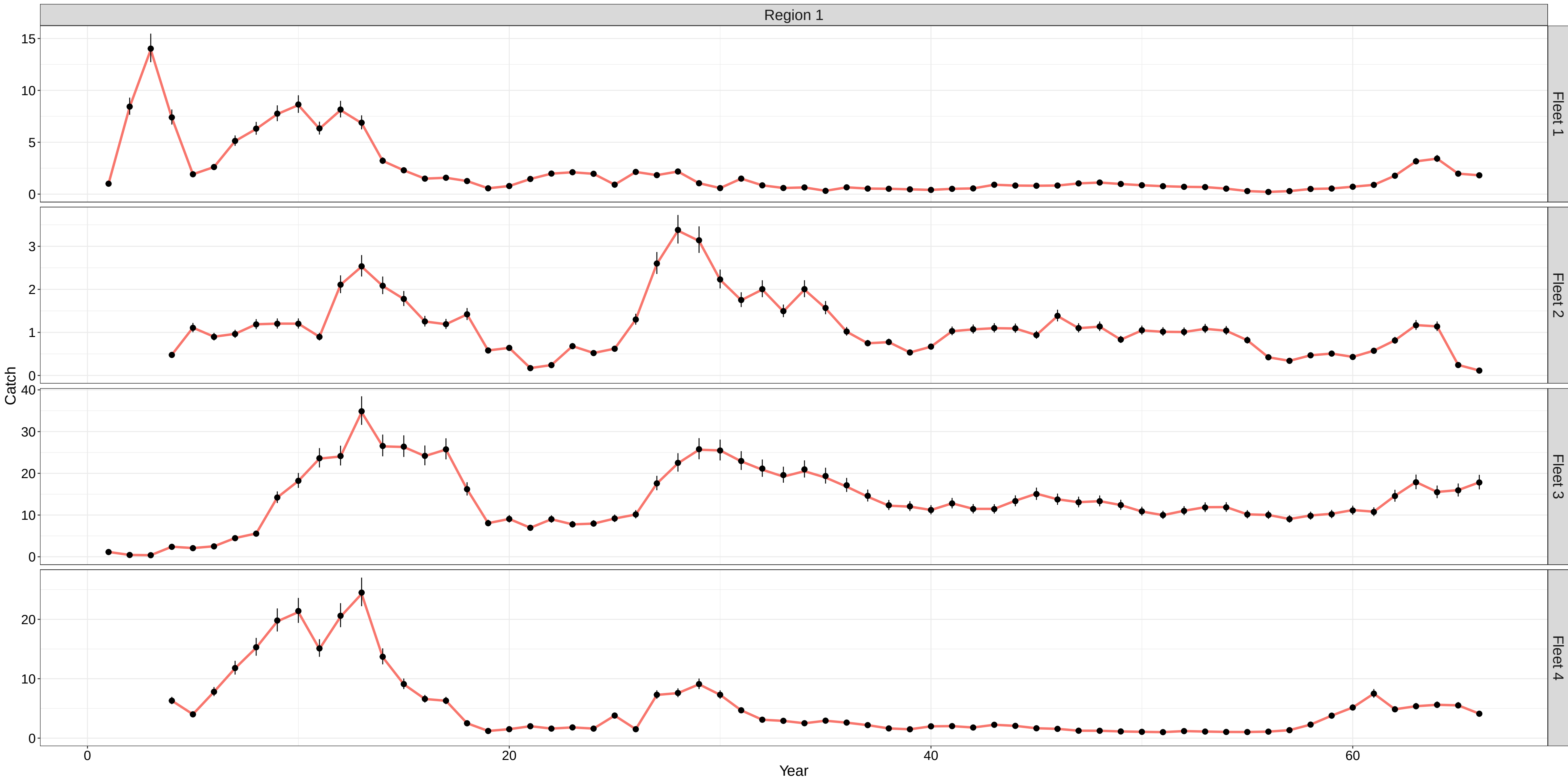
Year



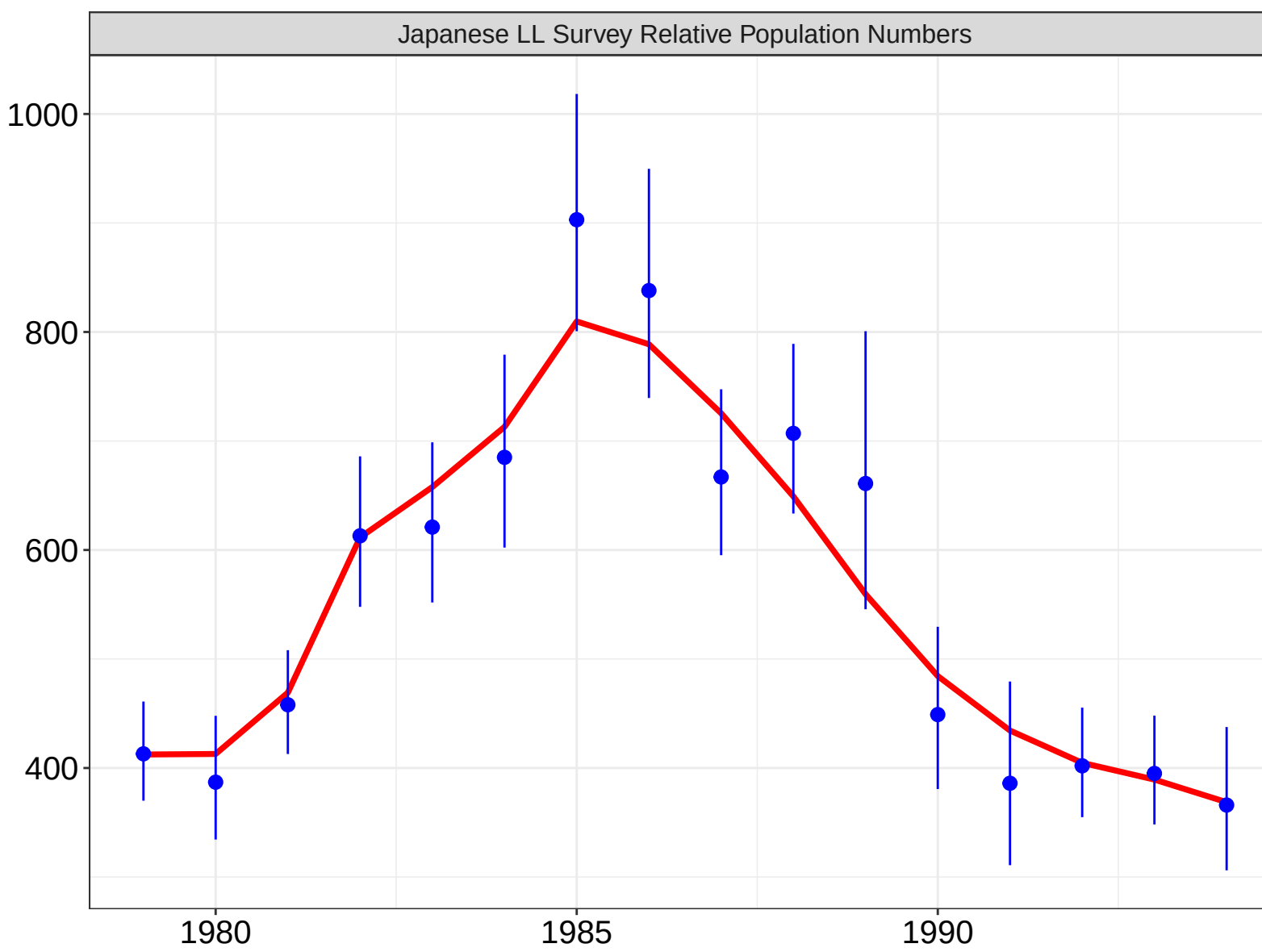
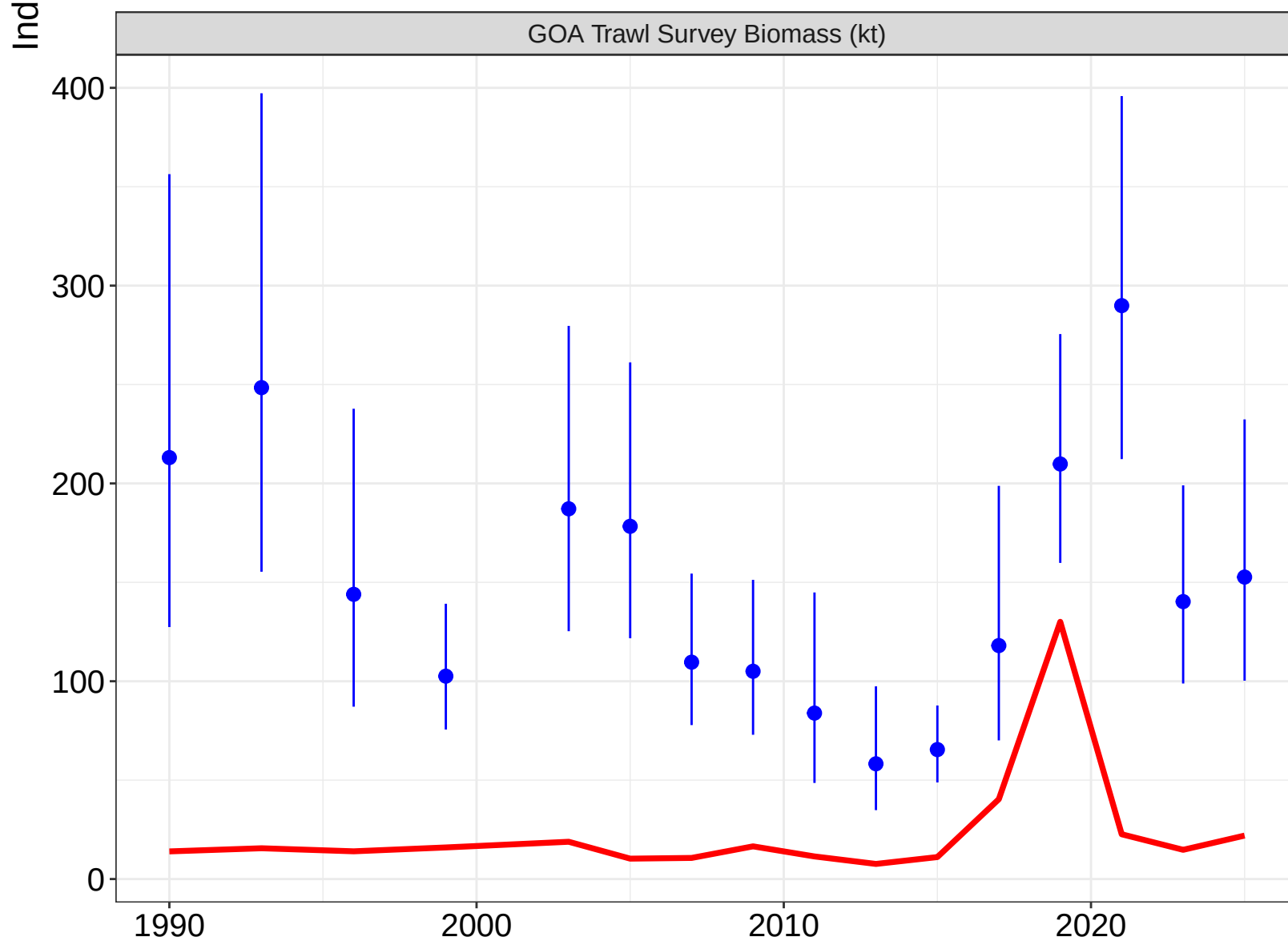
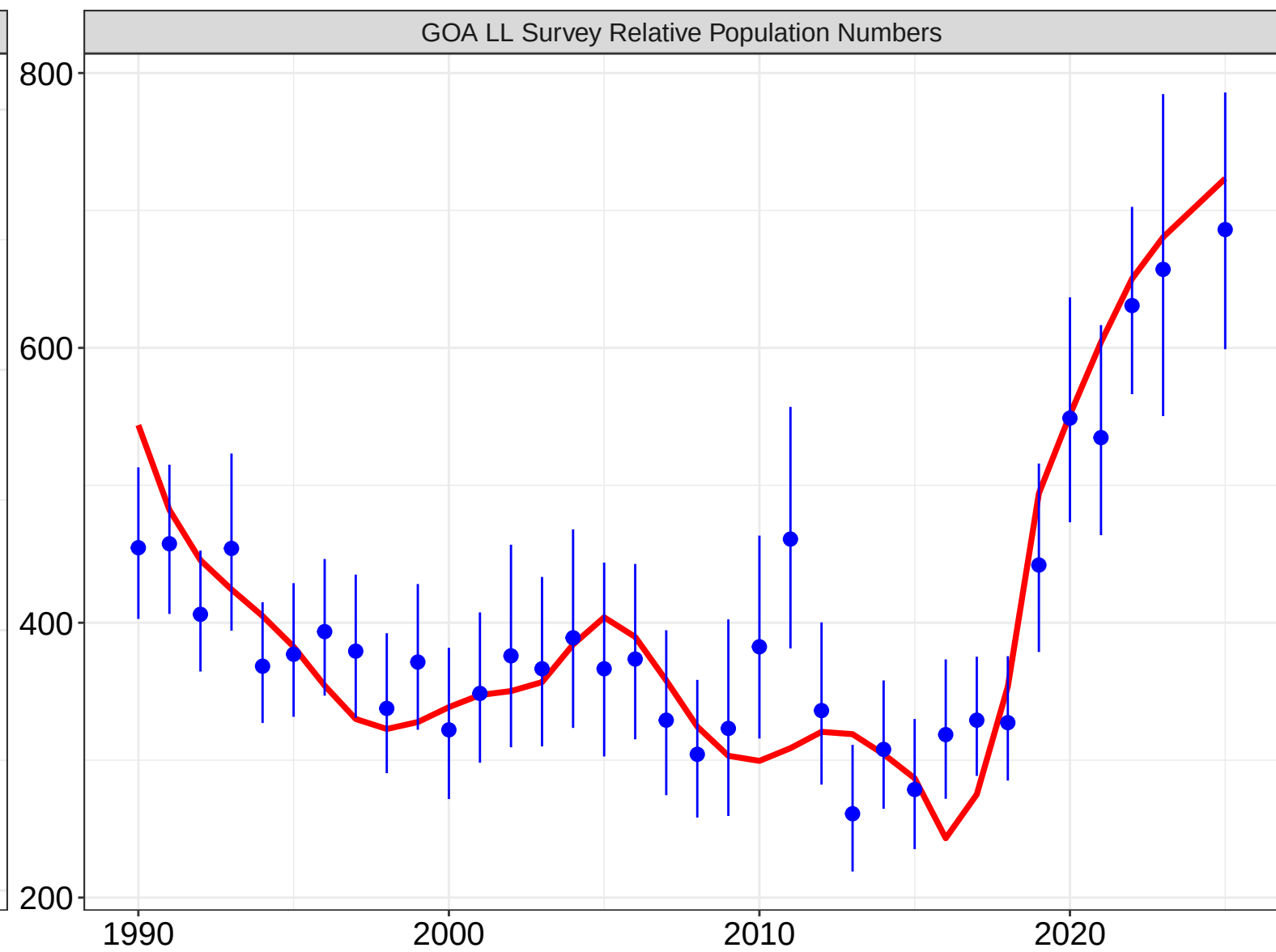
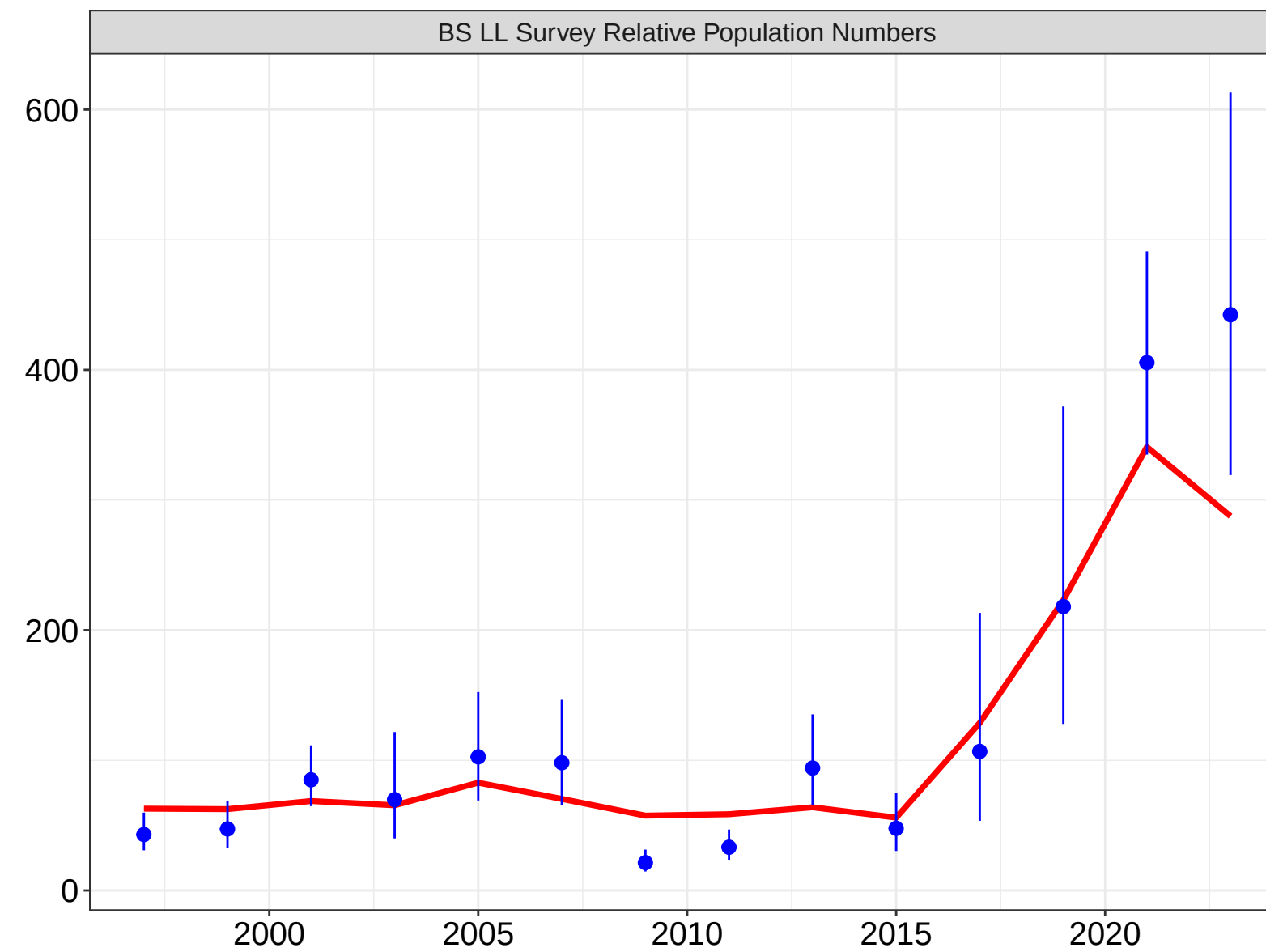
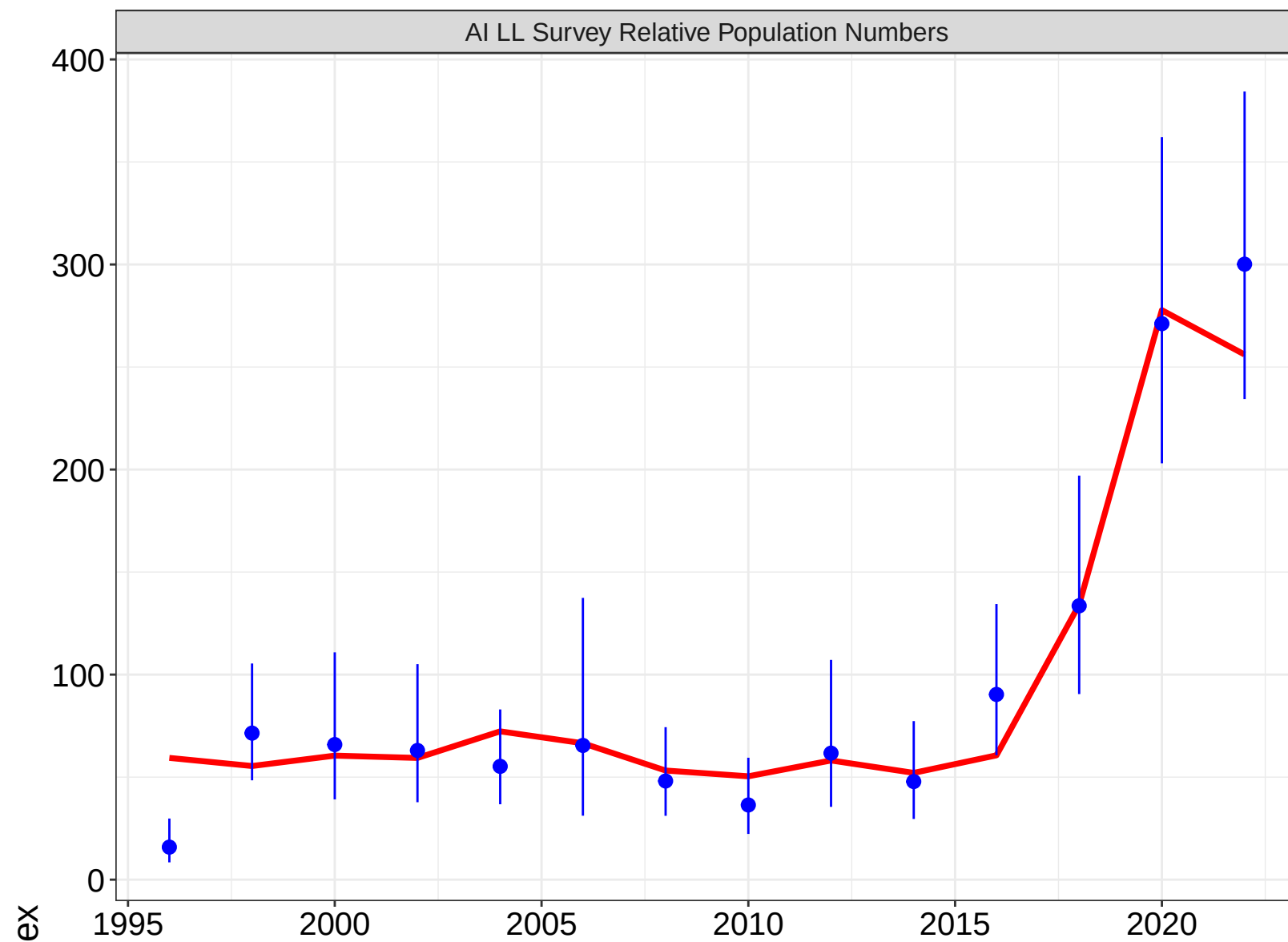


Model — 1

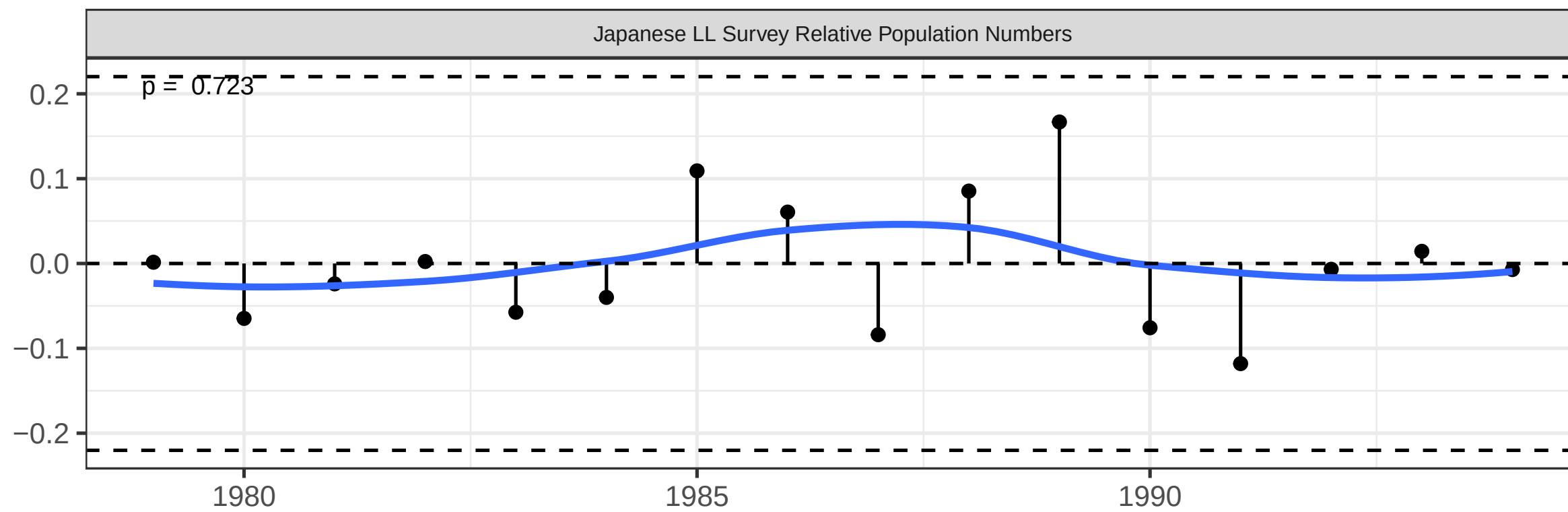
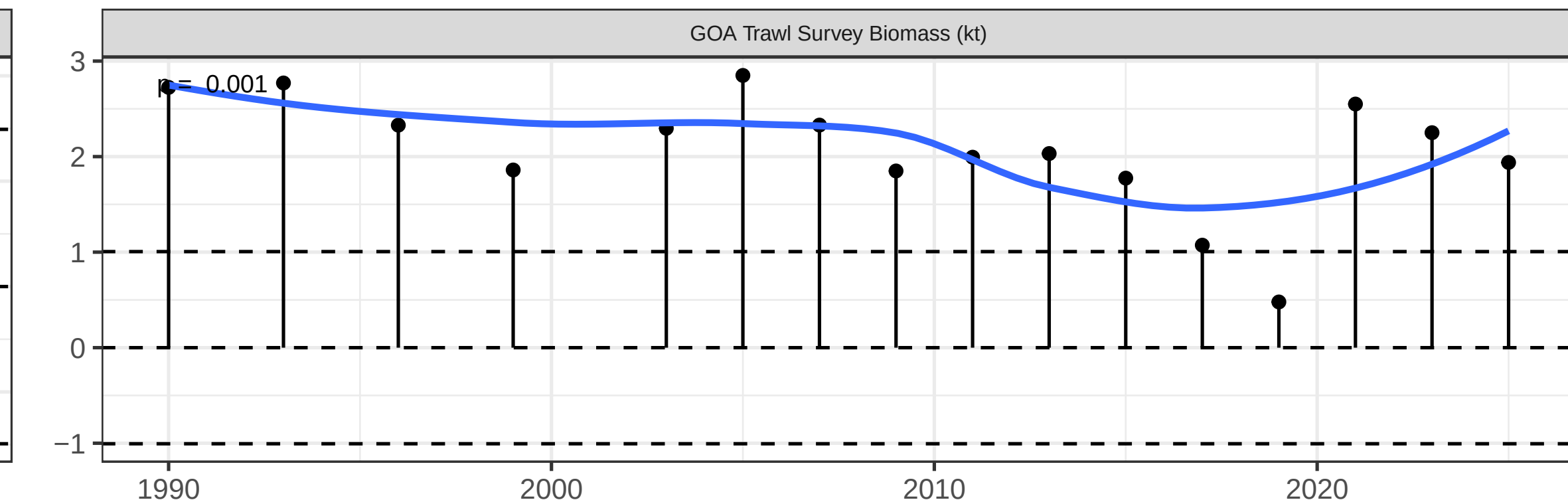
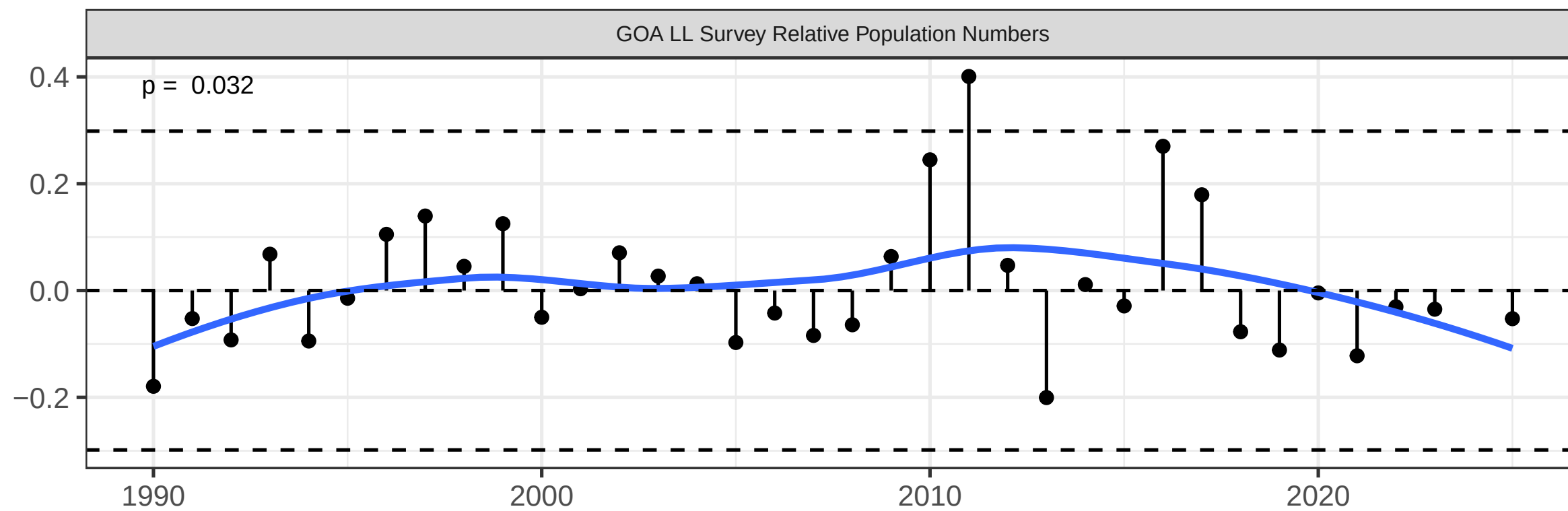
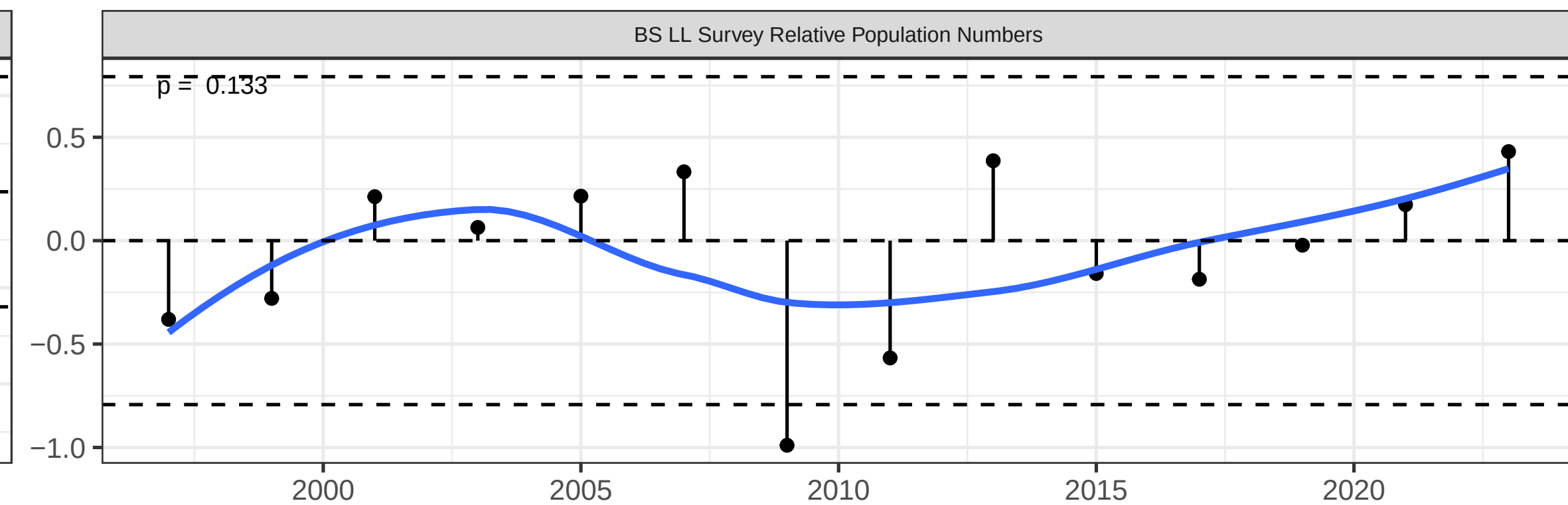
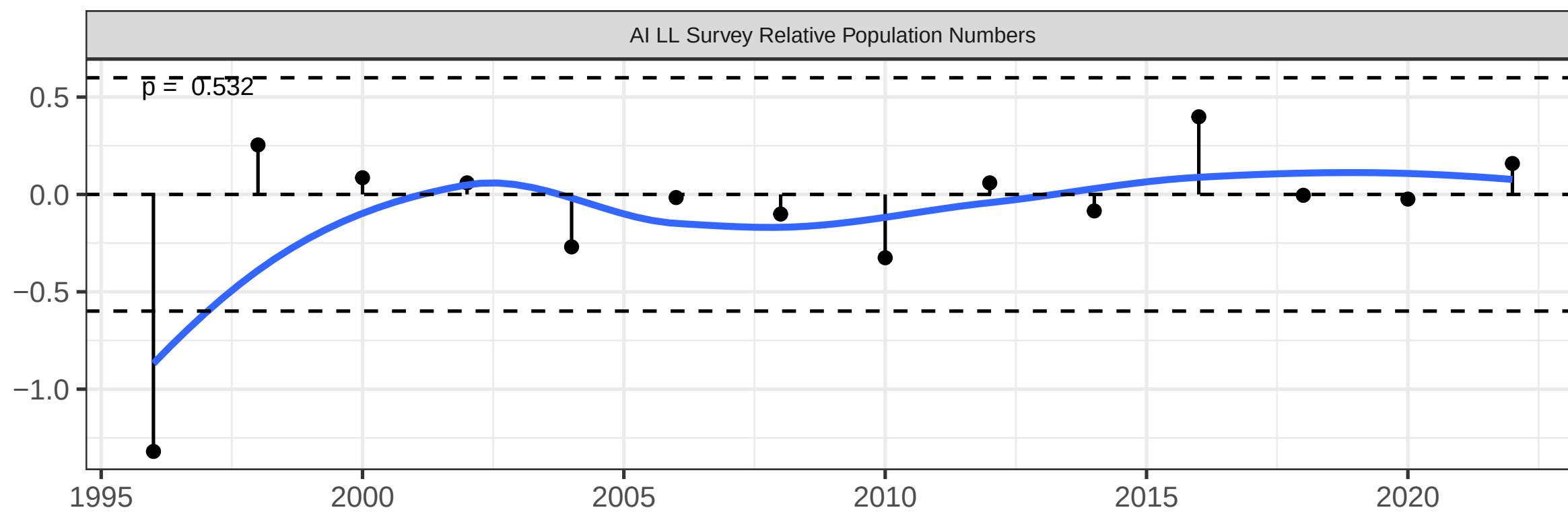
Region 1



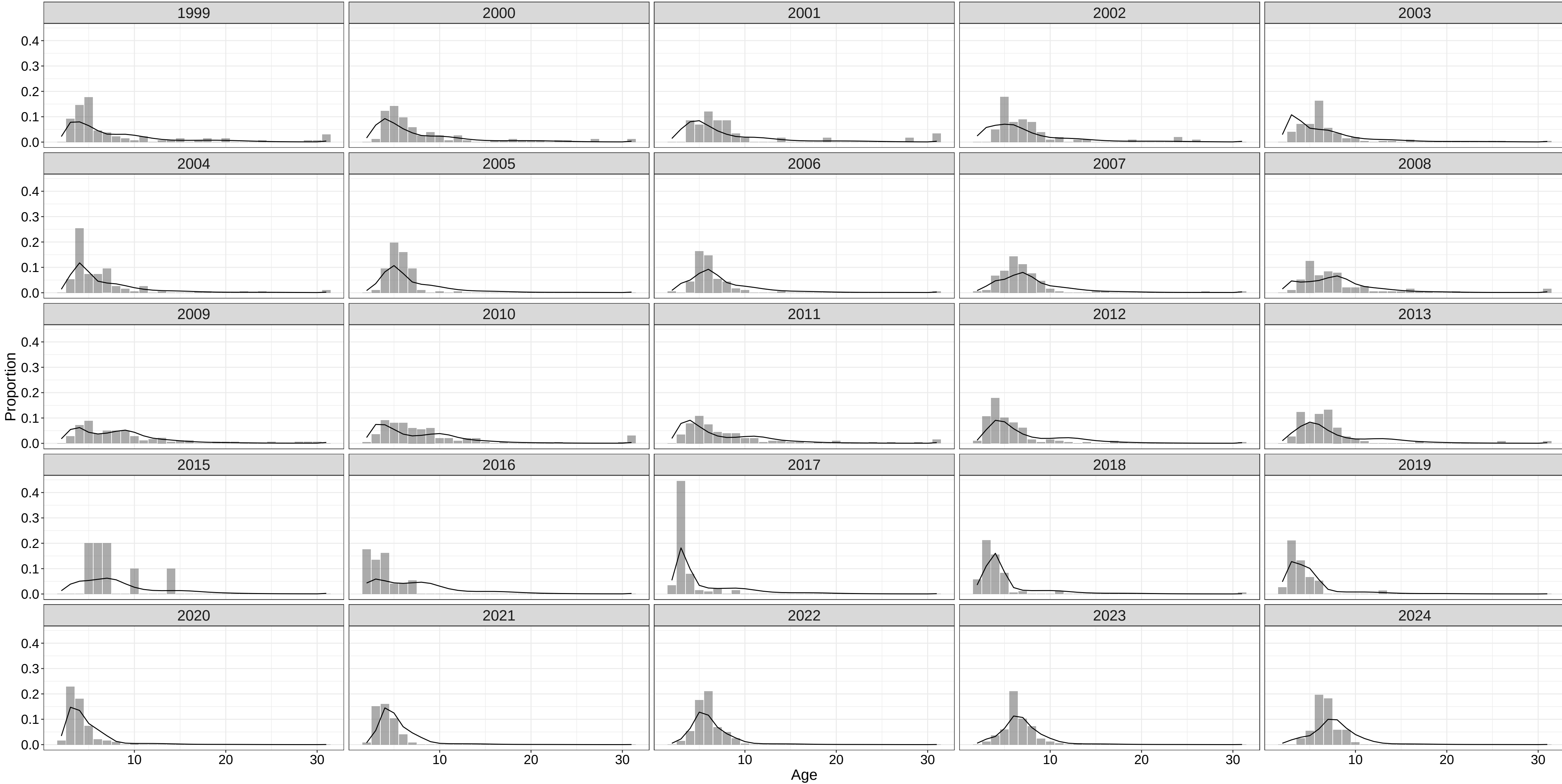
Index Fits



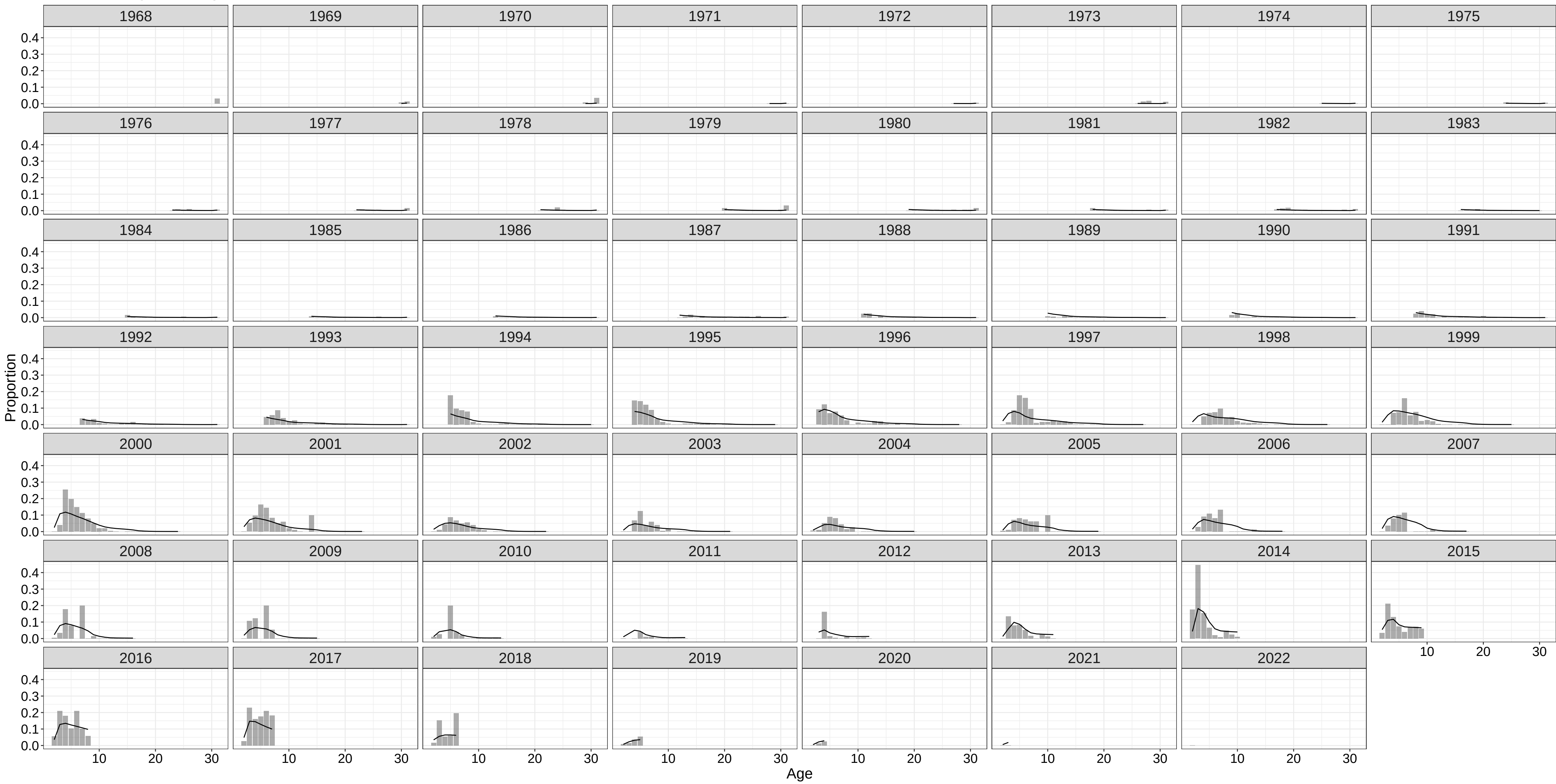
Index Residuals Runs Test



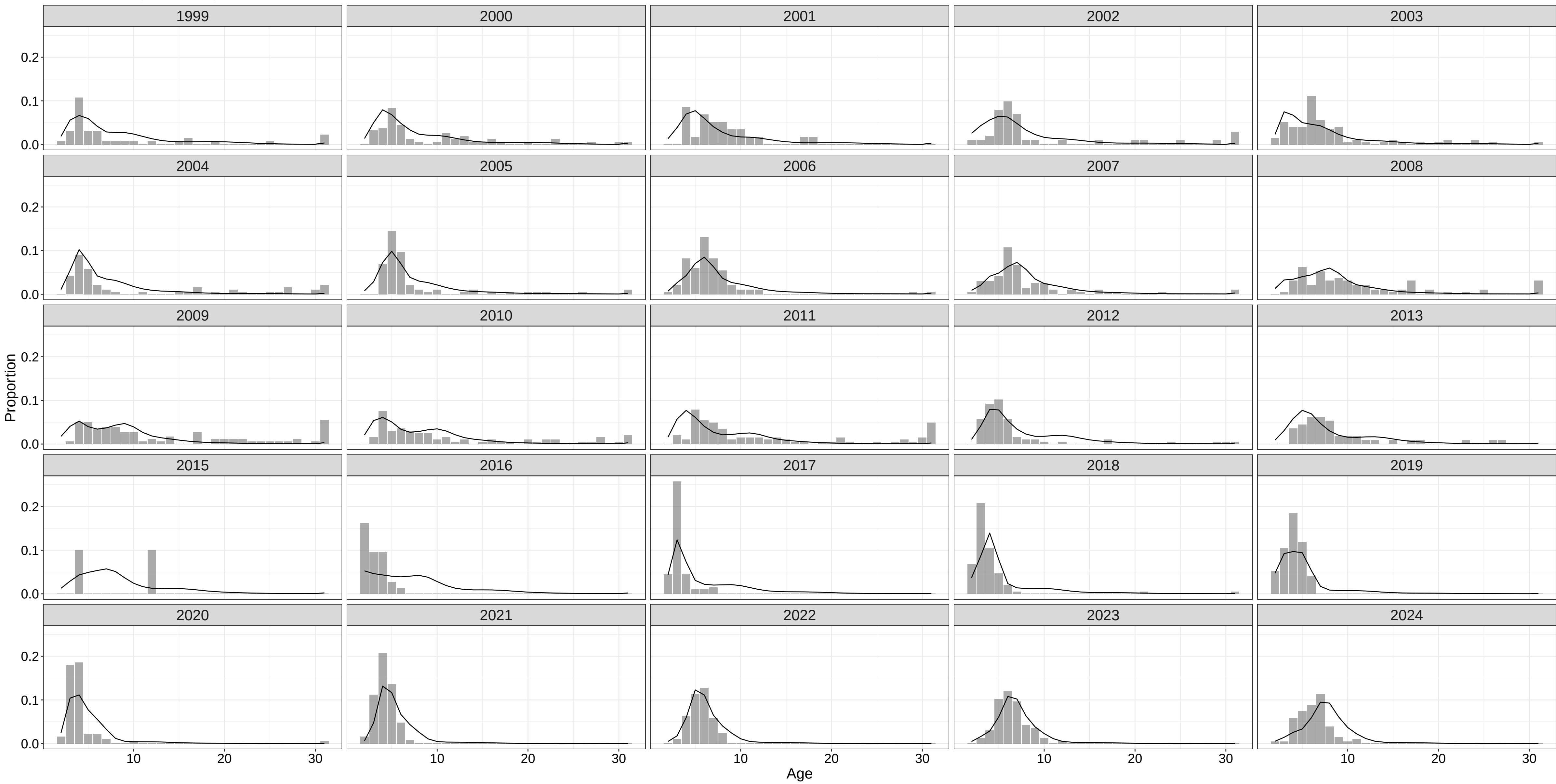
BS LLF Age Comps Female



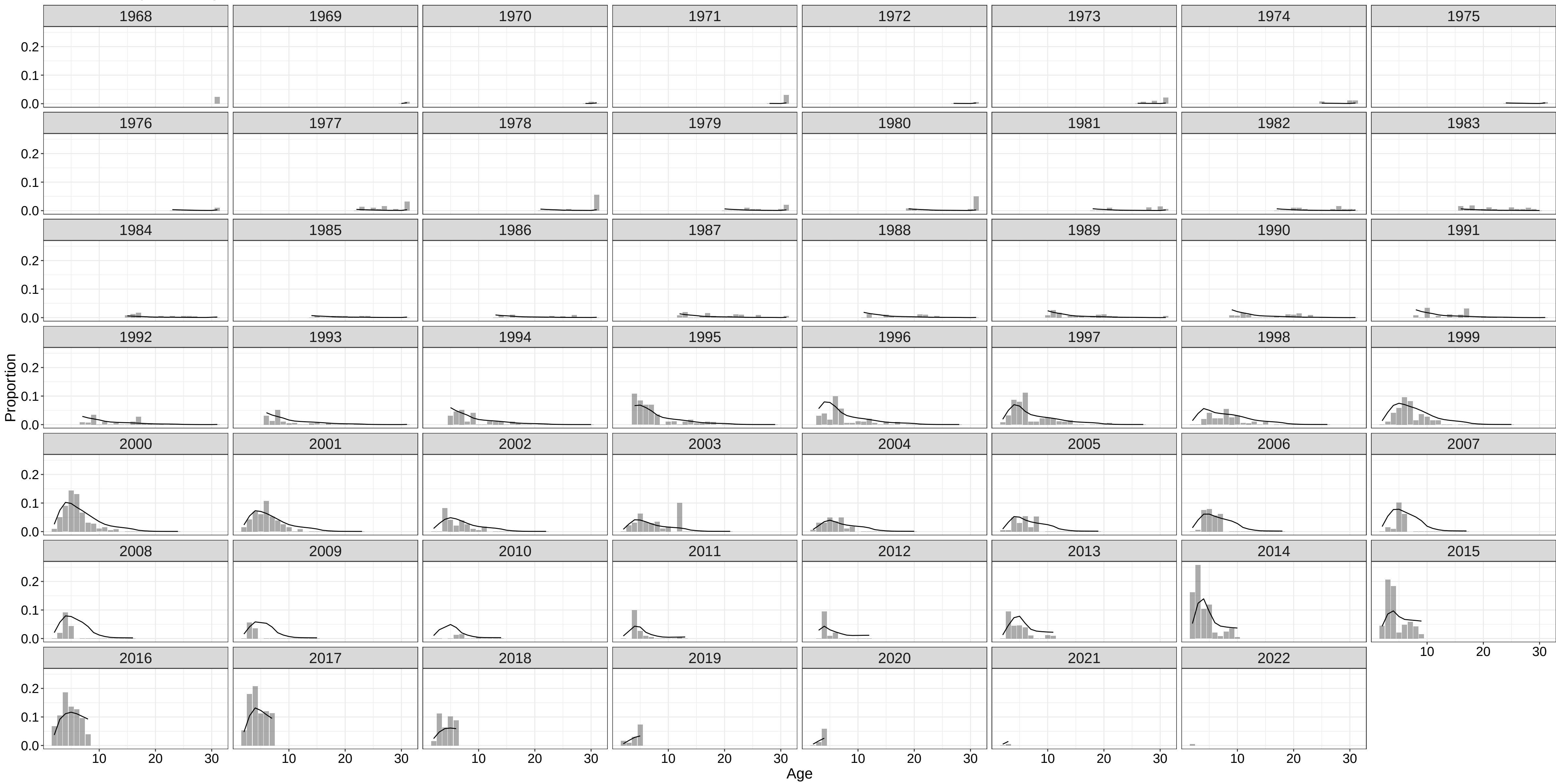
BS LLF Age Comps Cohort Fits Female



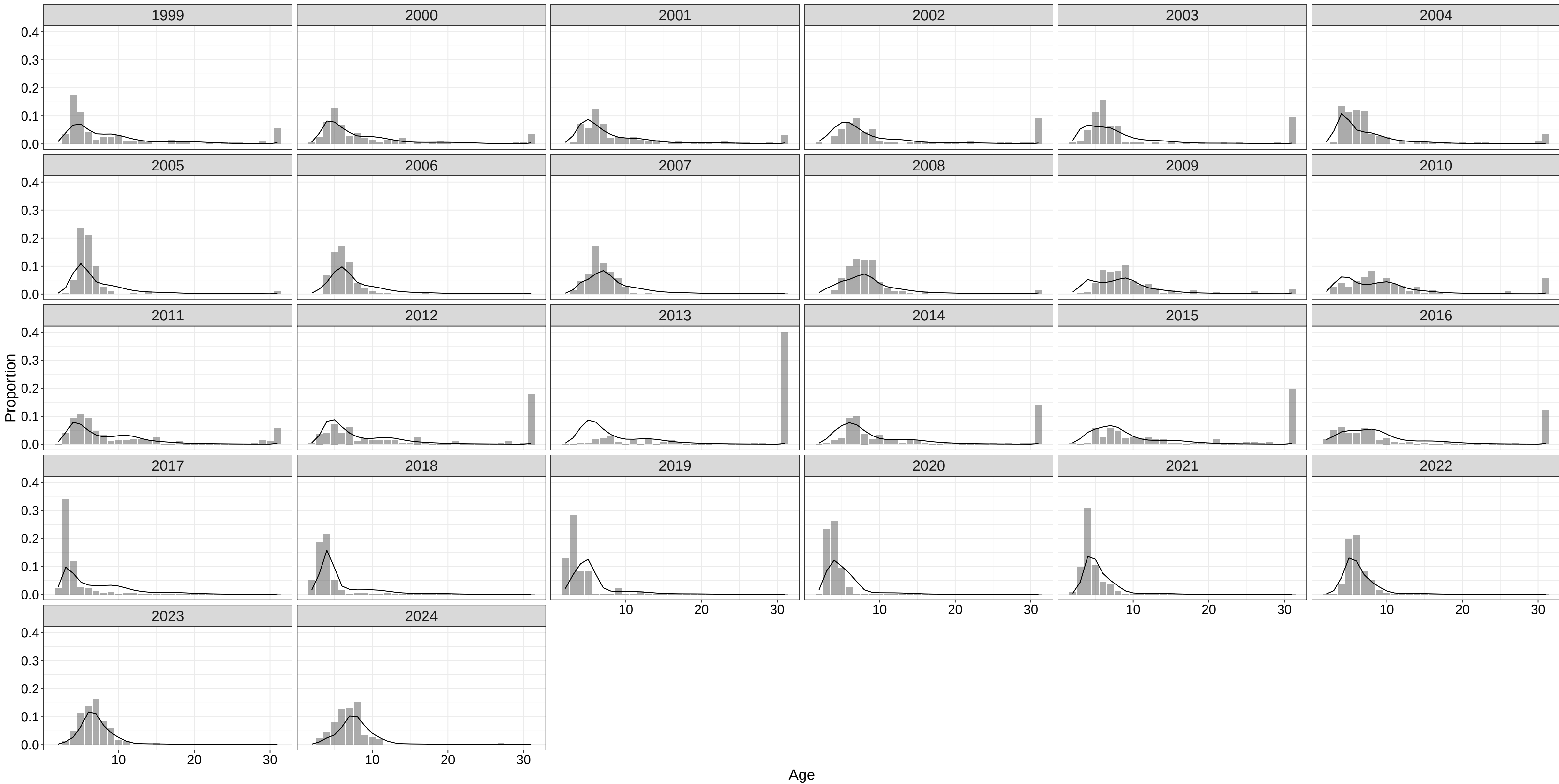
BS LLF Age Comps Male



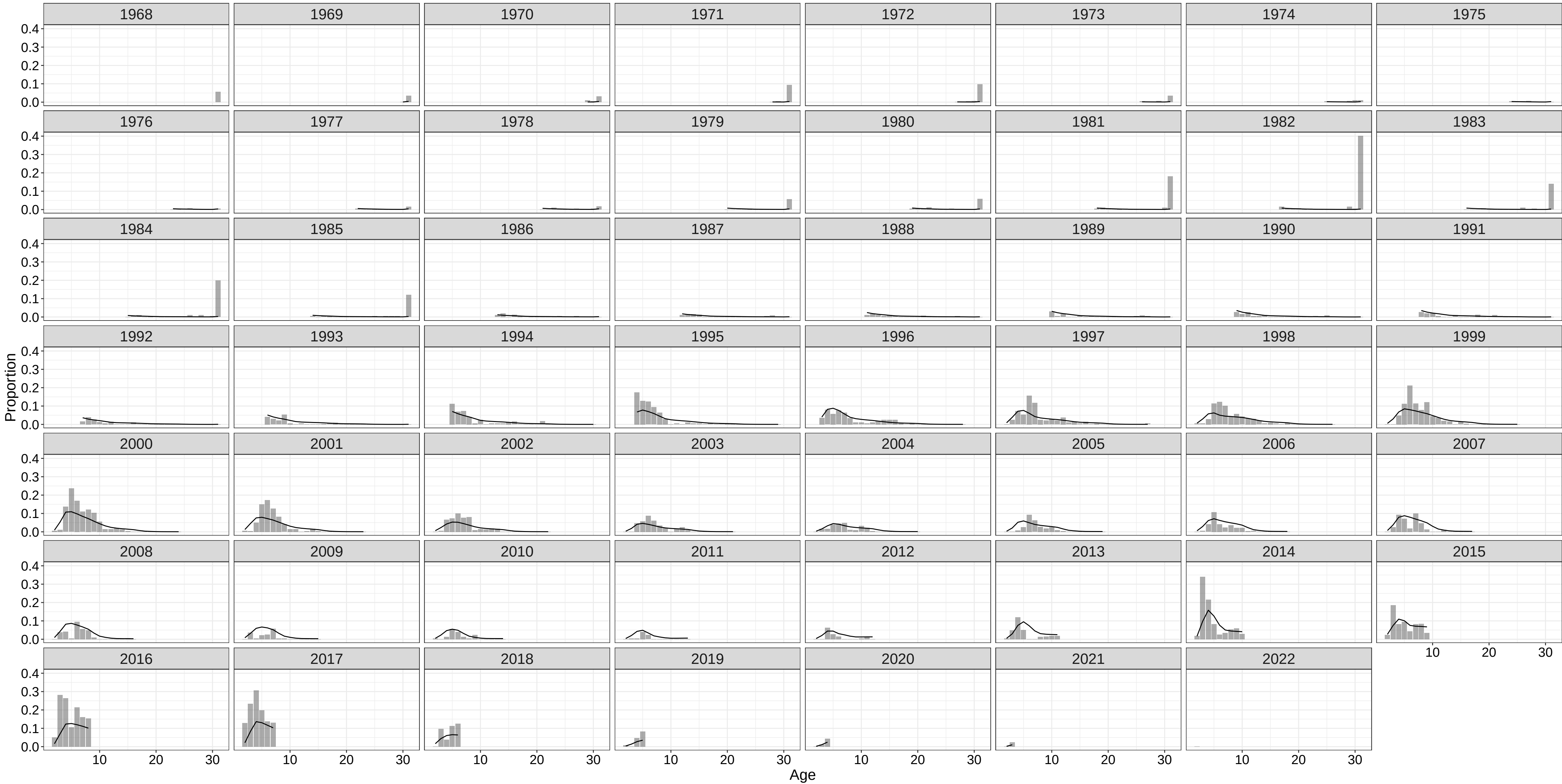
BS LLF Age Comps Cohort Fits Male



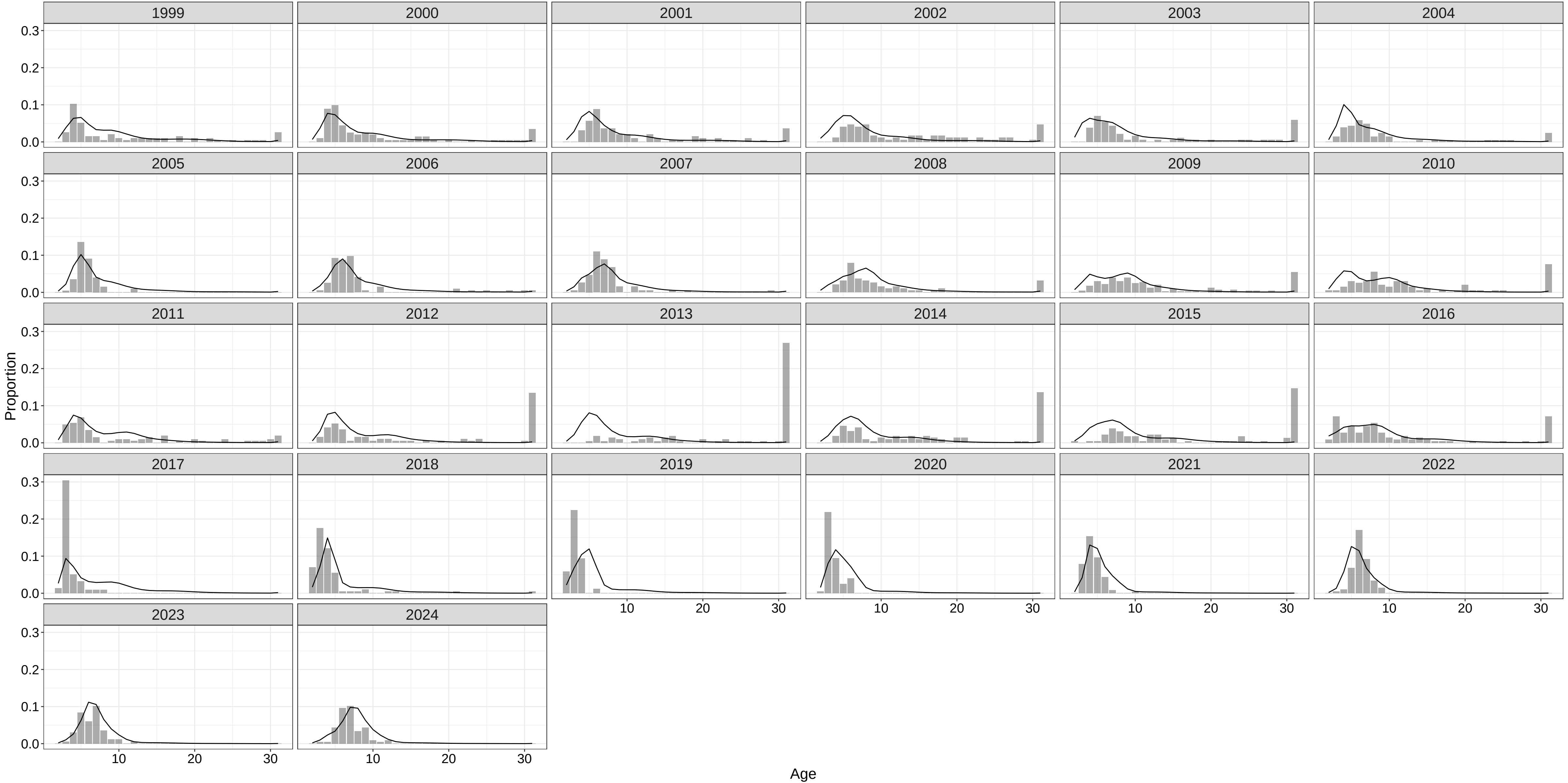
AI LLF Age Comps Female



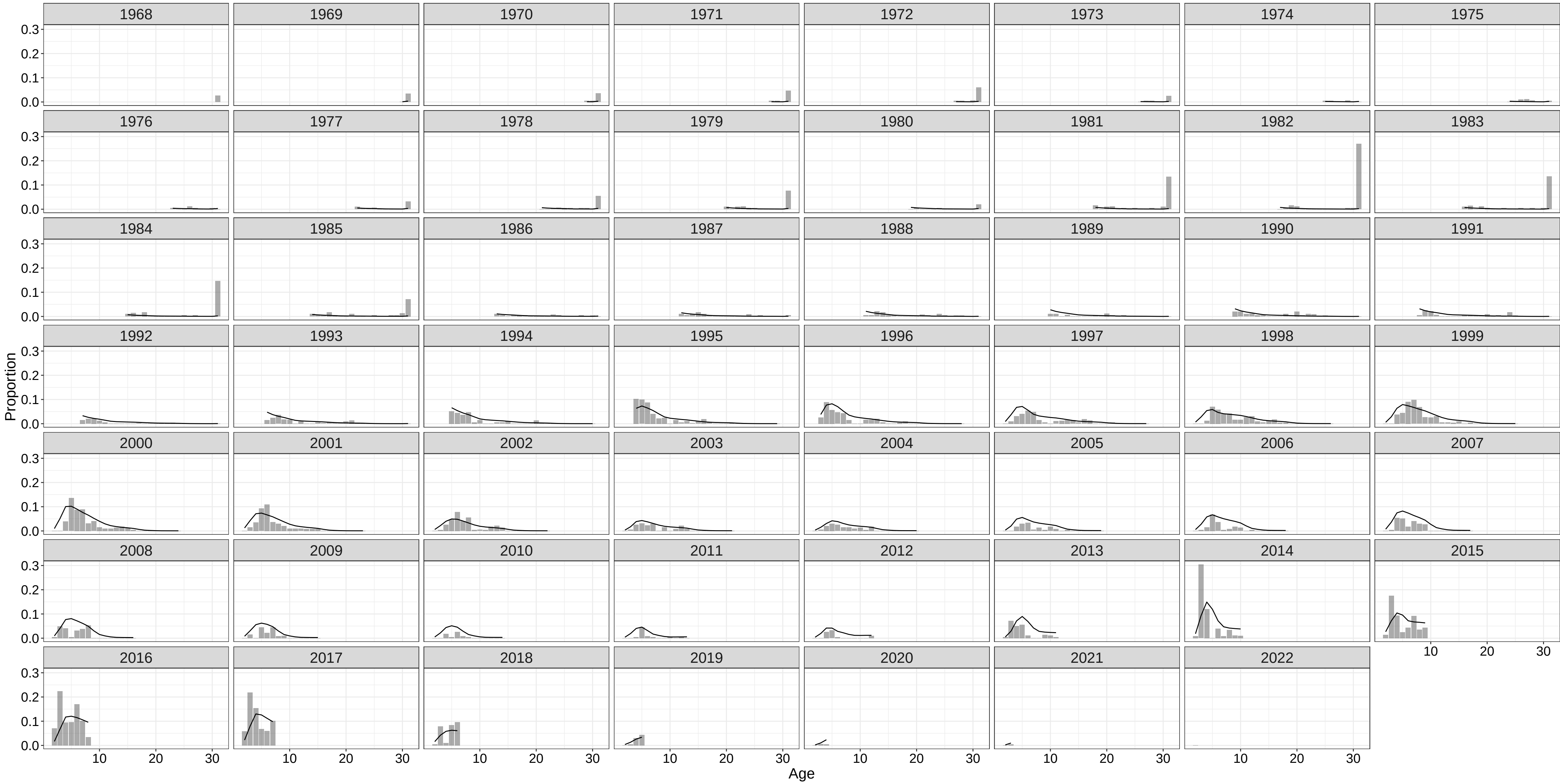
AI LLF Age Comps Cohort Fits Female



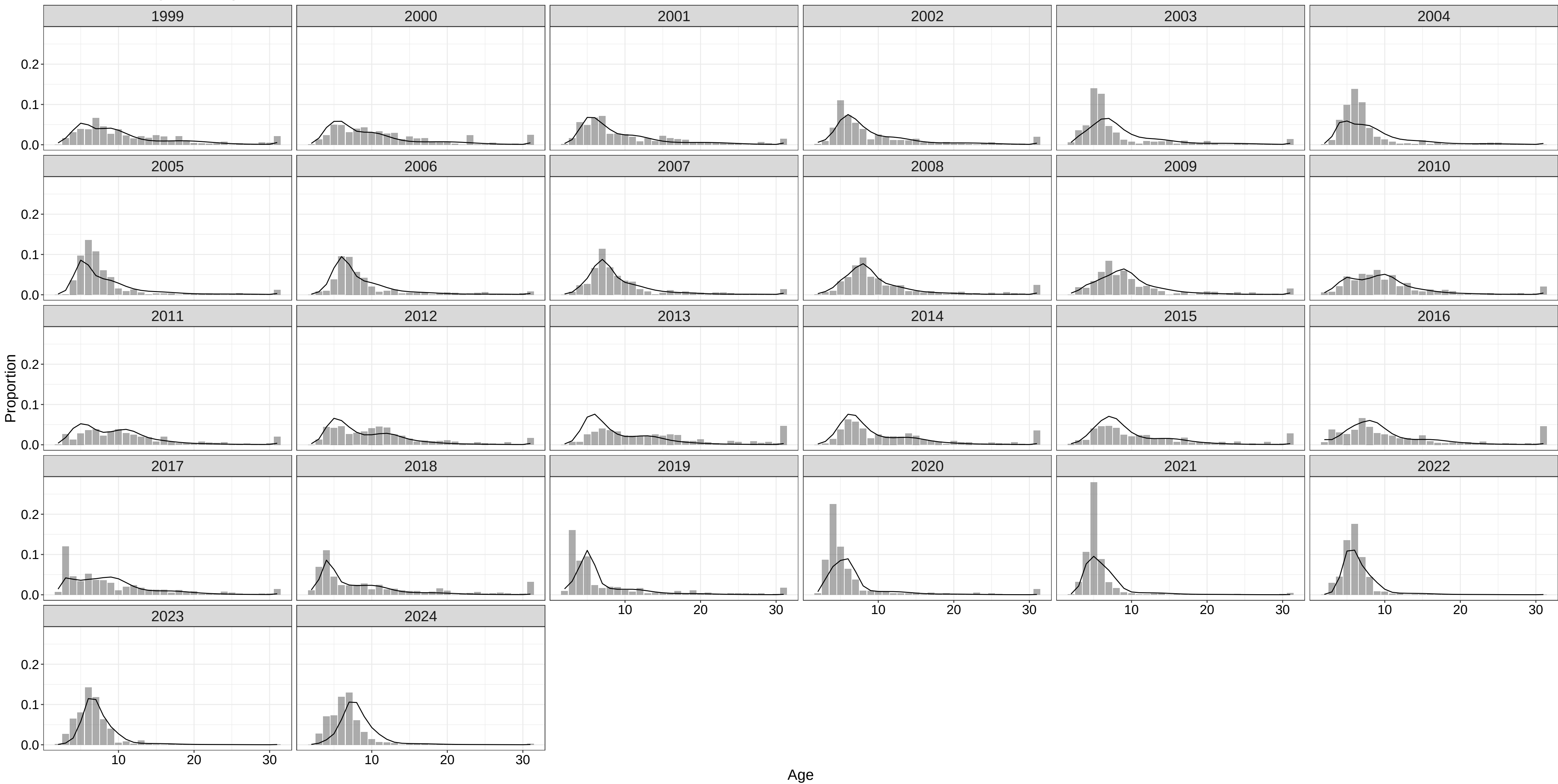
AI LLF Age Comps Male



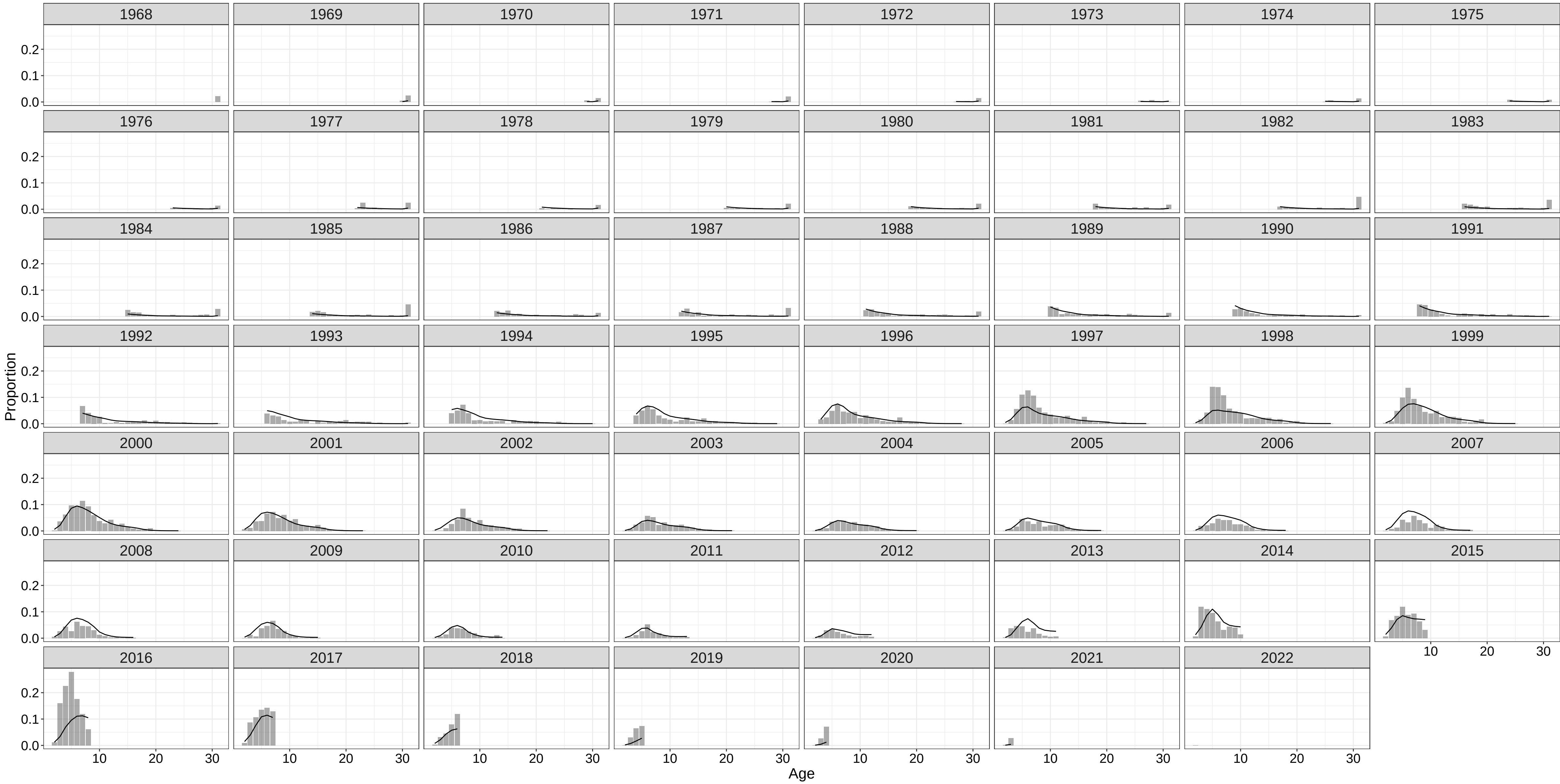
AI LLF Age Comps Cohort Fits Male



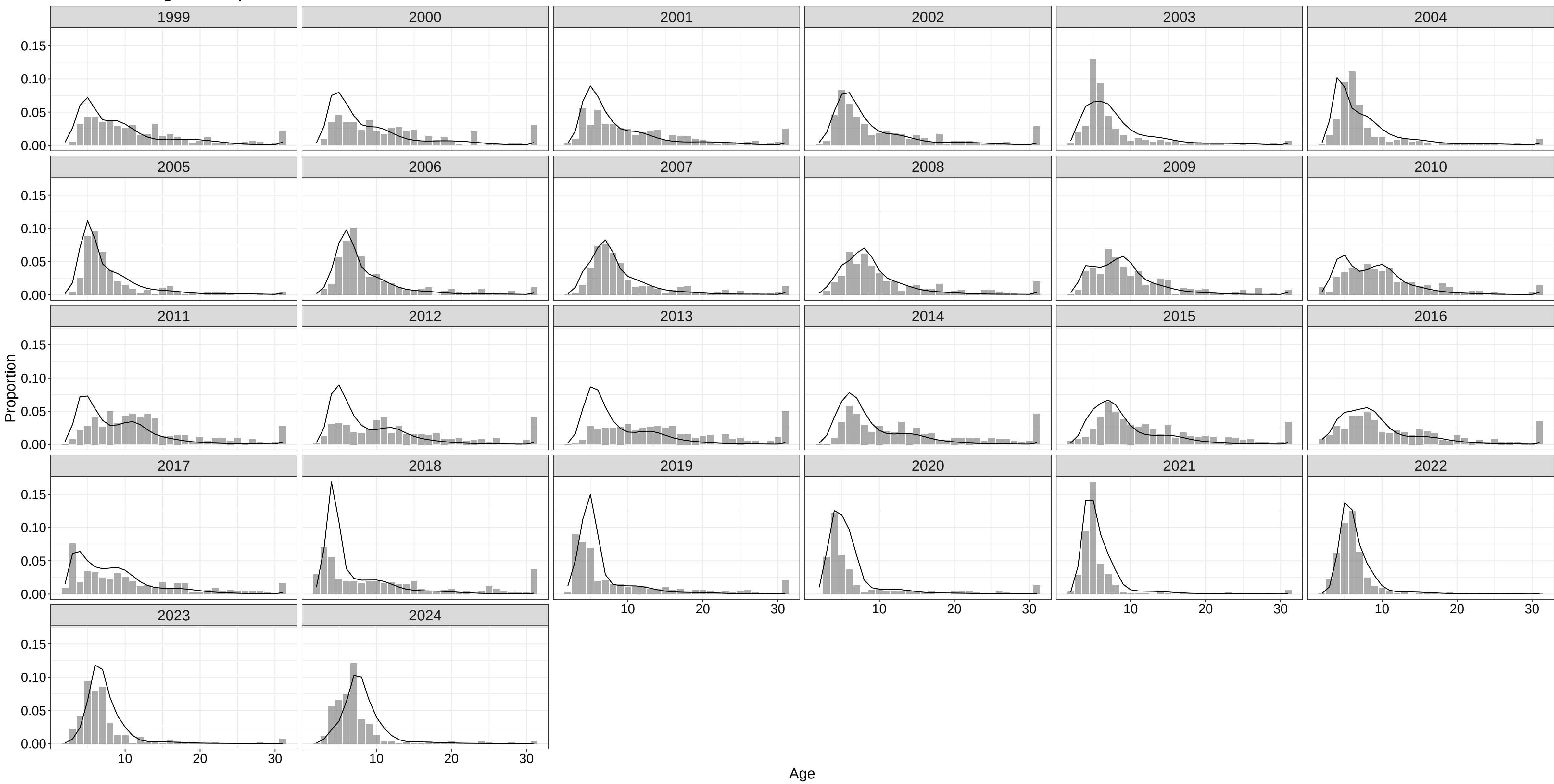
GOA LLF Age Comps Female



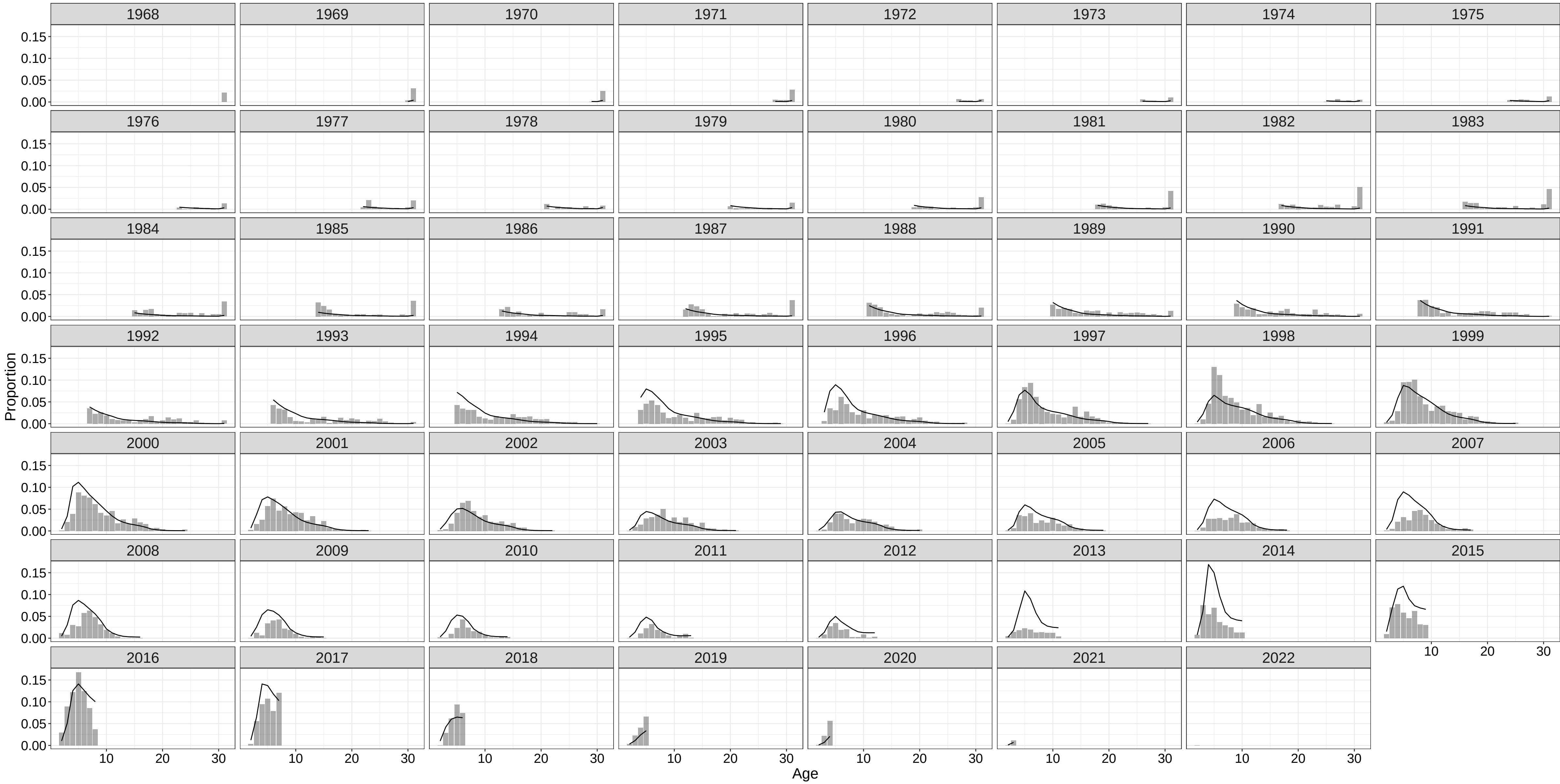
GOA LLF Age Comps Cohort Fits Female



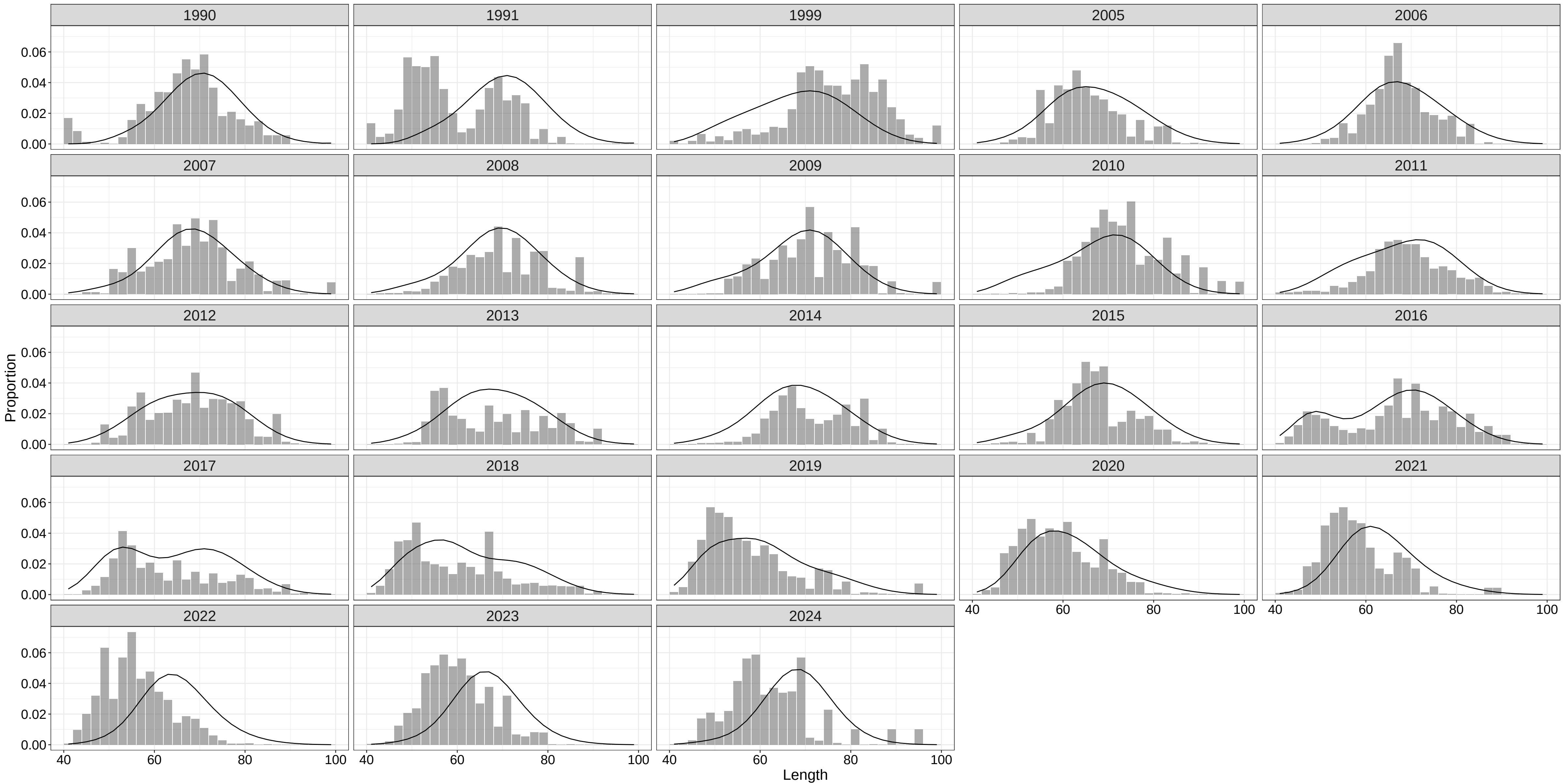
GOA LLF Age Comps Male



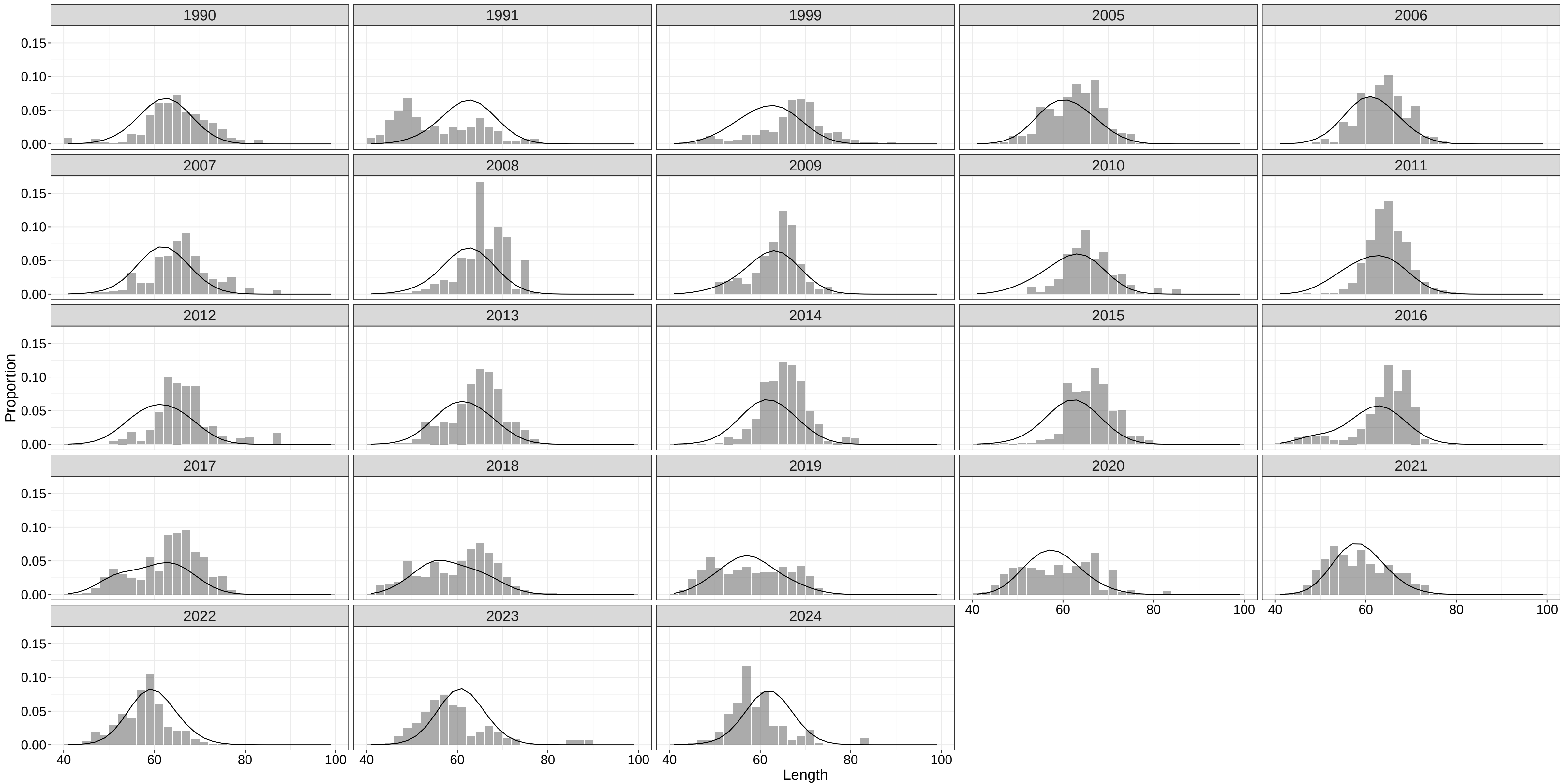
GOA LLF Age Comps Cohort Fits Male



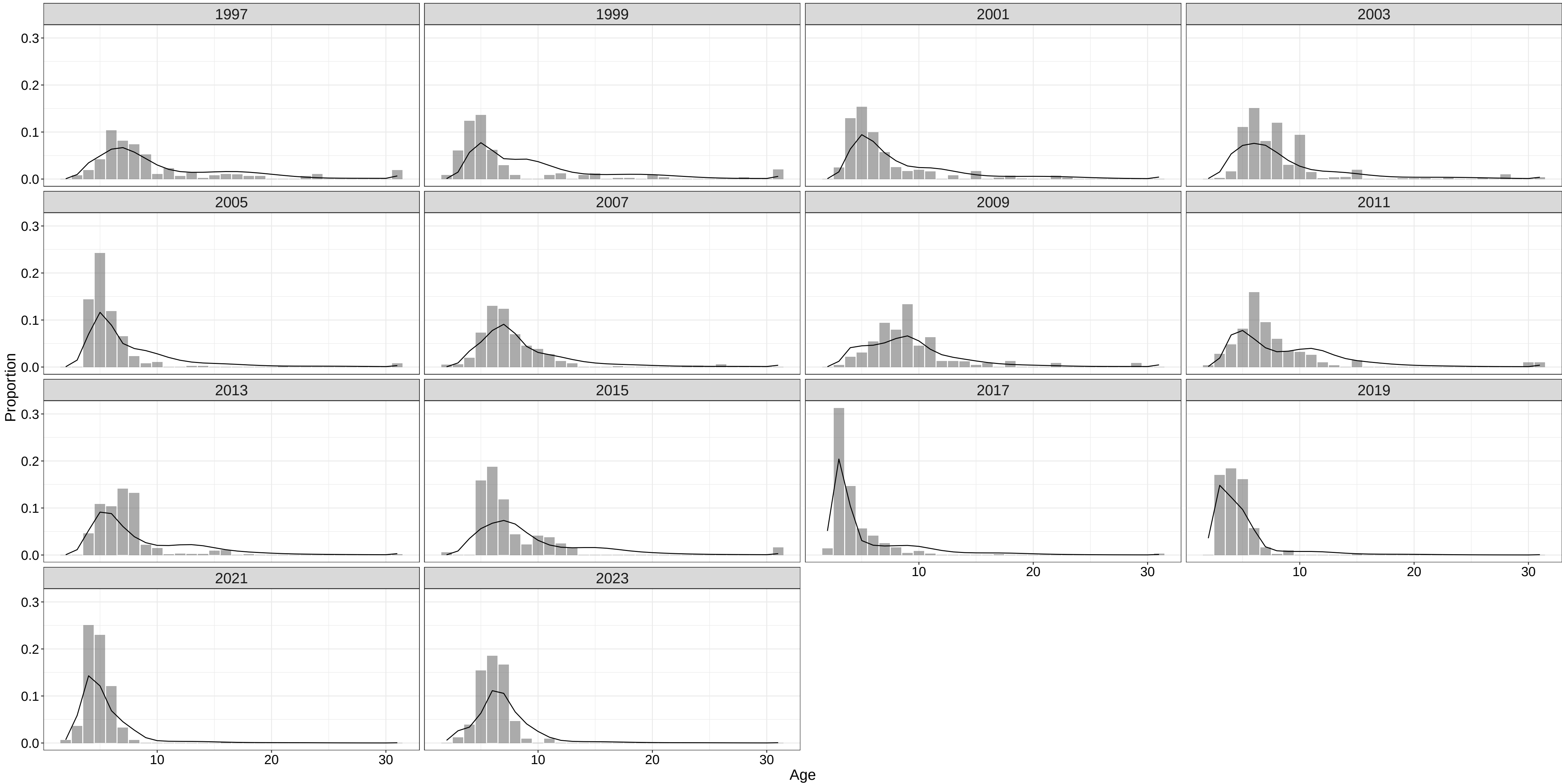
TF Length Comps Female



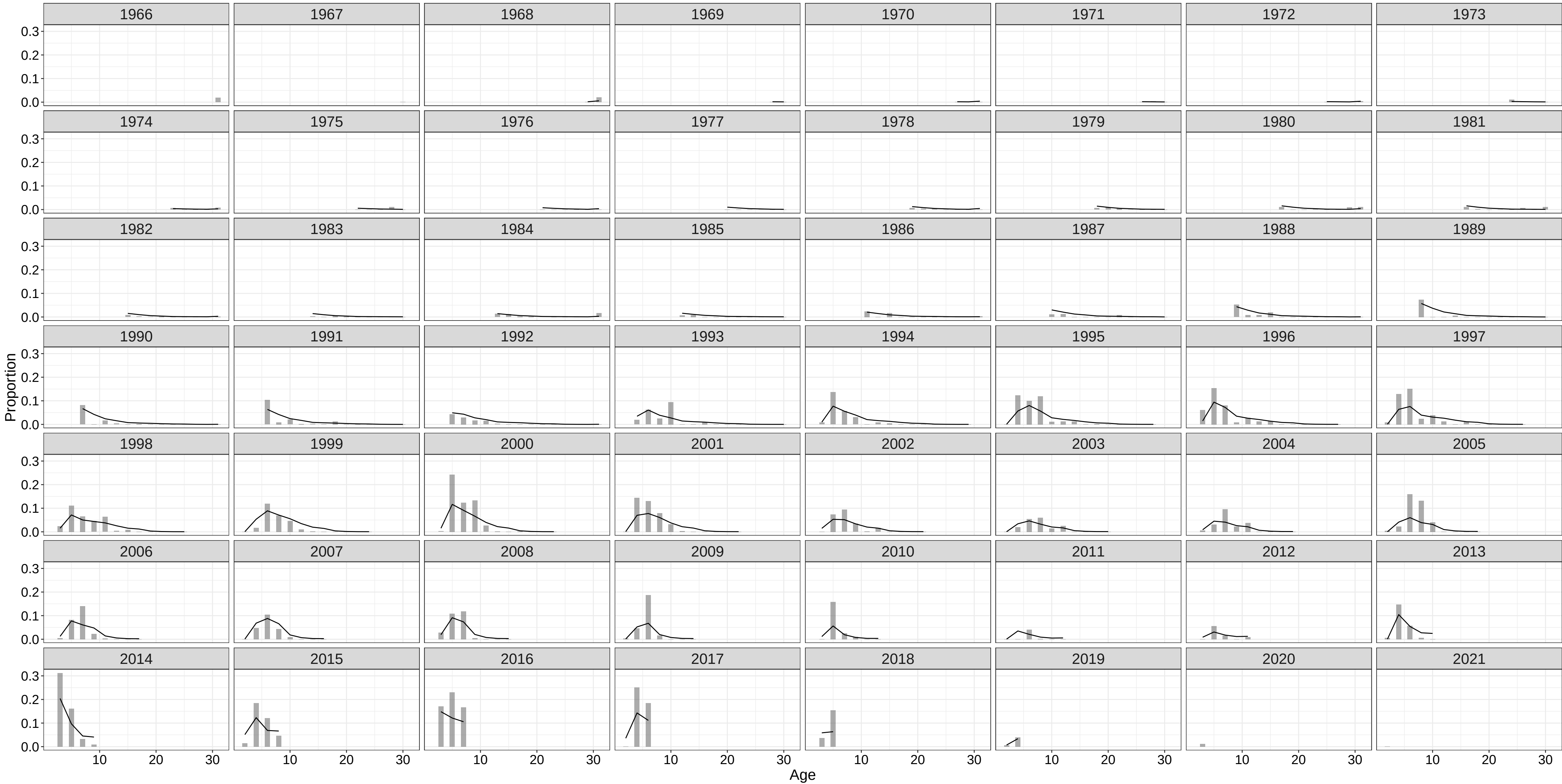
TF Length Comps Male



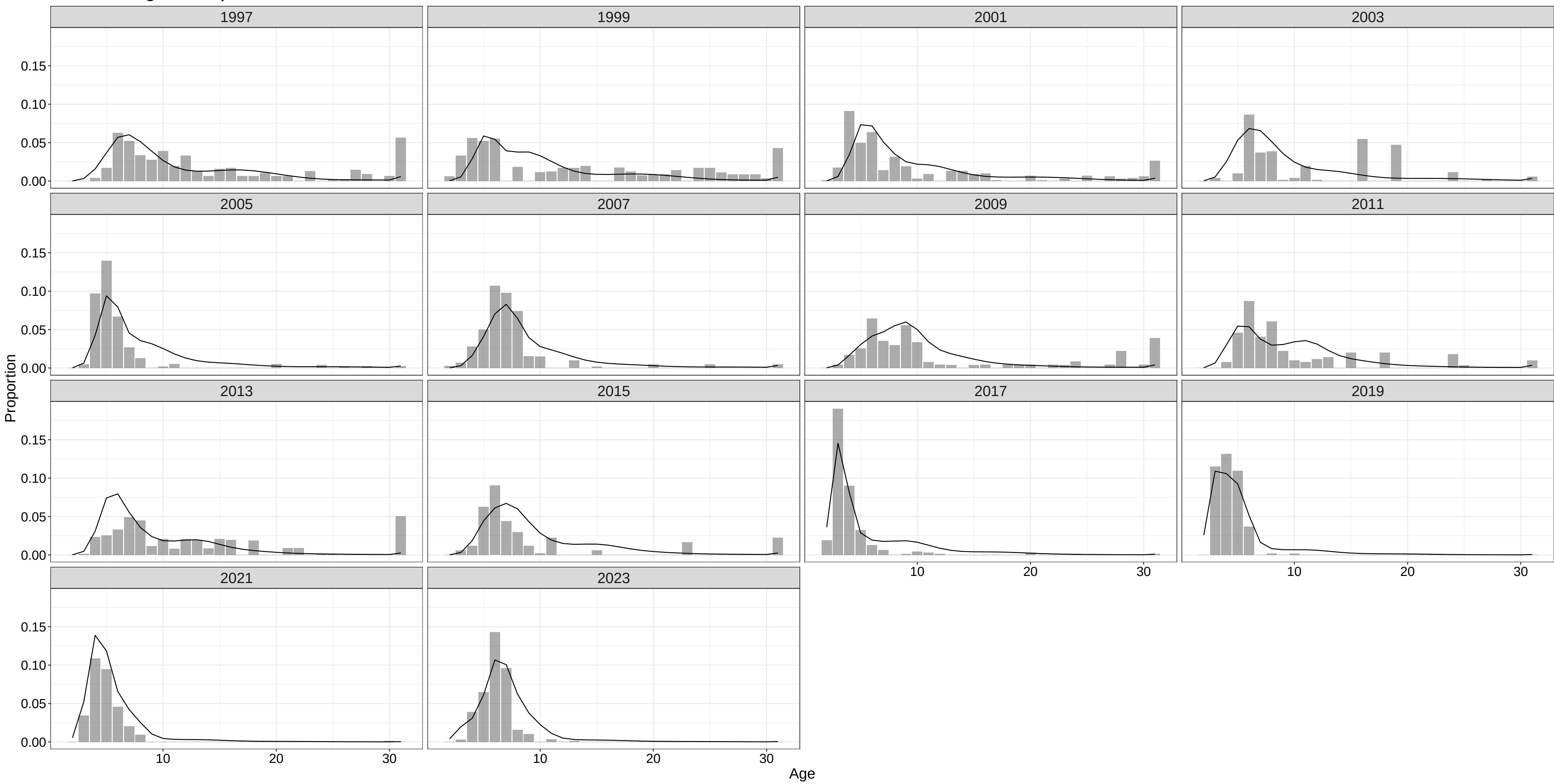
BS LLS Age Comps Female



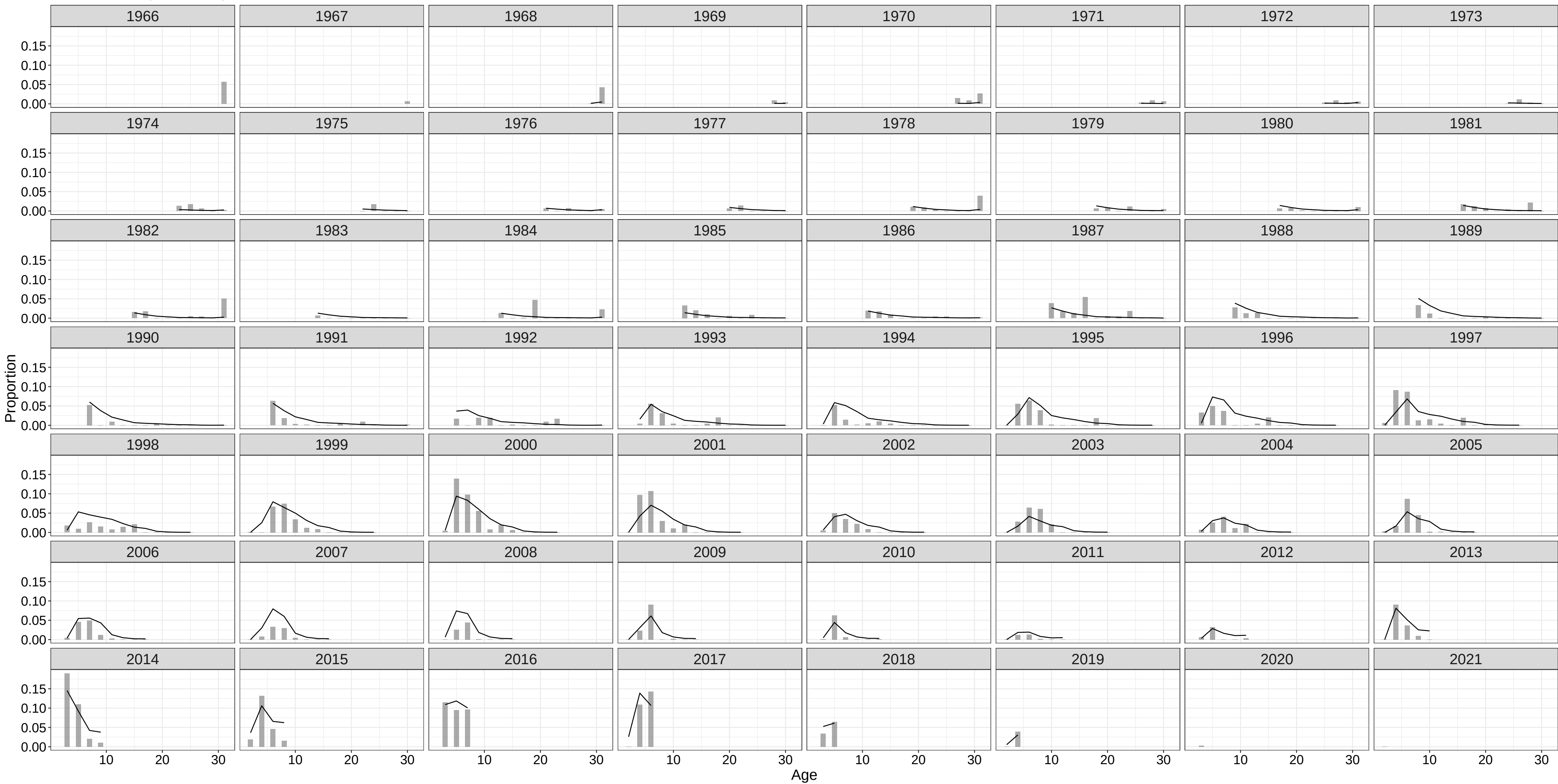
BS LLS Age Comps Cohort Fit Female



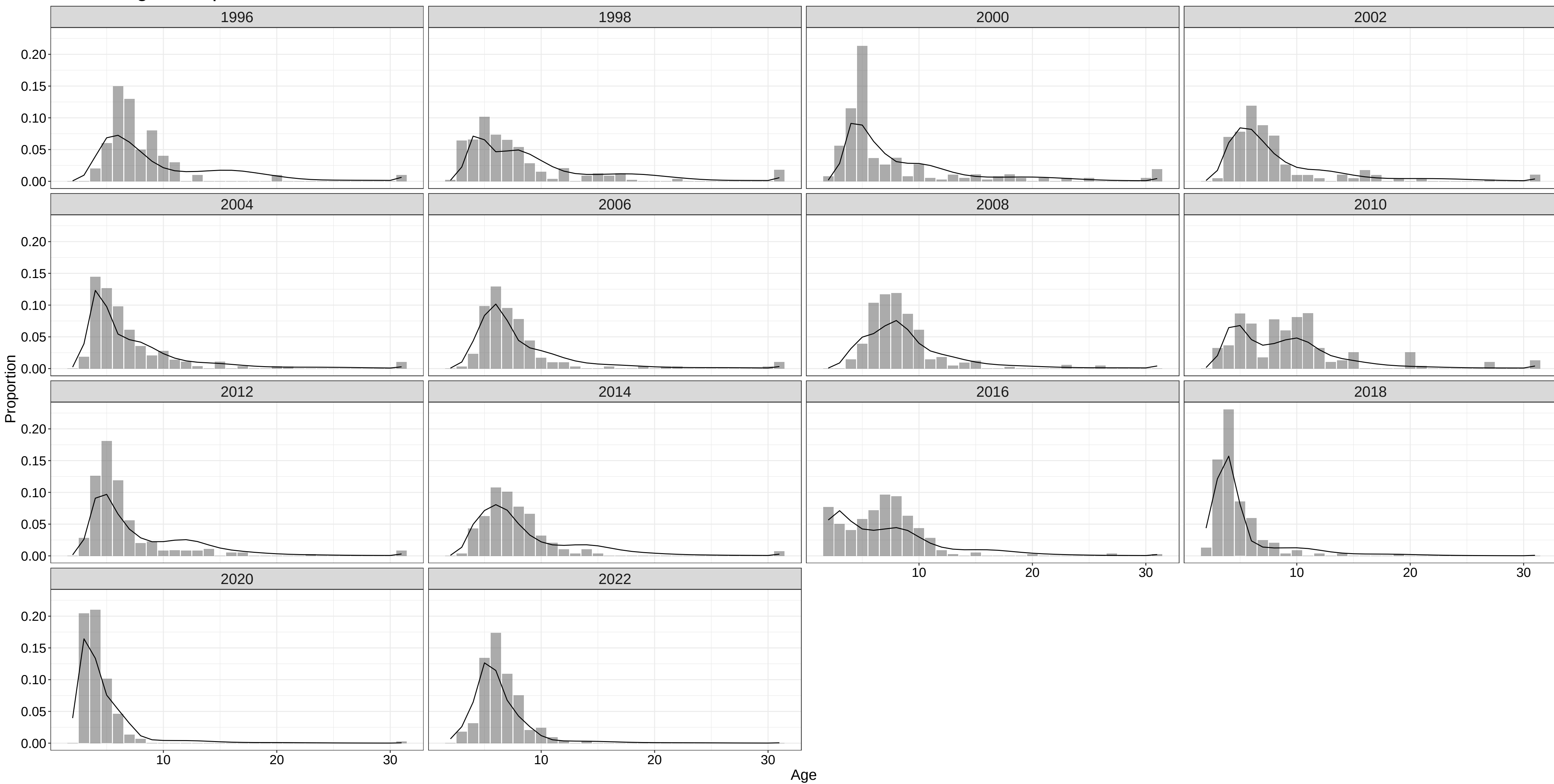
BS LLS Age Comps Male



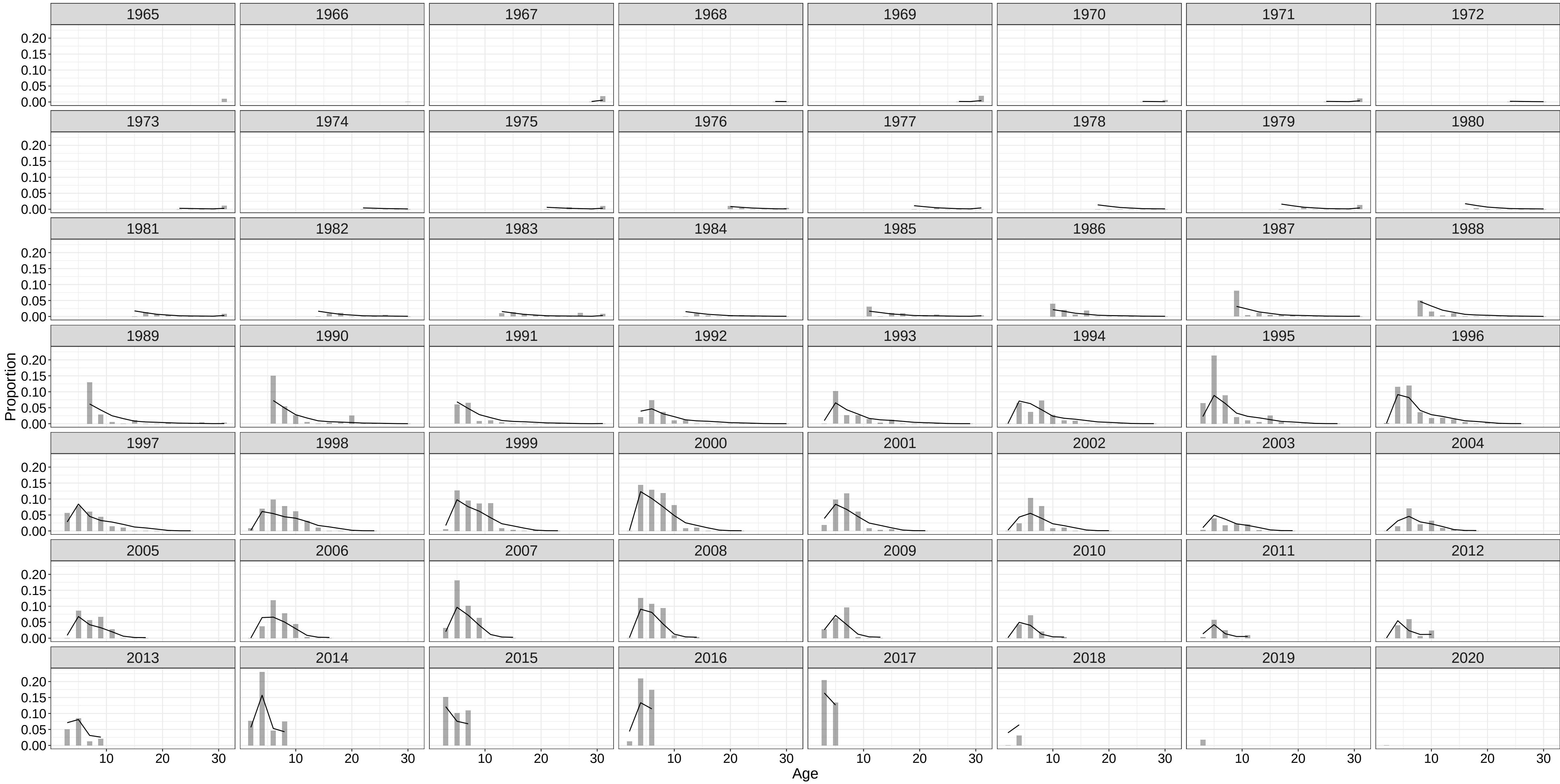
BS LLS Age Comps Cohort Fit Male



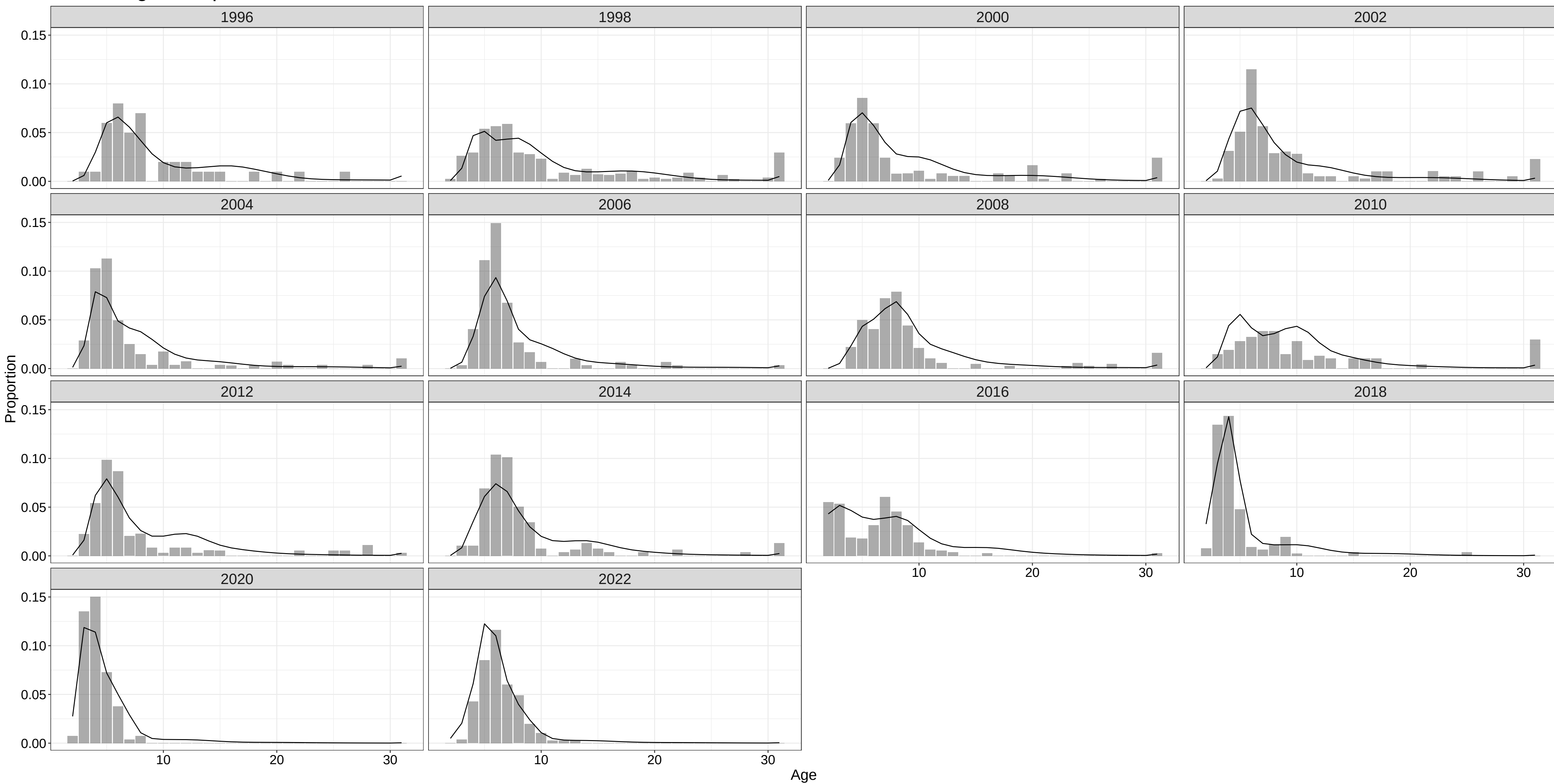
AI LLS Age Comps Female



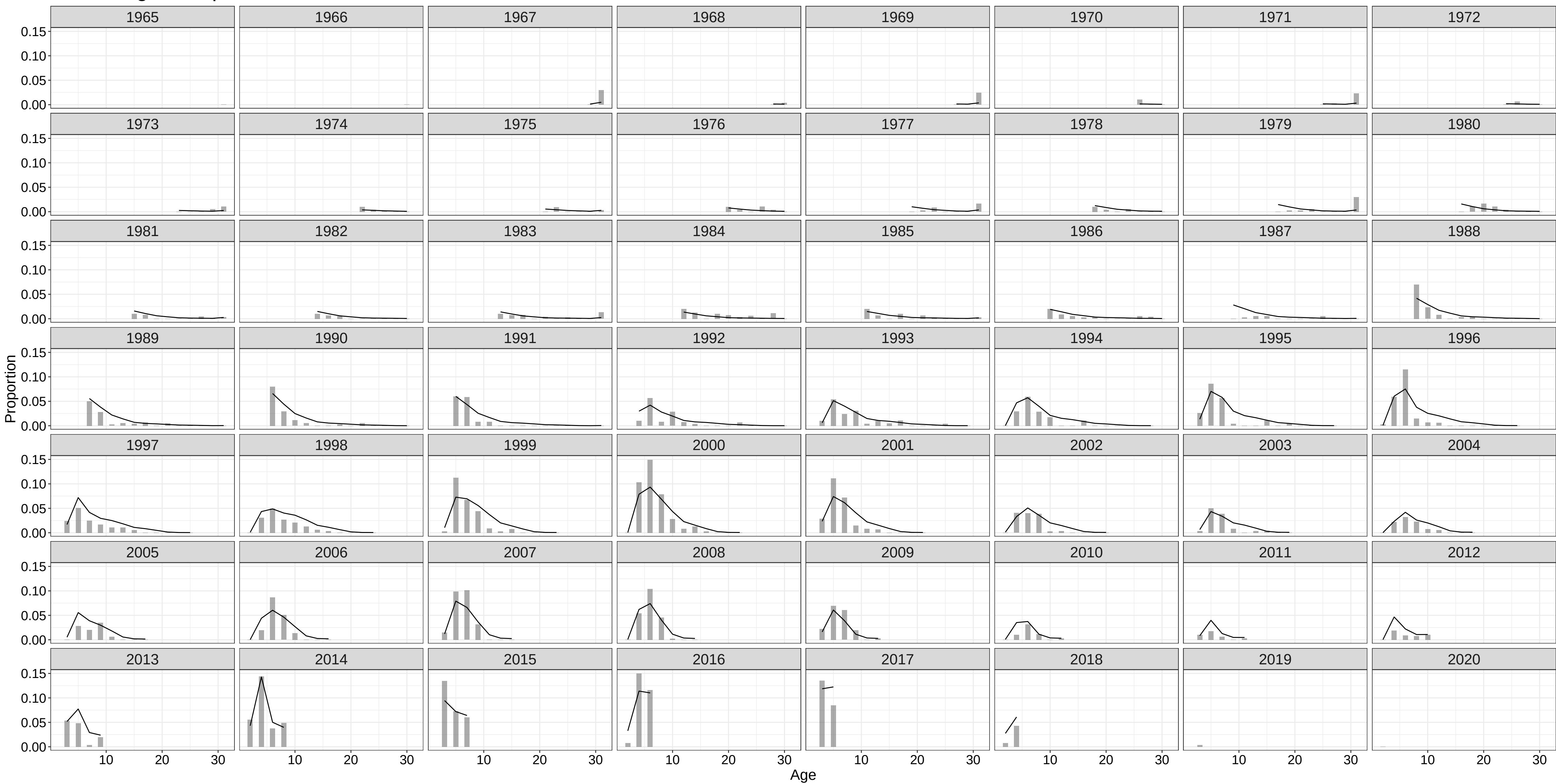
AI LLS Age Comps Cohort Fit Female



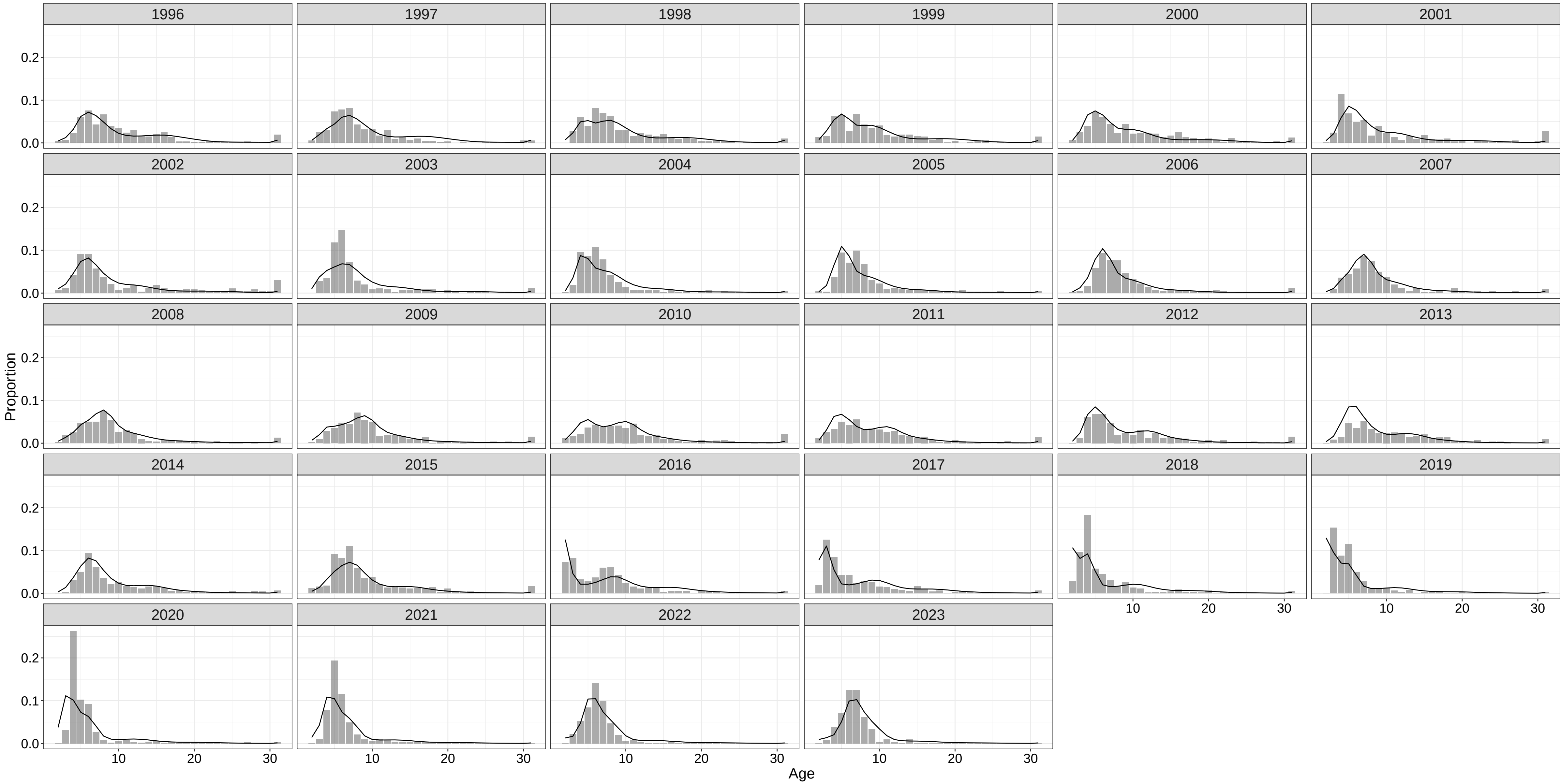
AI LLS Age Comps Male



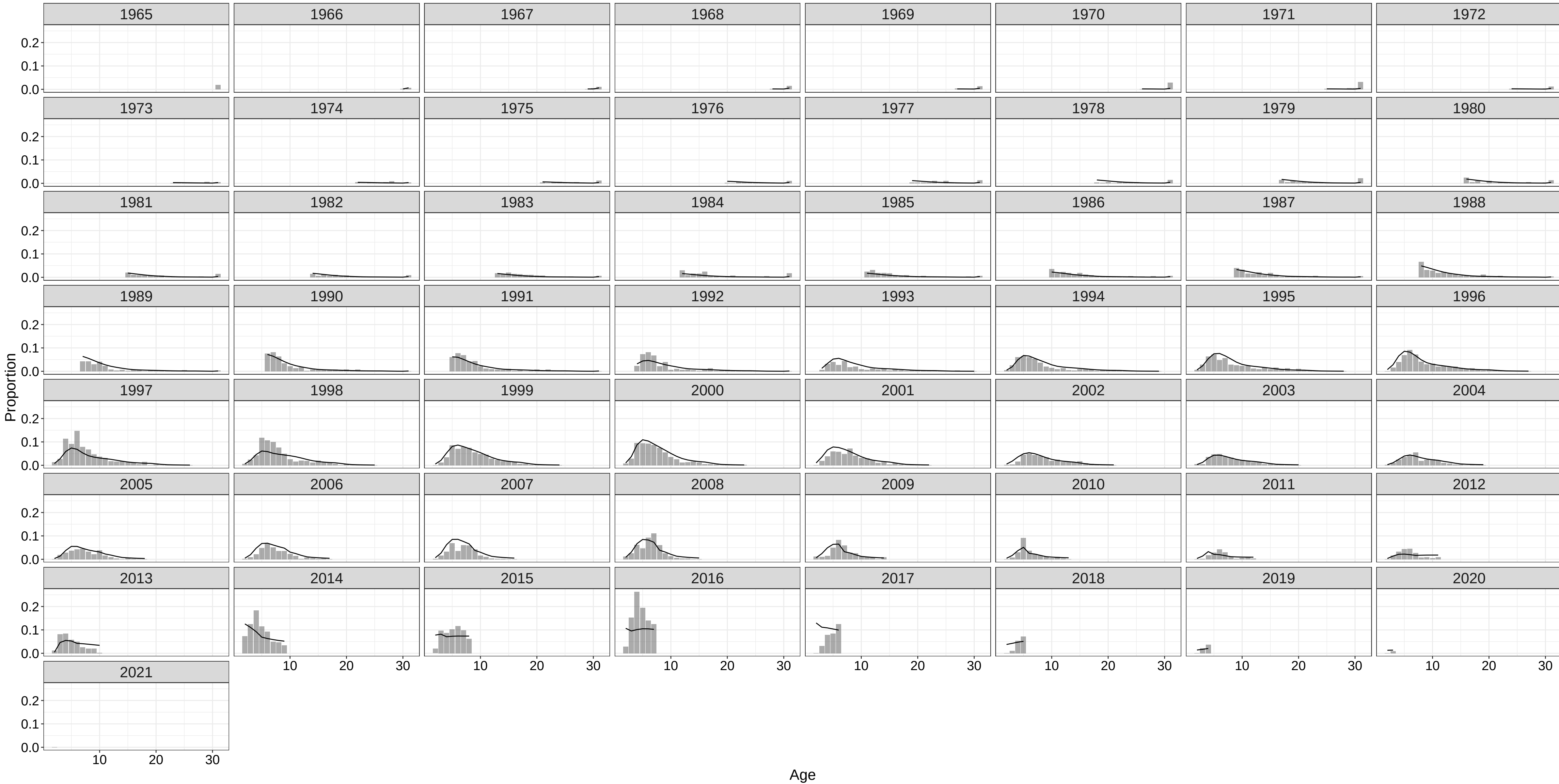
AI LLS Age Comps Cohort Fit Male



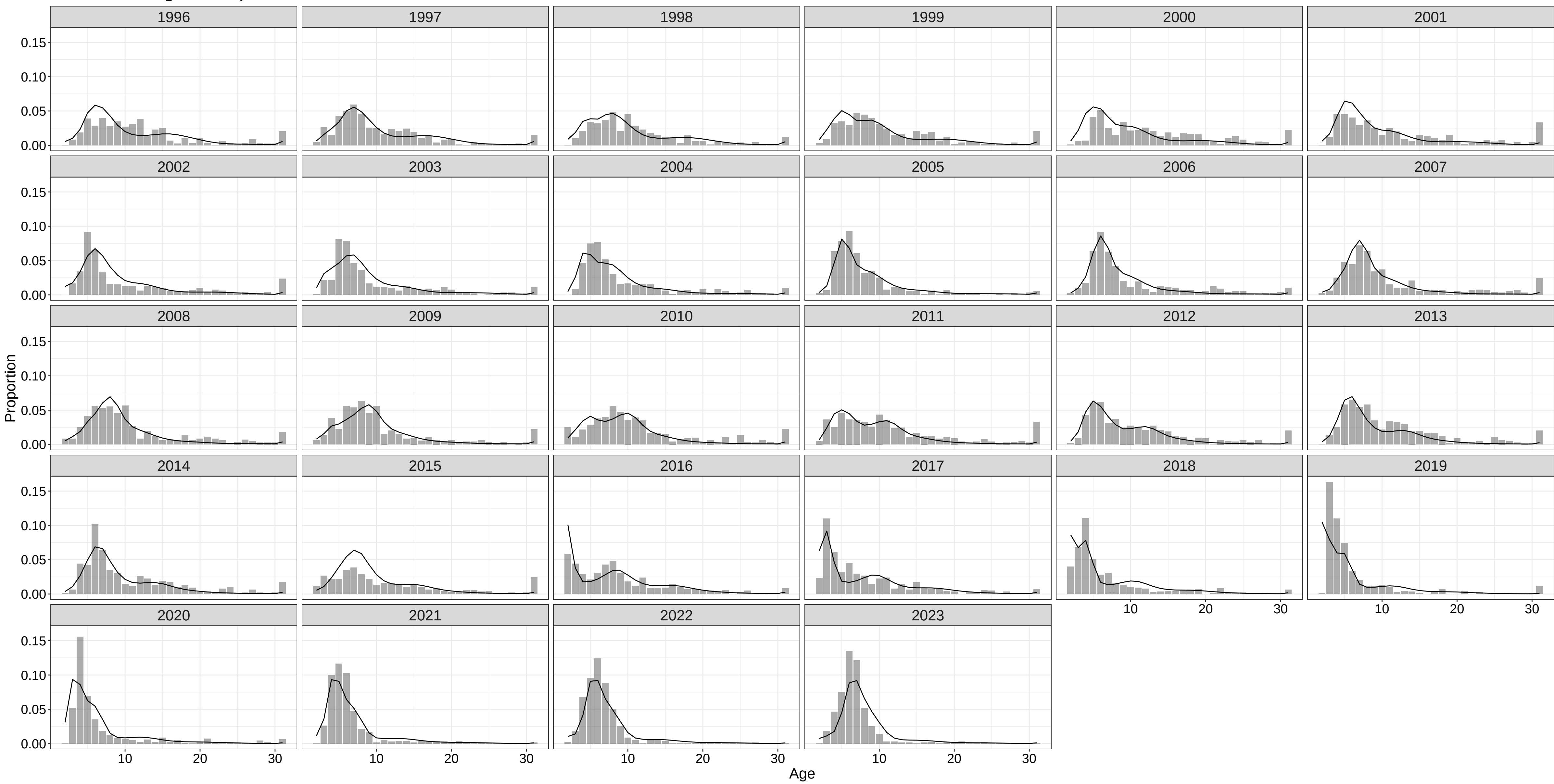
GOA LLS Age Comps Female



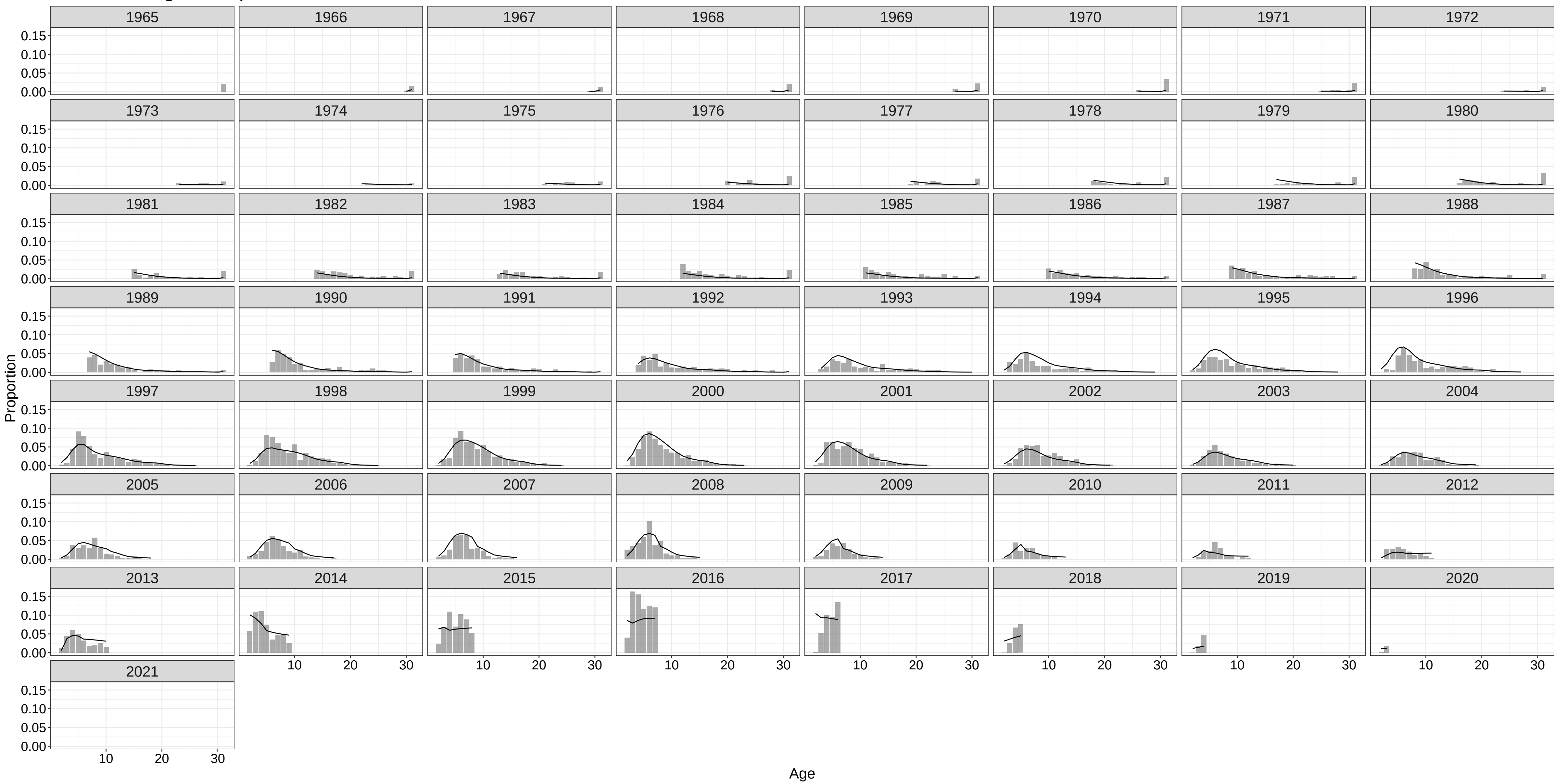
GOA LLS Age Comps Cohort Fit Female



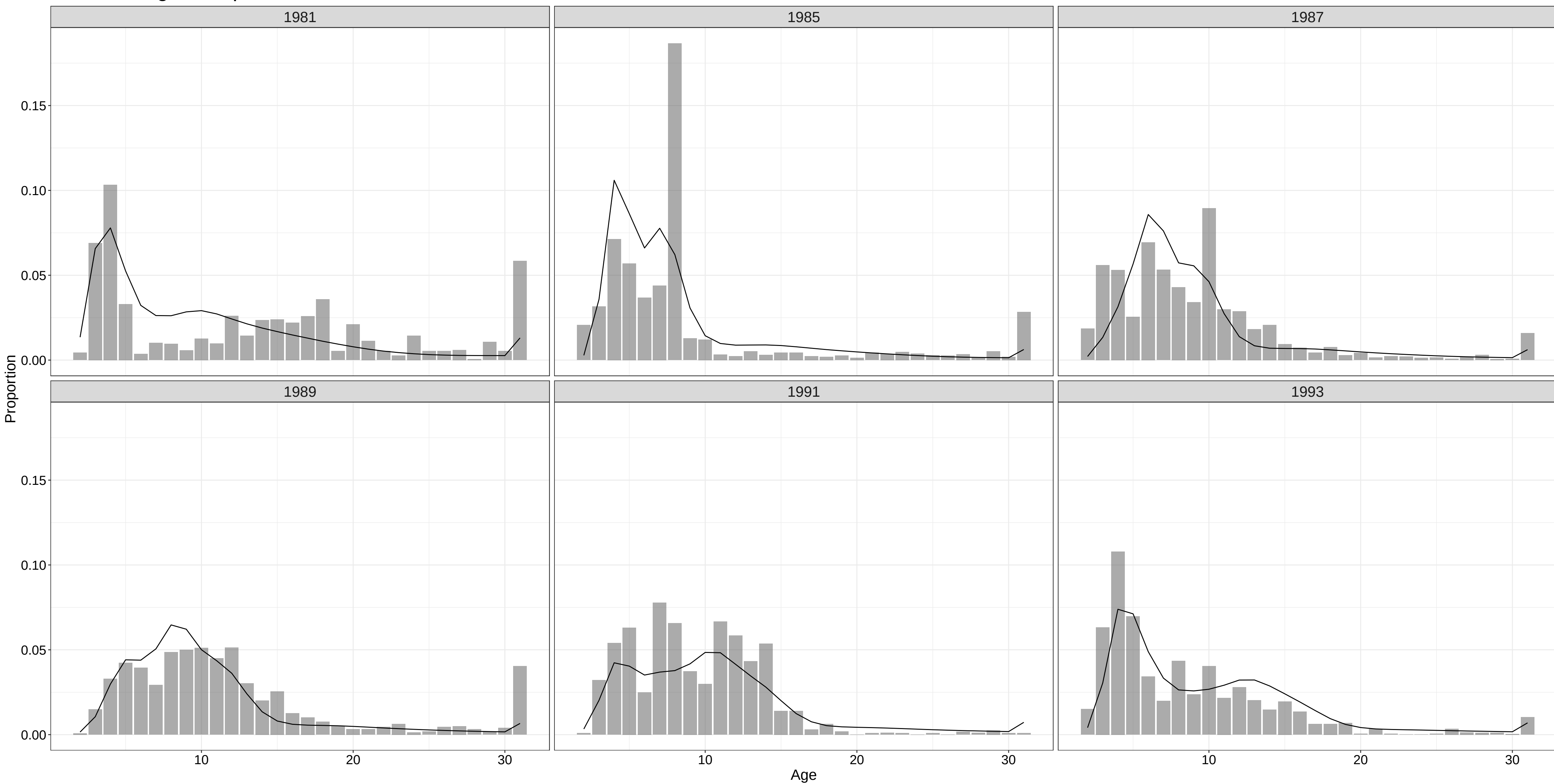
GOA LLS Age Comps Male



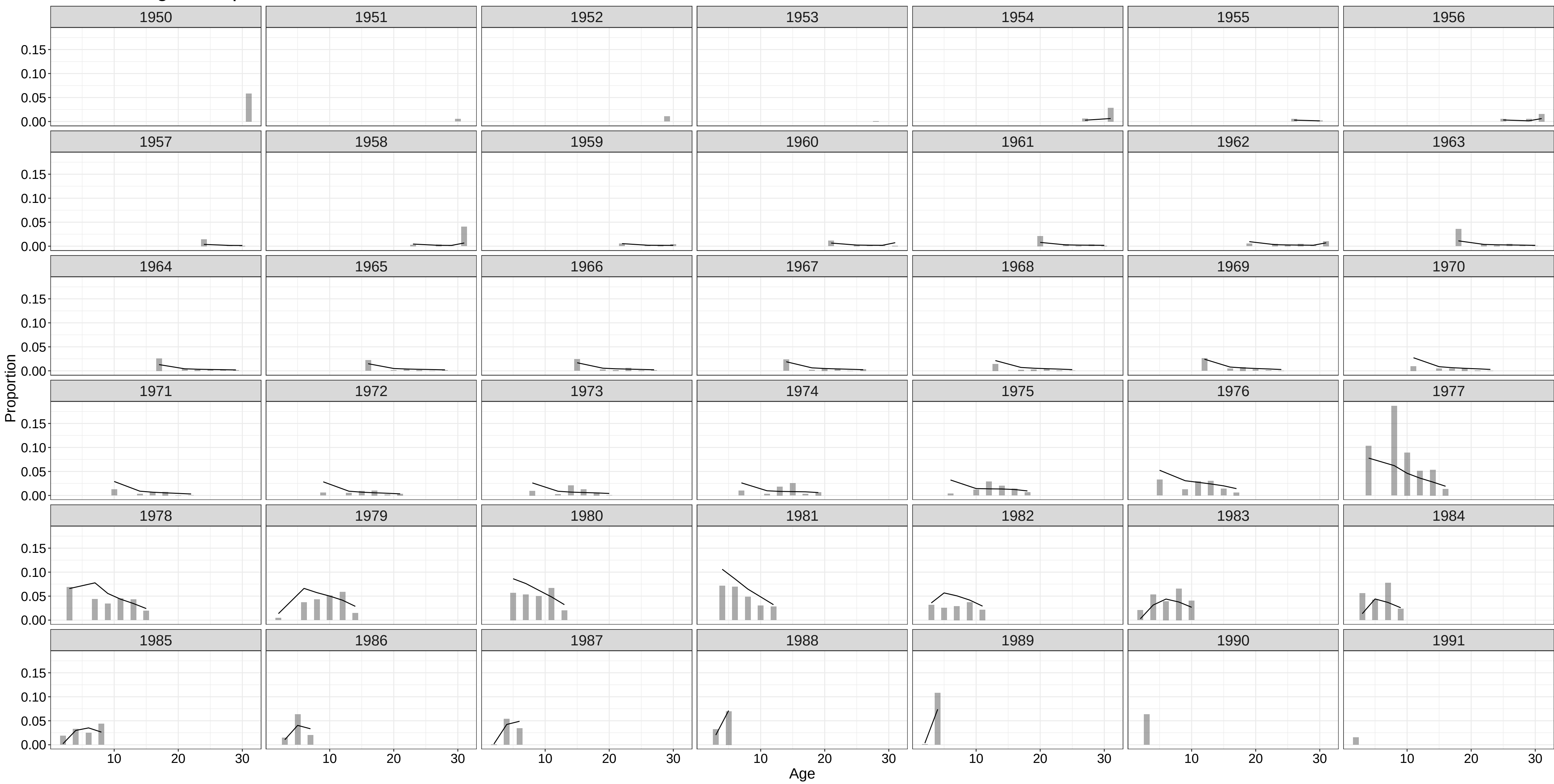
GOA LLS Age Comps Cohort Fit Male



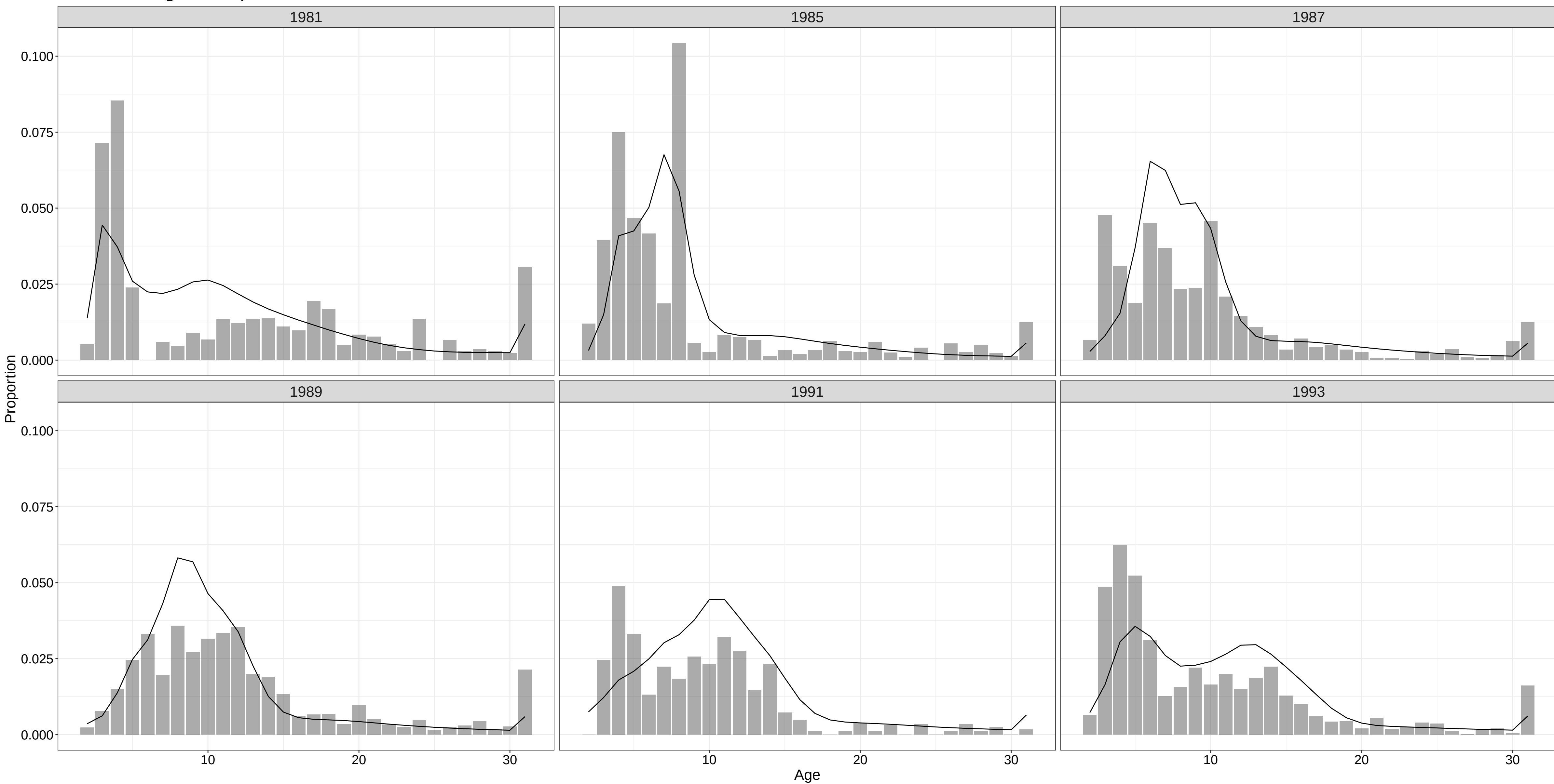
JPN LLS Age Comps Female



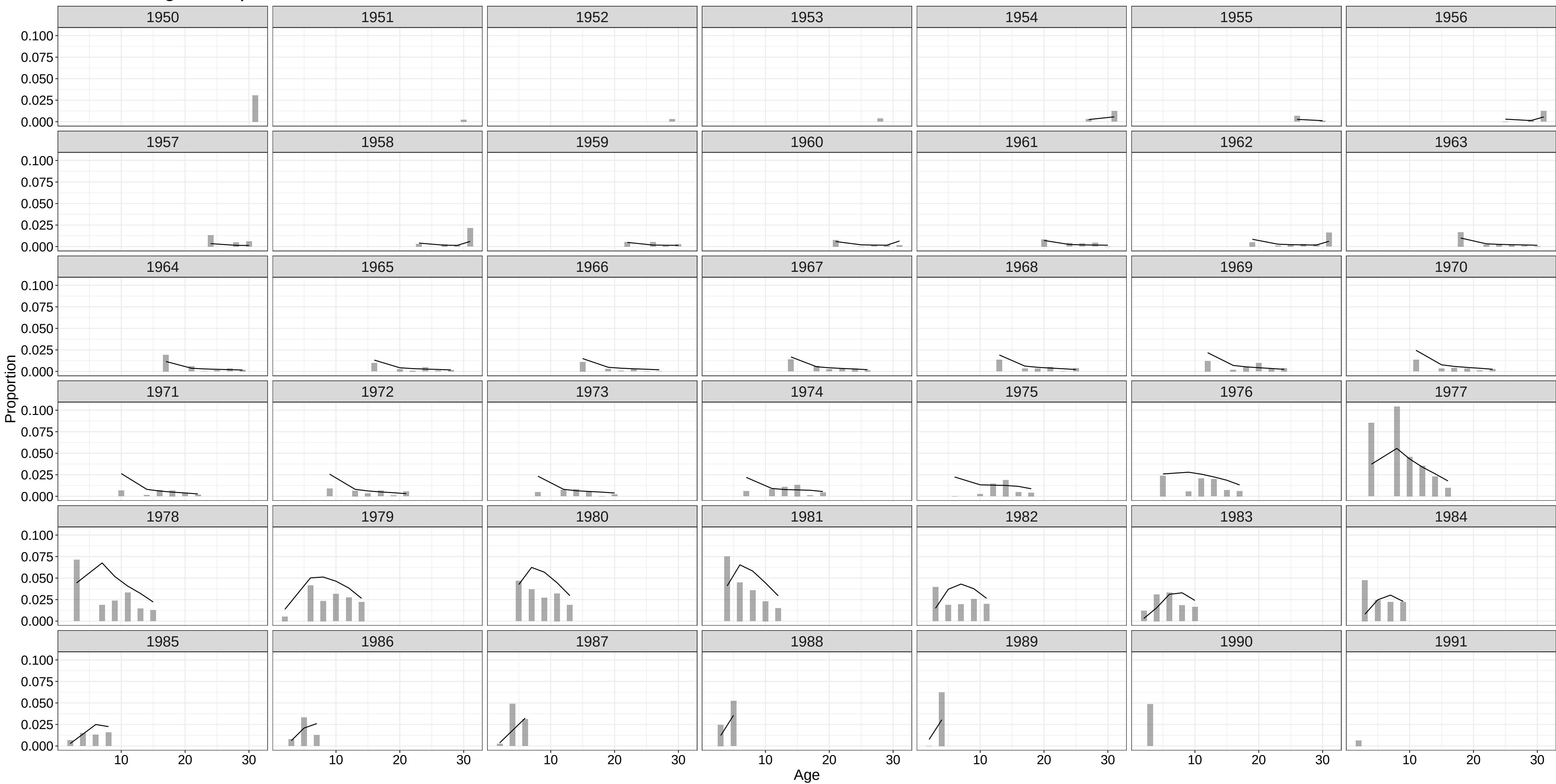
JPN LLS Age Comps Cohort Fit Female



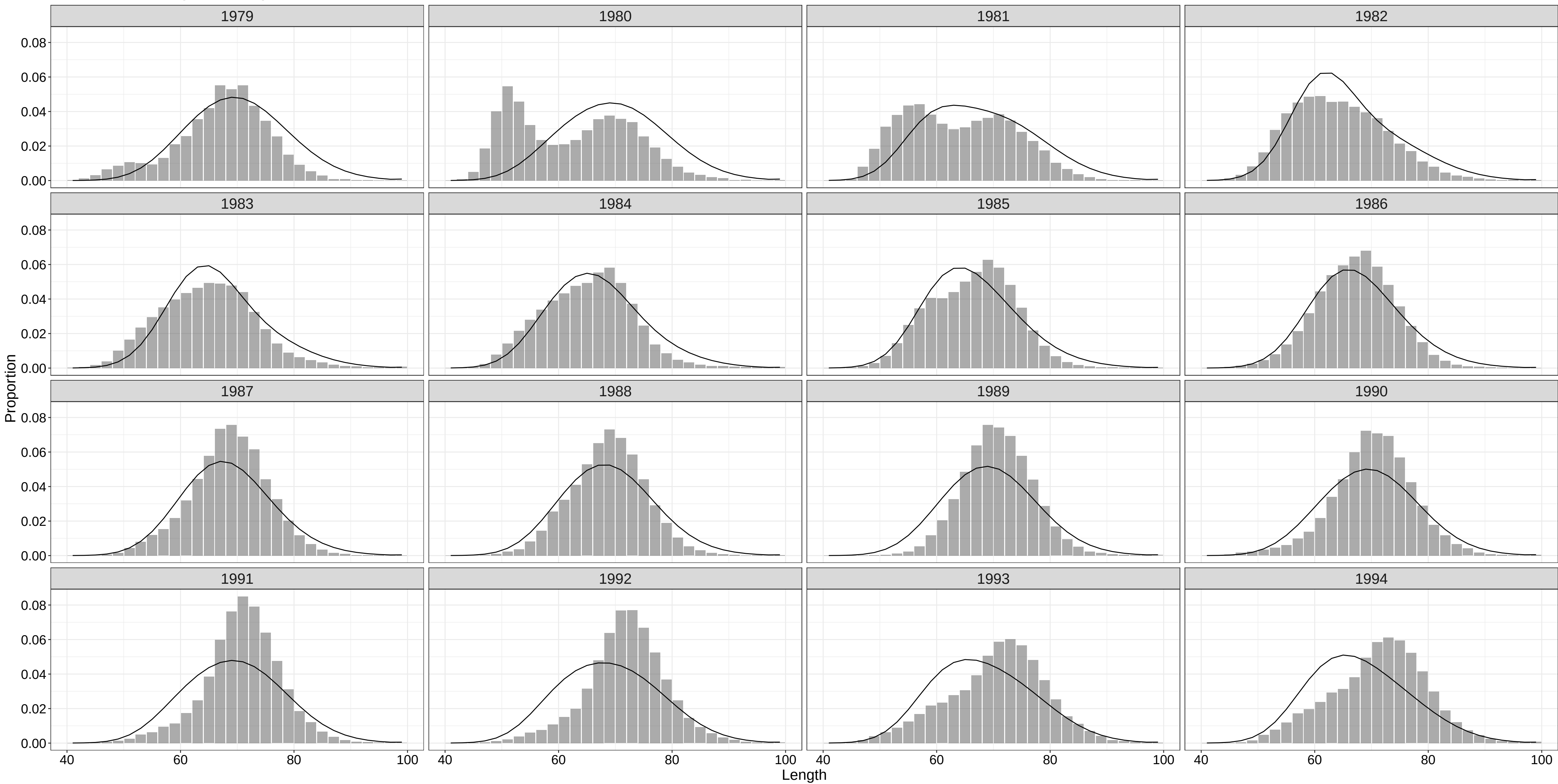
JPN LLS Age Comps Male



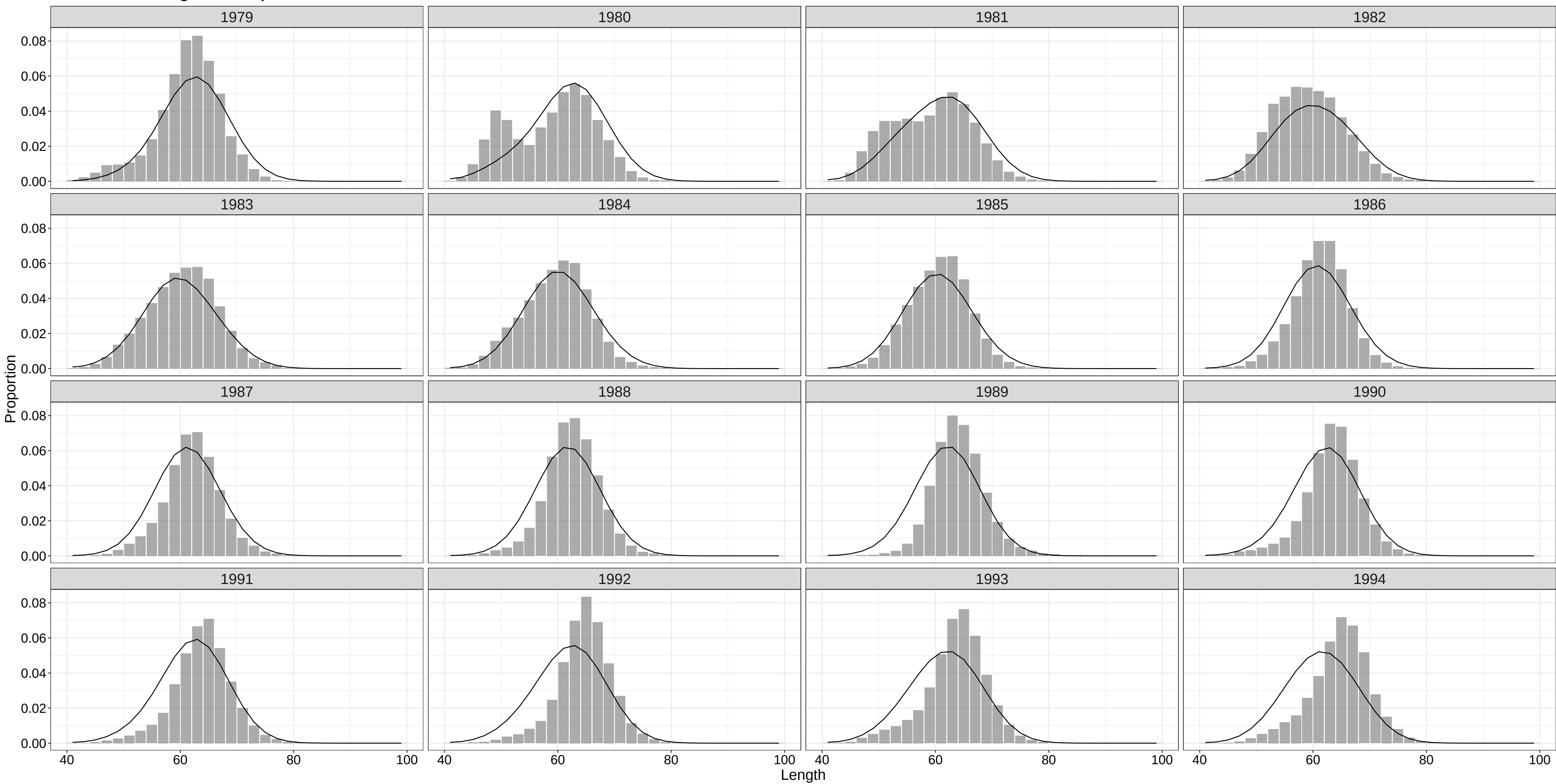
JPN LLS Age Comps Cohort Fit Male



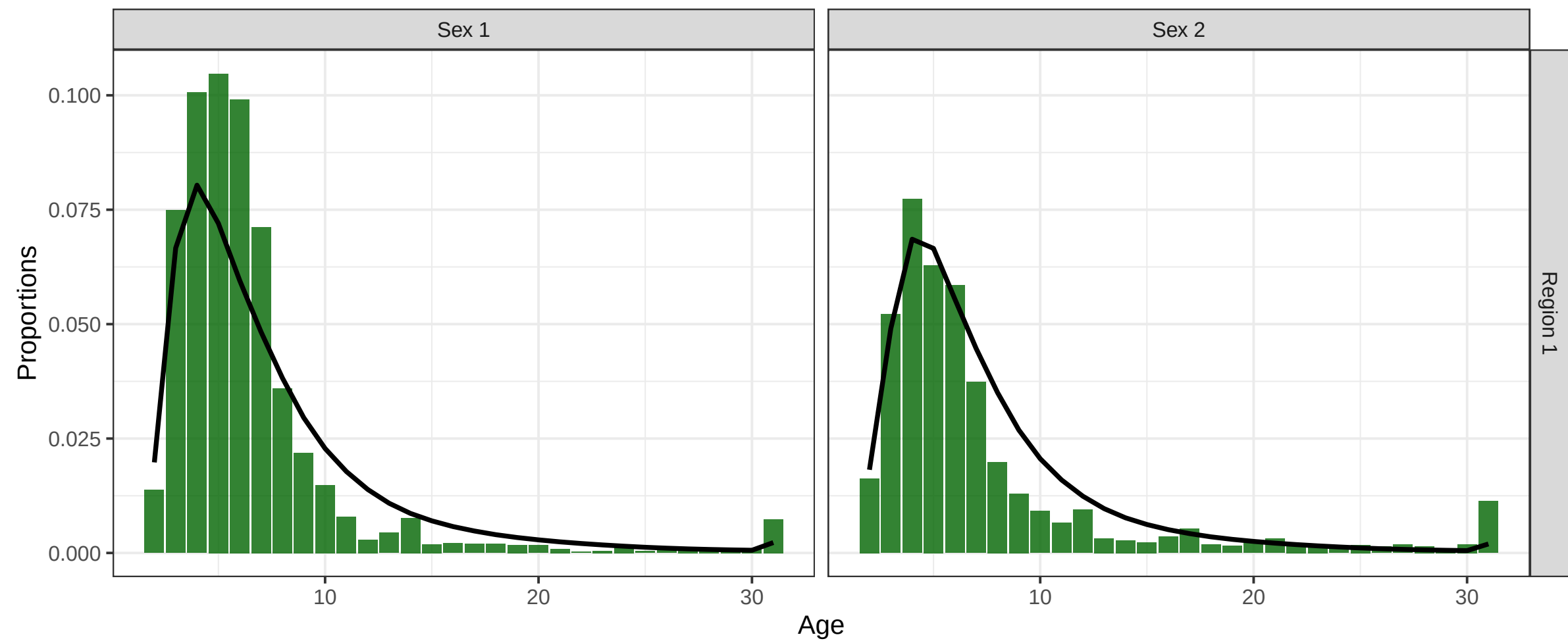
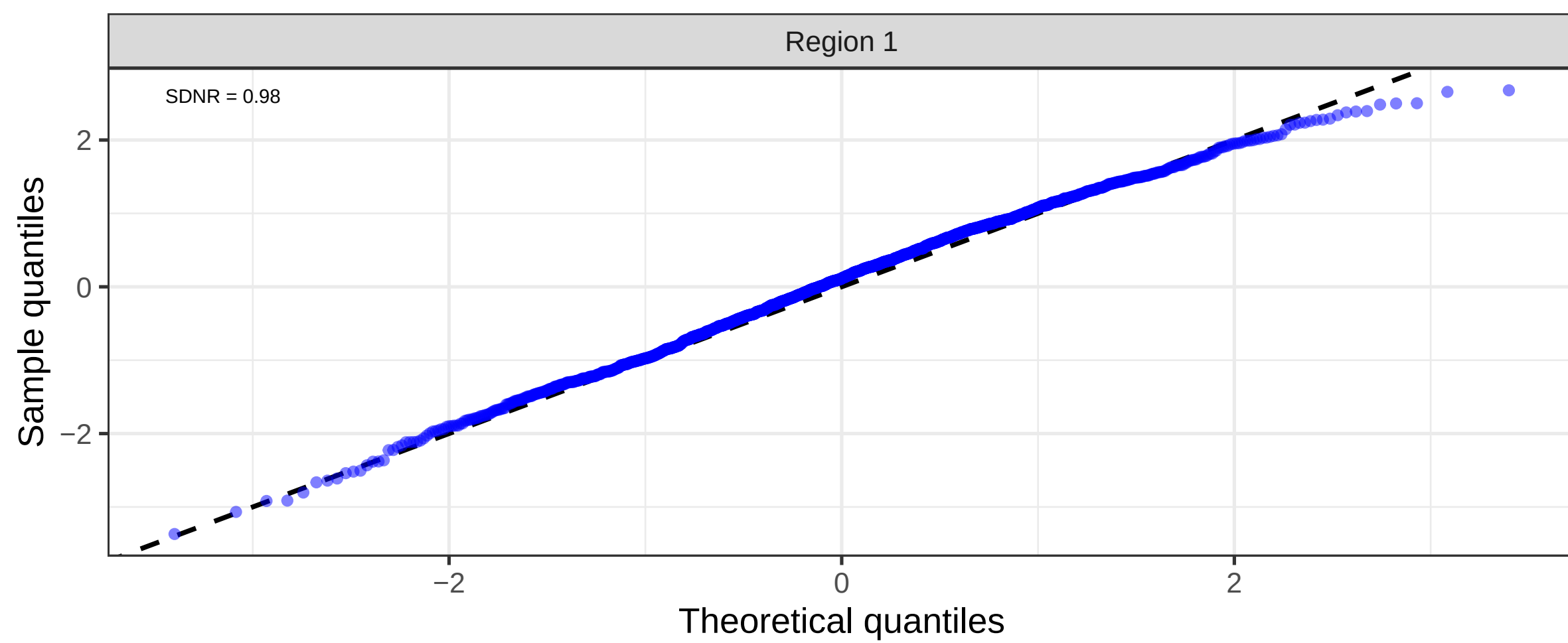
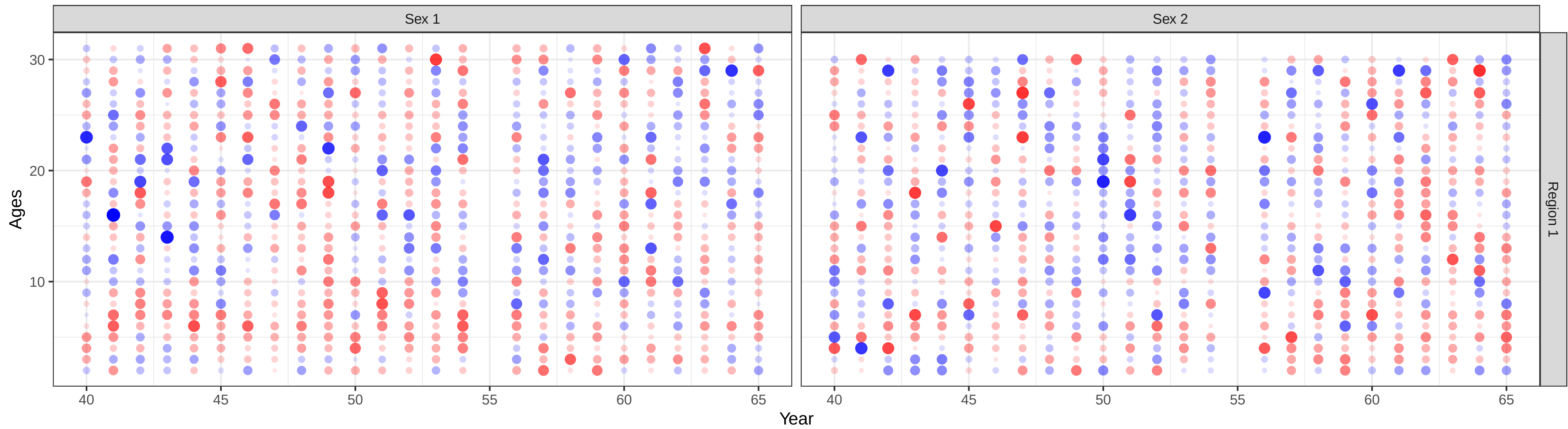
JPN LLS Length Comps Female



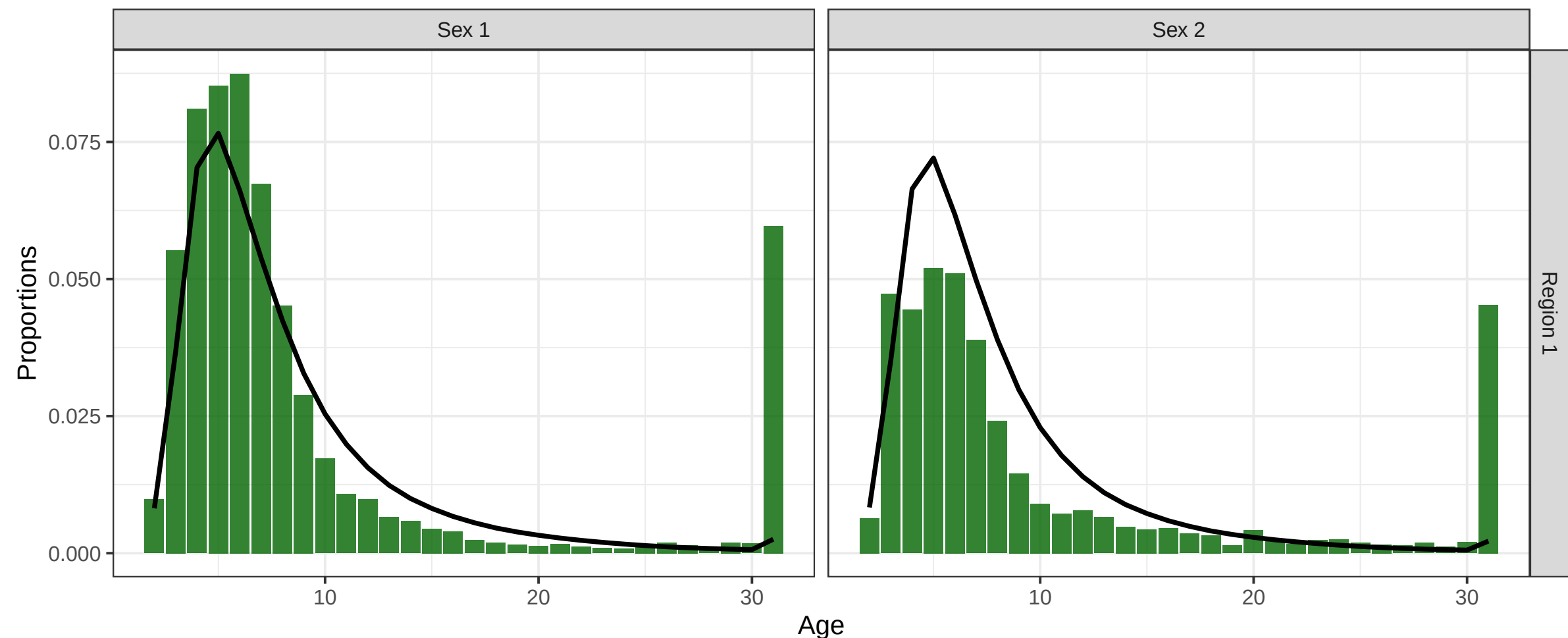
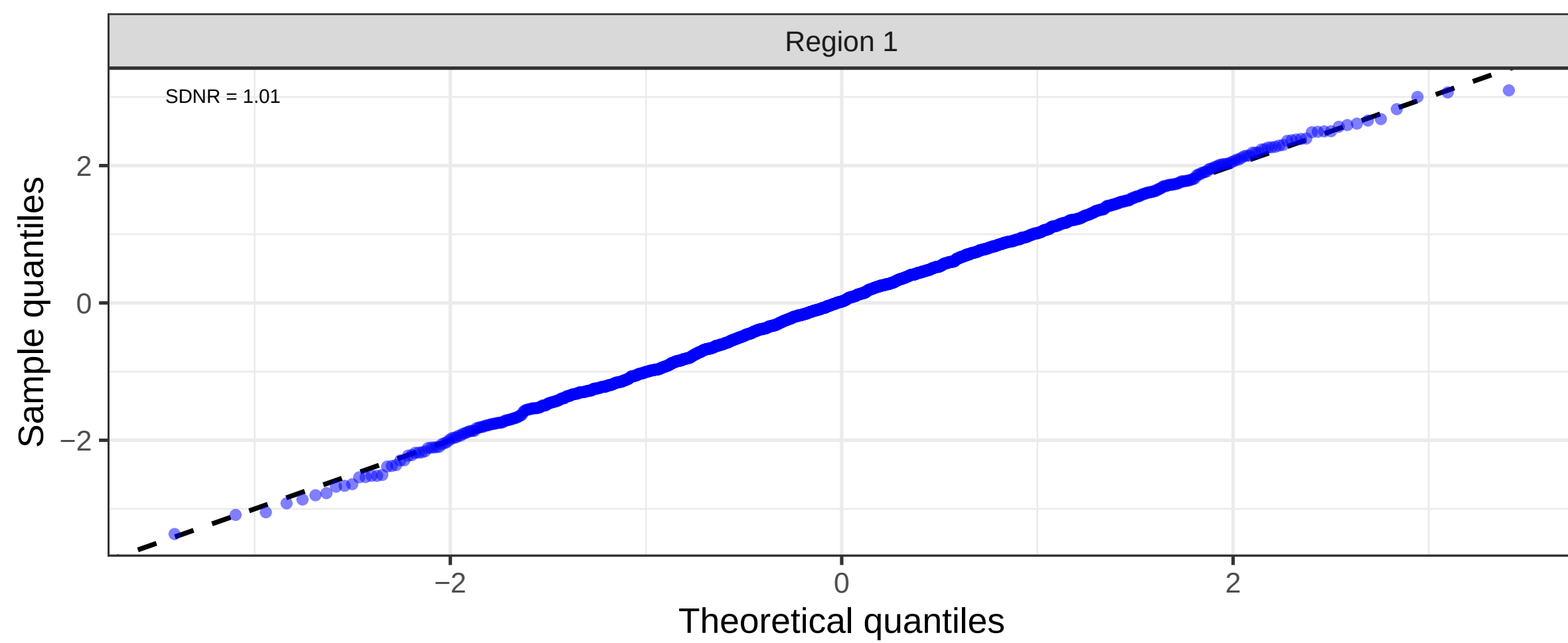
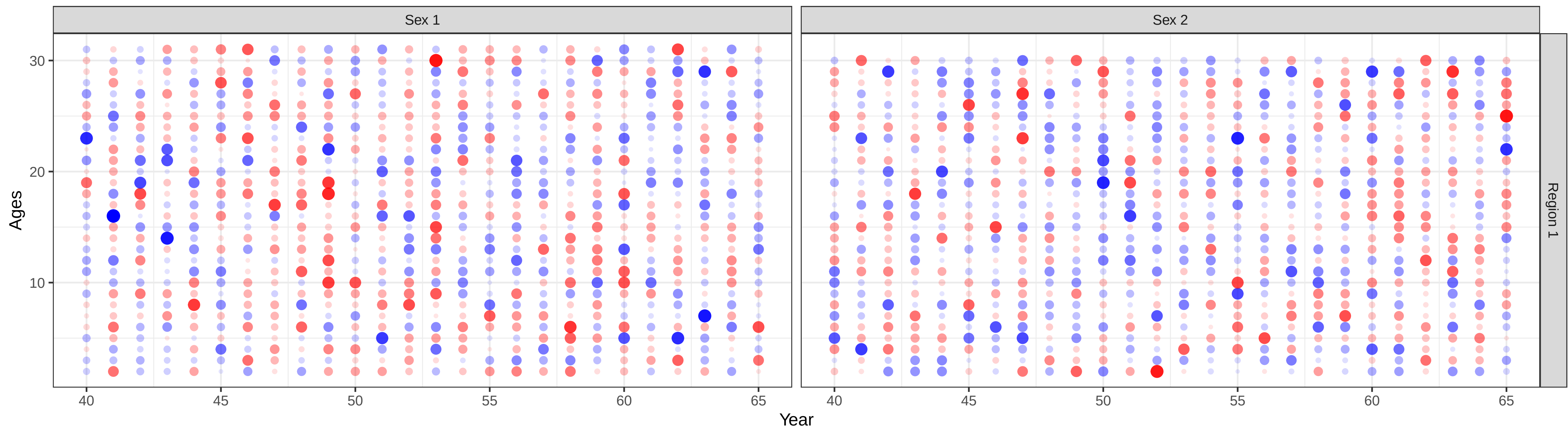
JPN LLS Length Comps Male



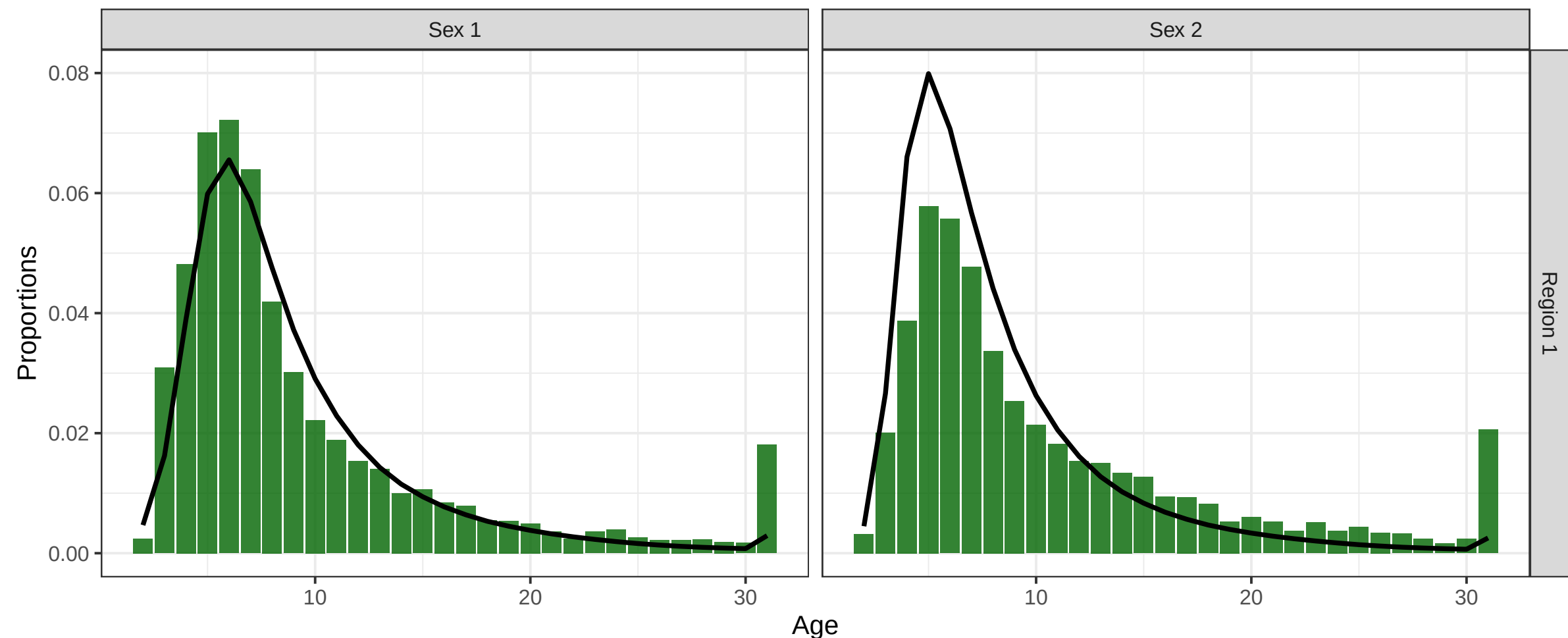
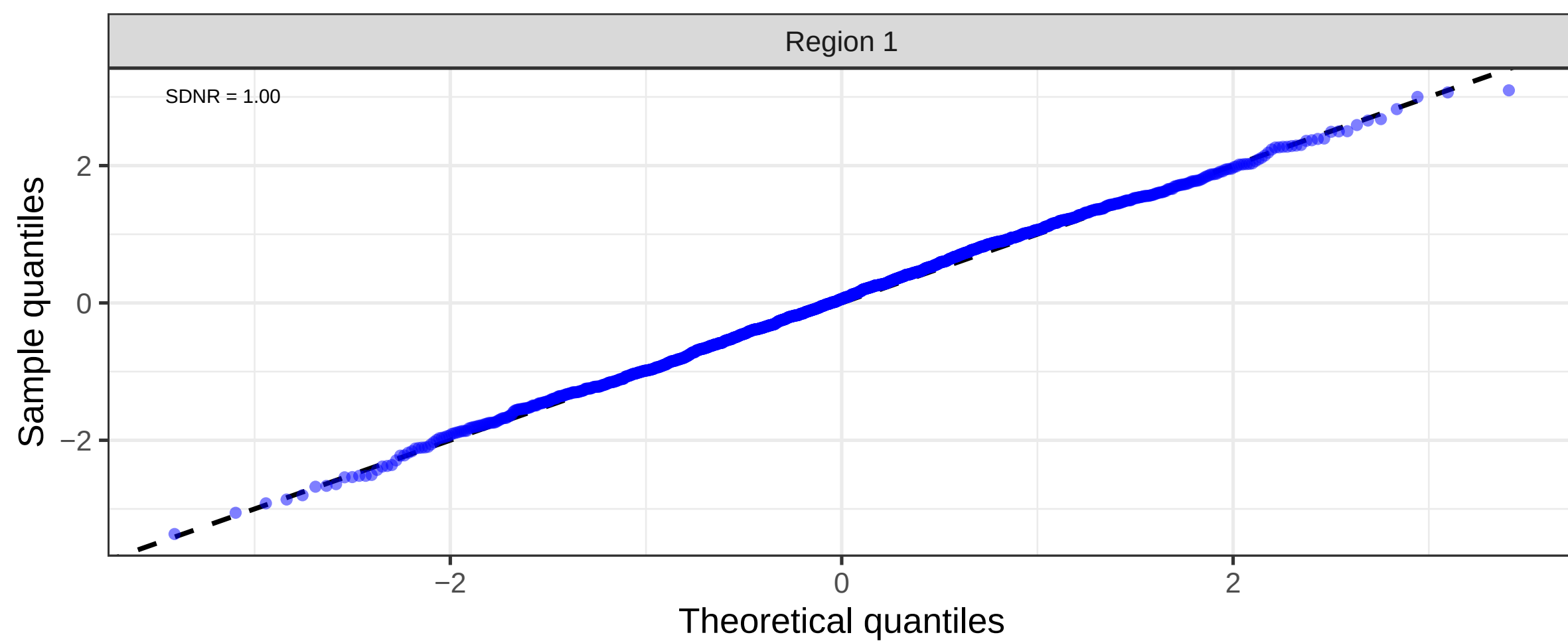
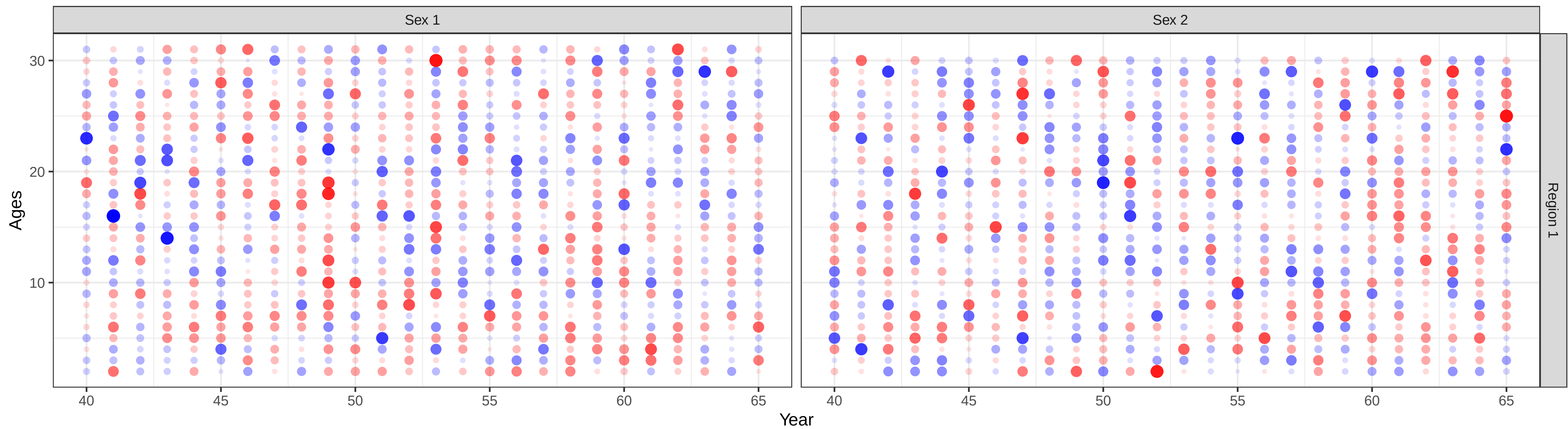
abs(Resid) 1 2 3 Sign ● Neg ● Pos

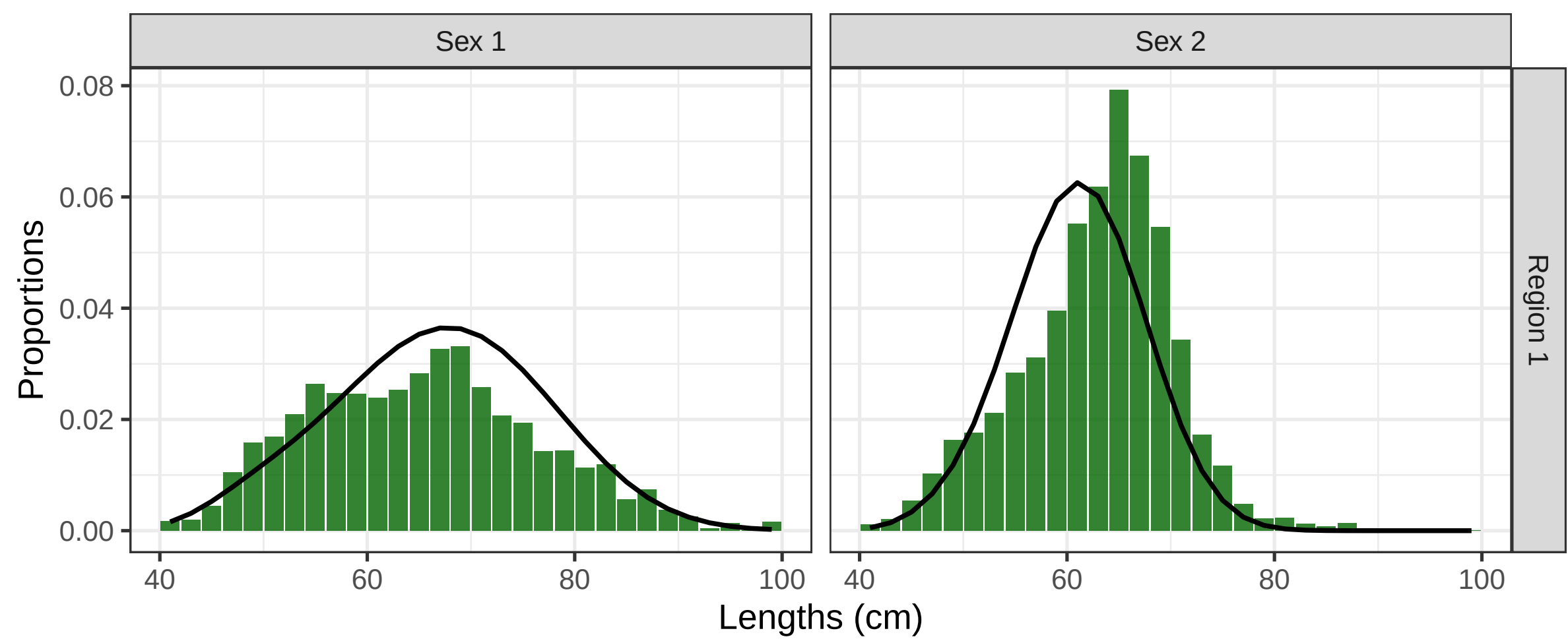
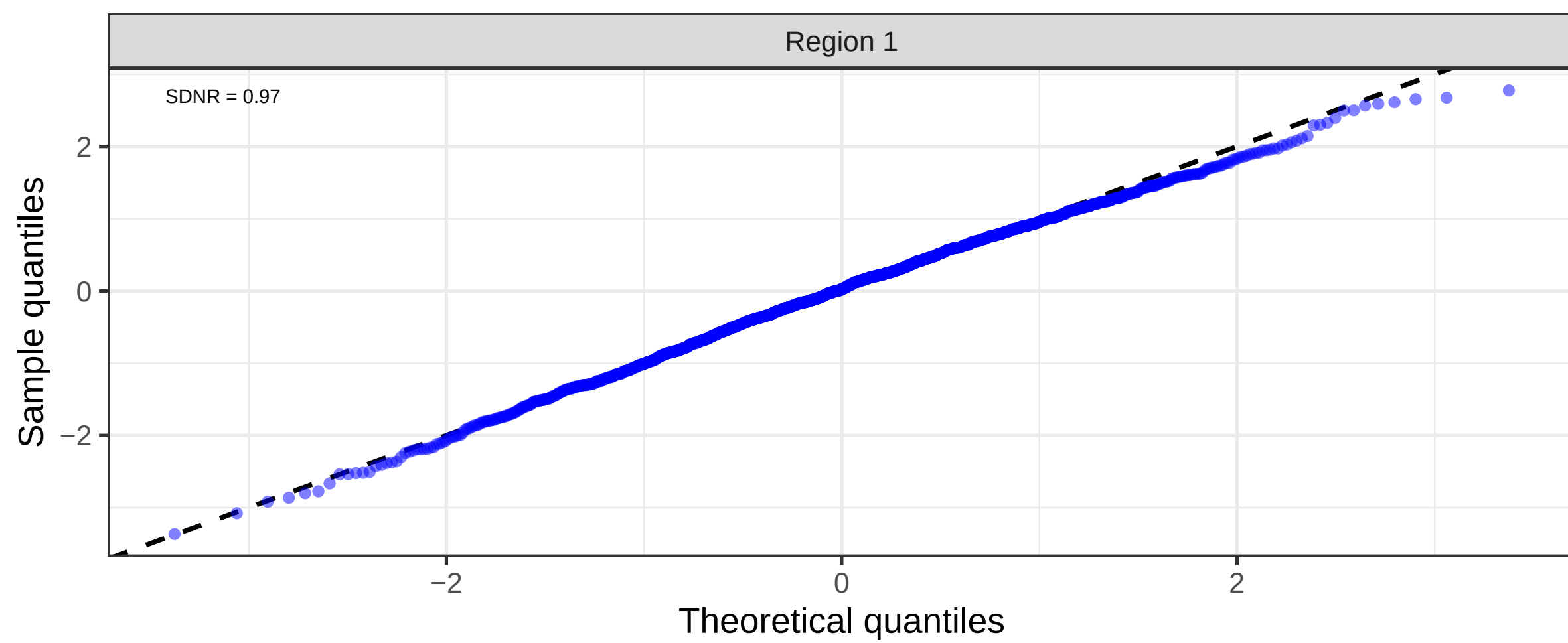
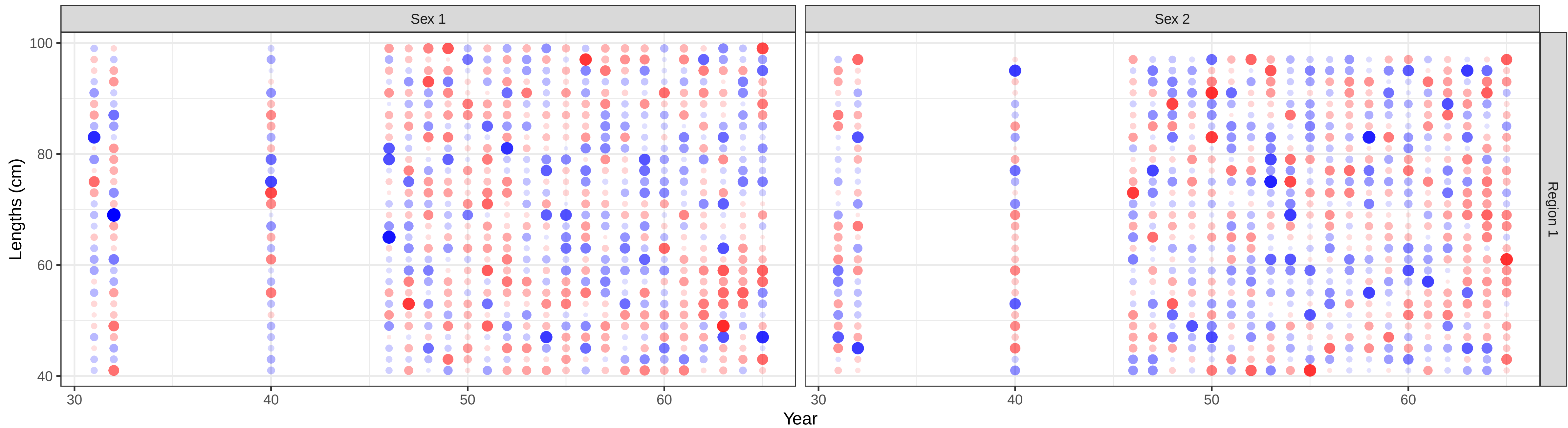


abs(Resid) 1 2 3 Sign ● Neg ● Pos

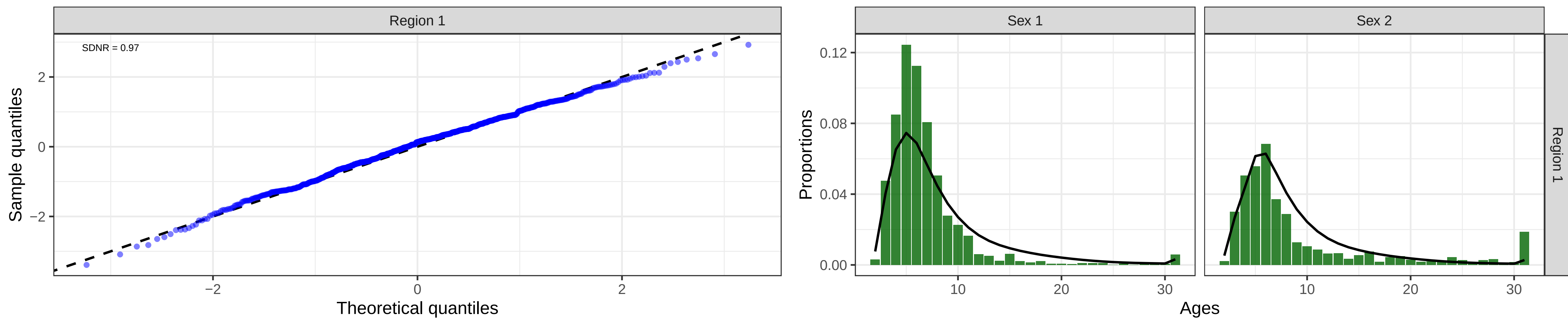
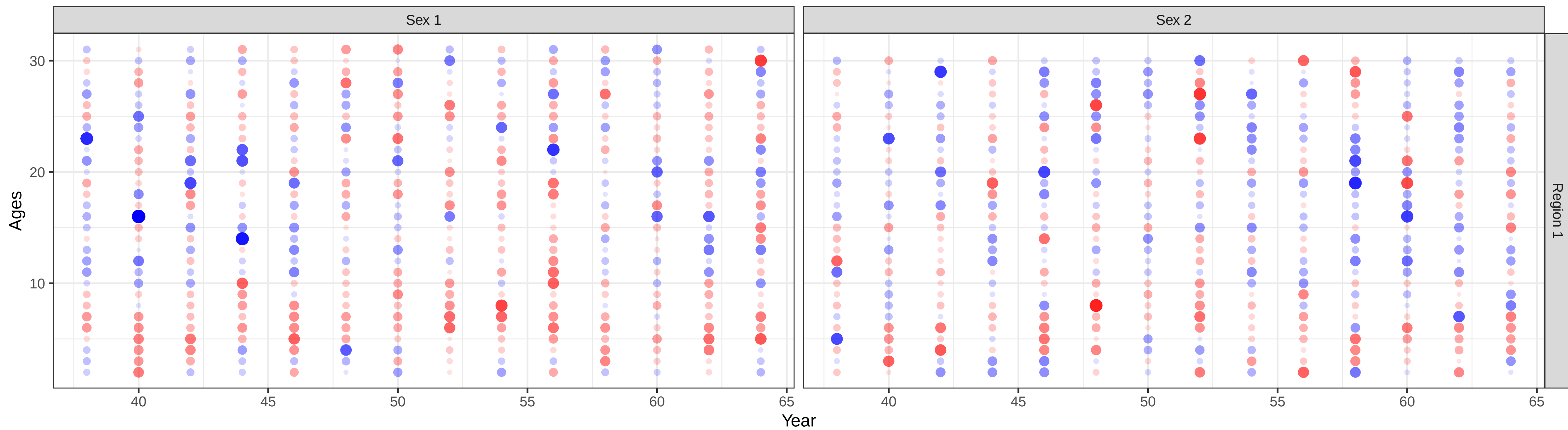


abs(Resid) ● 1 ● 2 ● 3 Sign ● Neg ● Pos

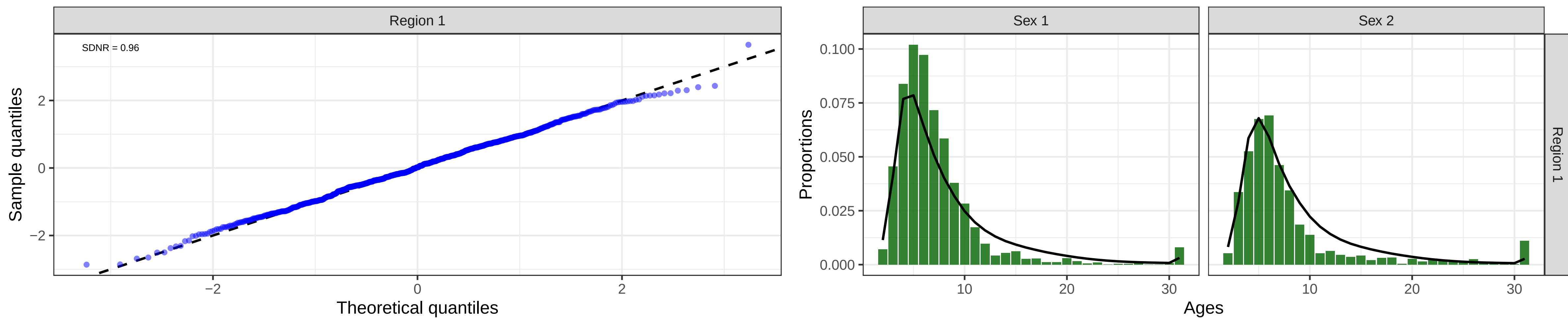
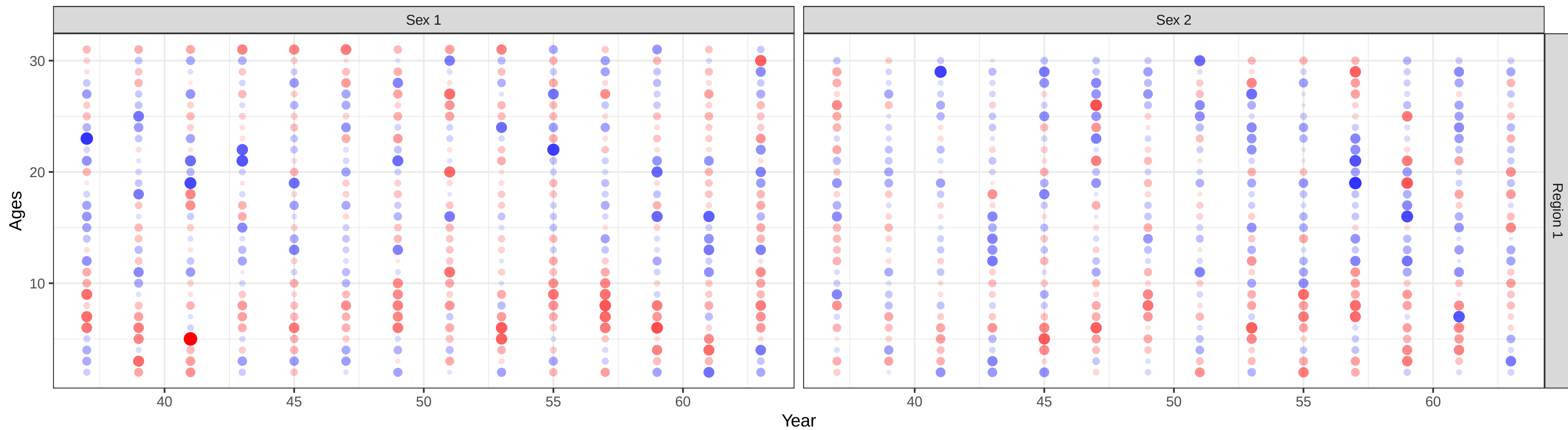




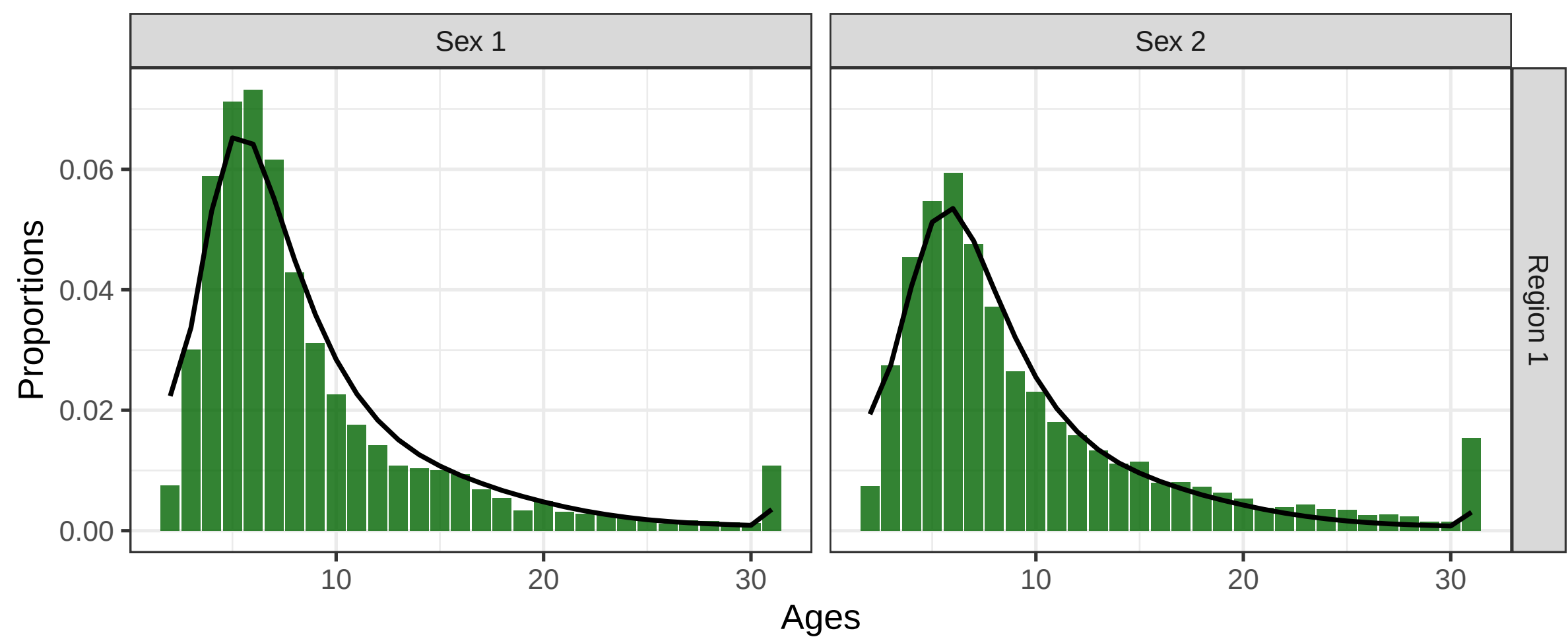
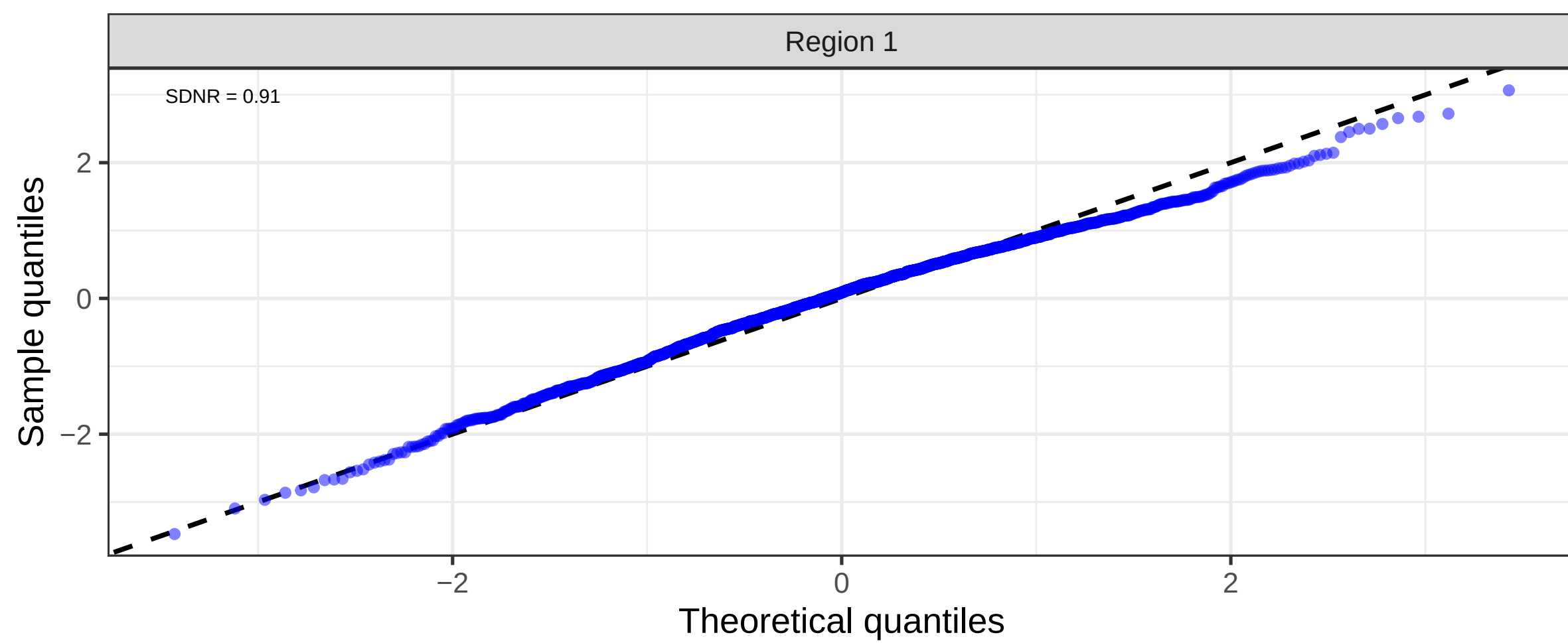
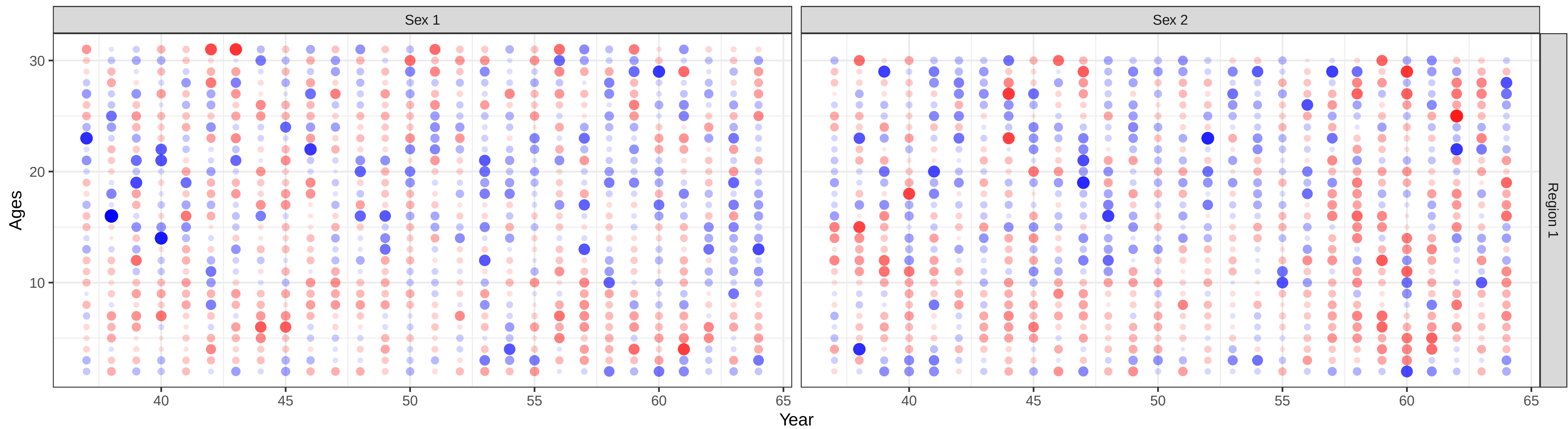
abs(Resid) 1 2 3 Sign ● Neg ● Pos



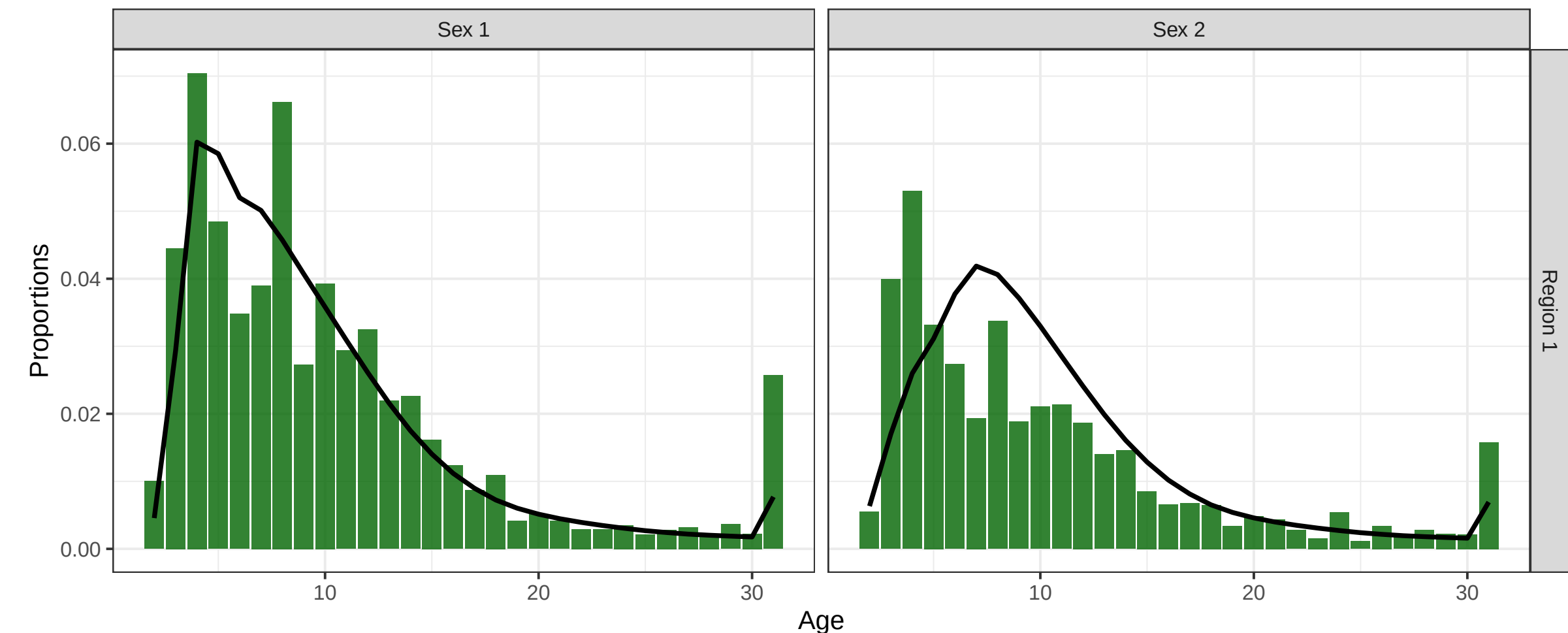
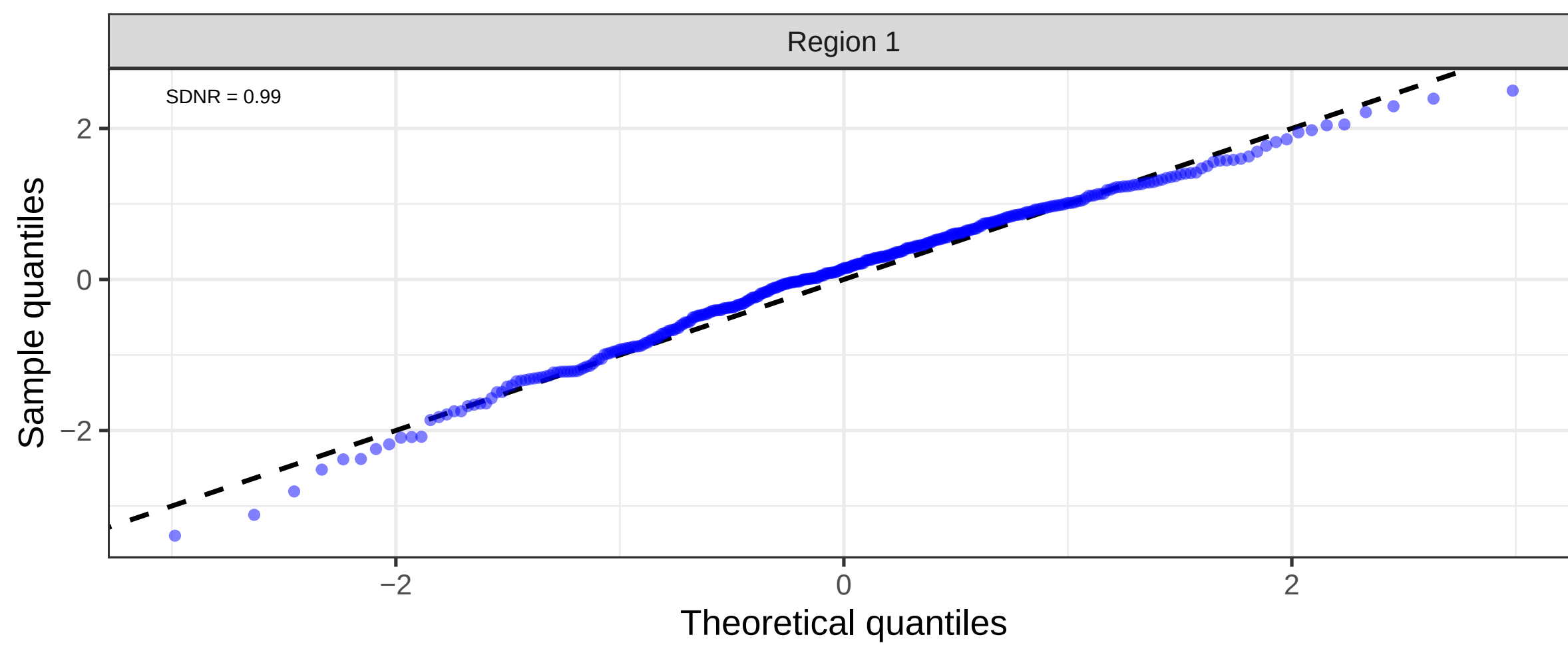
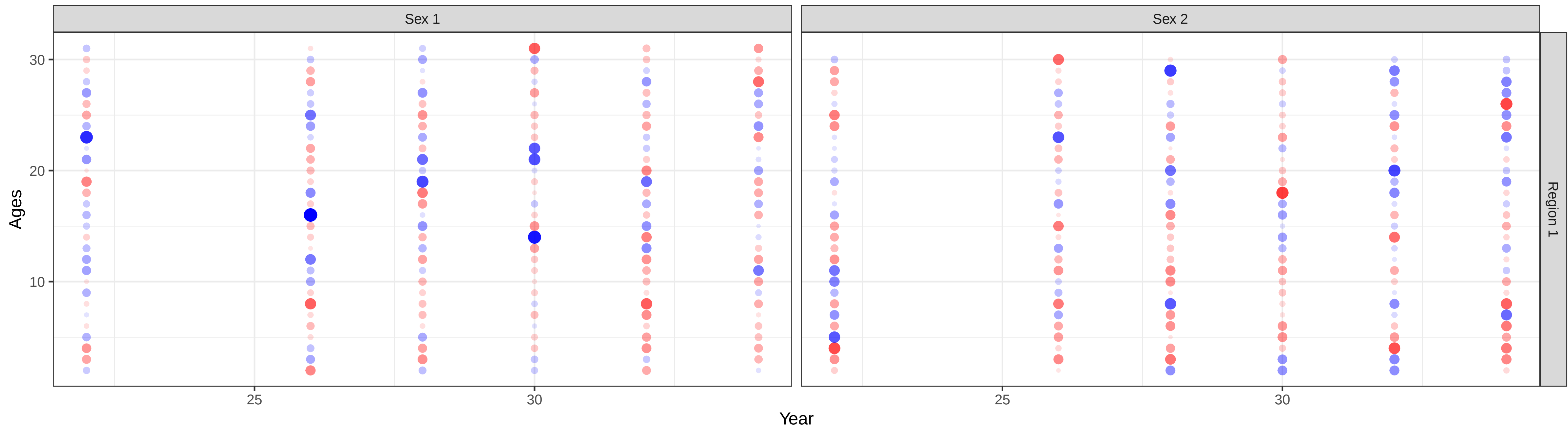
abs(Resid) ● 1 ● 2 ● 3 Sign ● Neg ● Pos



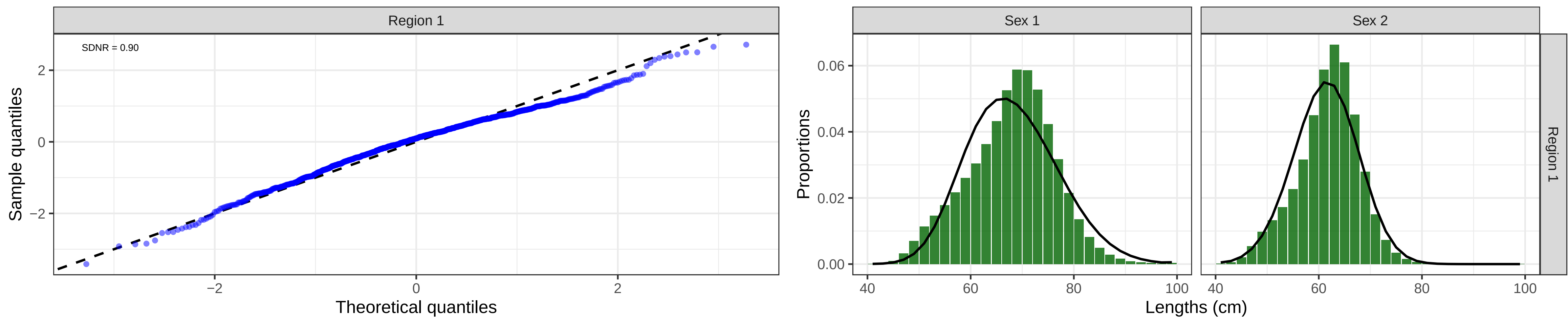
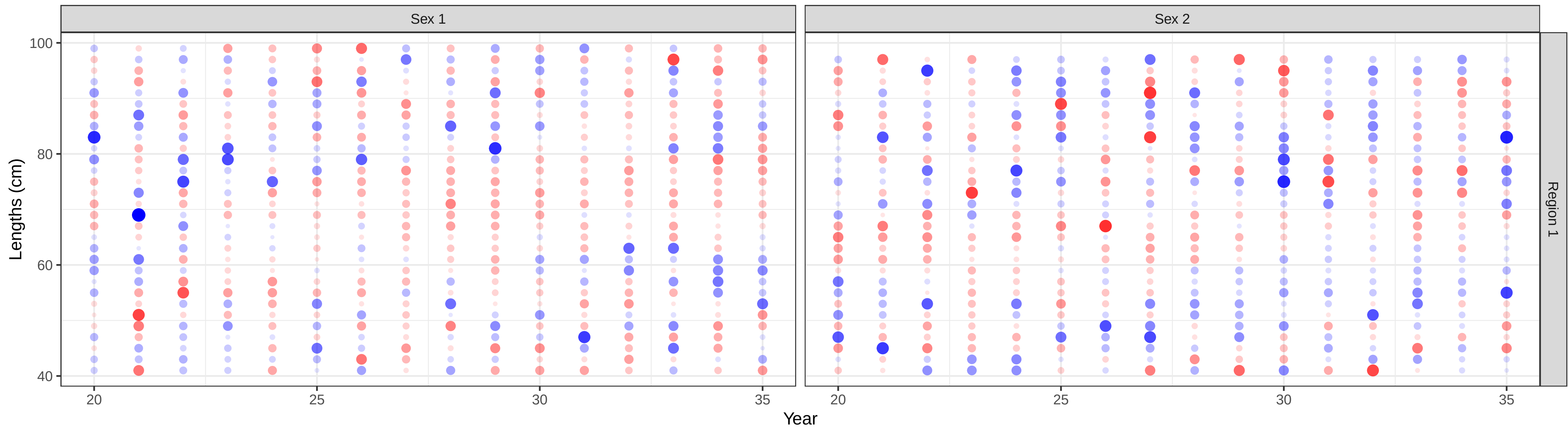
abs(Resid) ● 1 ● 2 ● 3 Sign ● Neg ● Pos



abs(Resid) 1 2 3 Sign ● Neg ● Pos

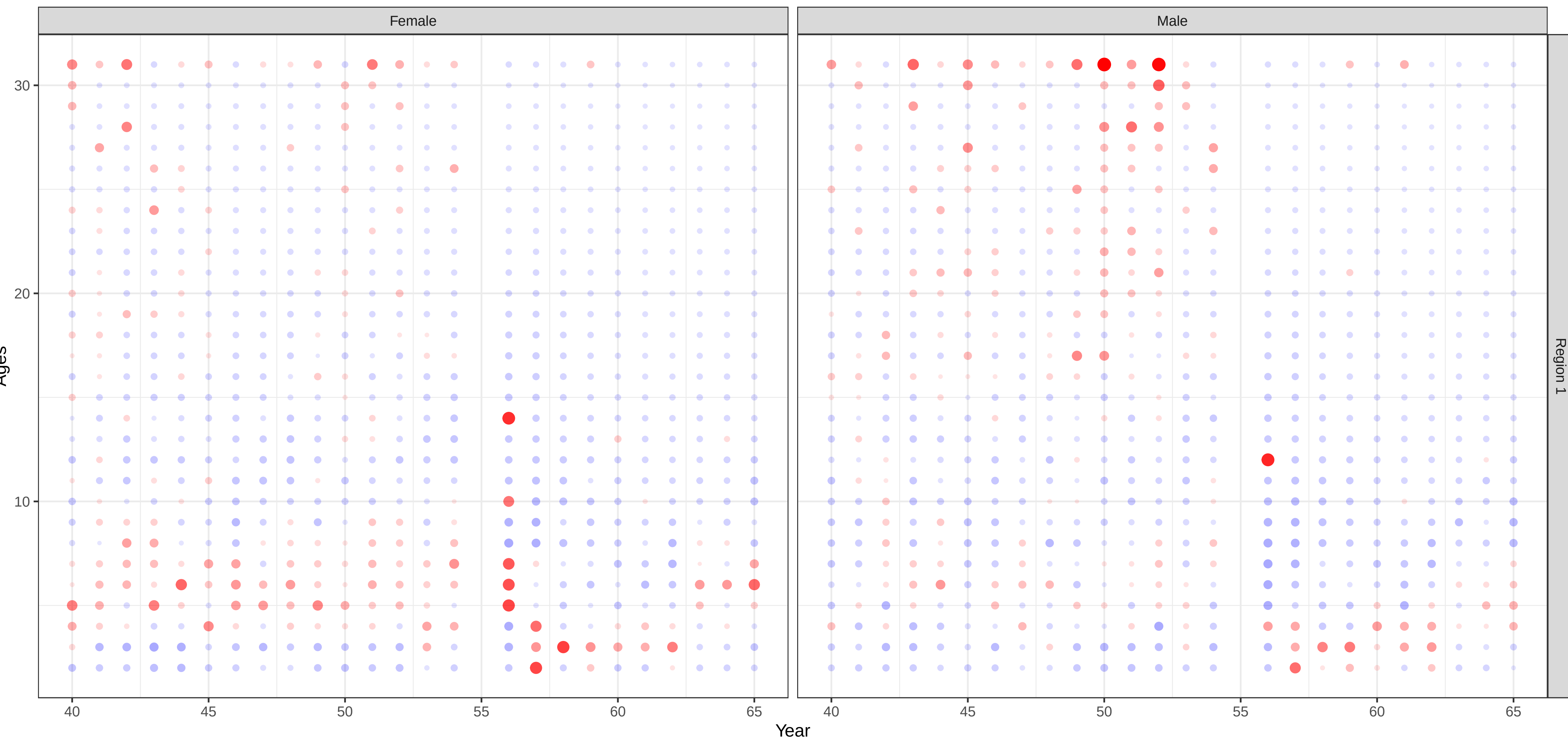


abs(Resid) 1 2 3 Sign ● Neg ● Pos



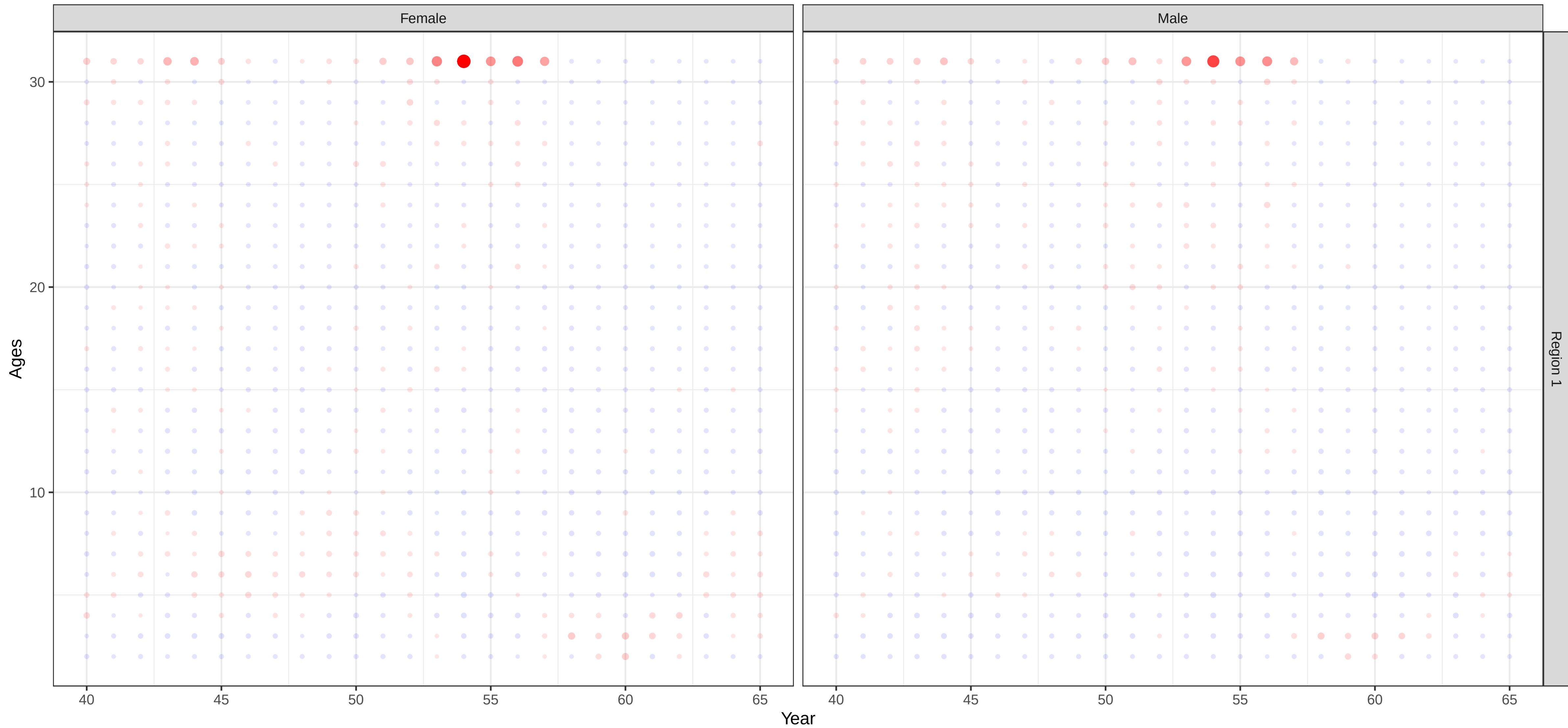
BS Fixed Gear Fishery Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 1 ● 2



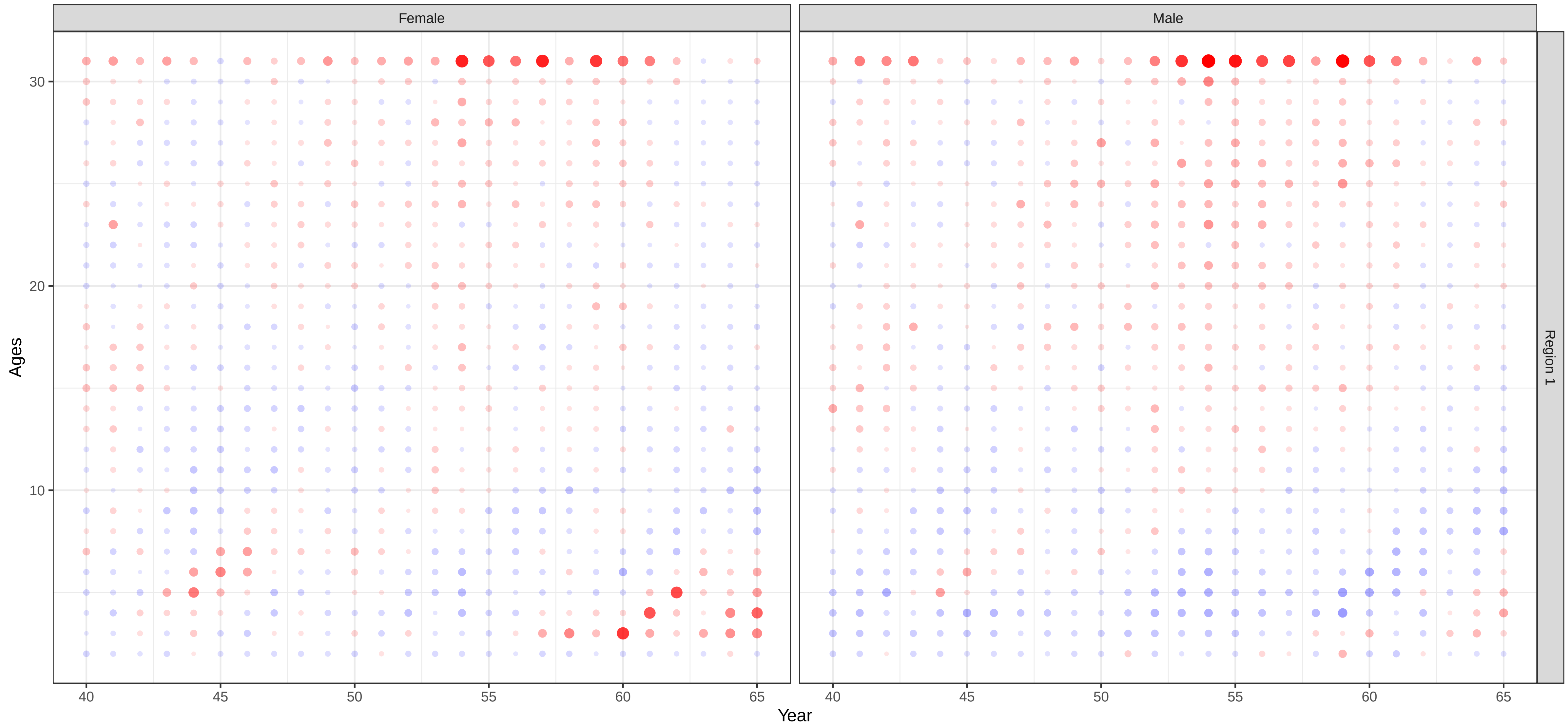
AI Fixed Gear Fishery Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 2 ● 4 ● 6



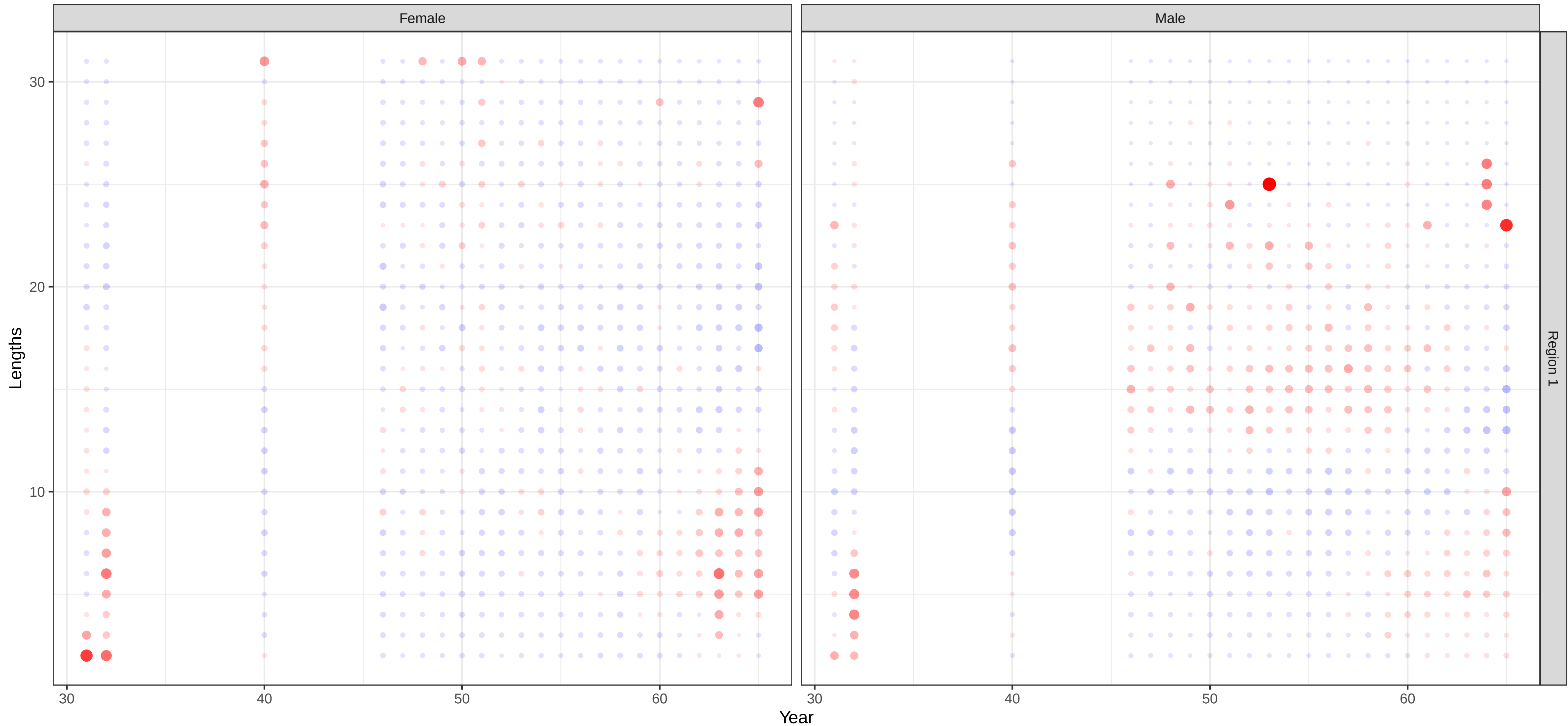
GOA Fixed Gear Fishery Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0



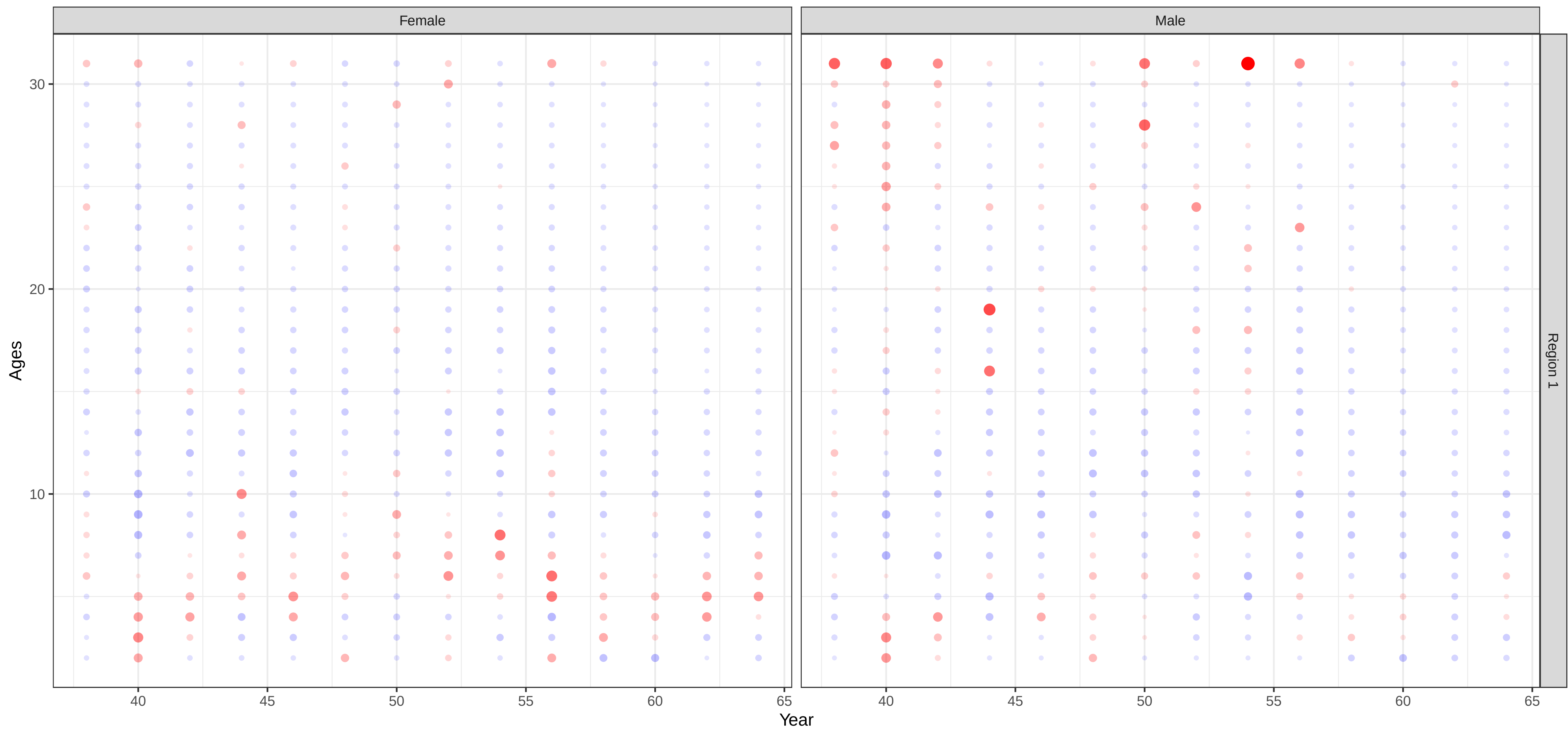
Trawl Fishery Length Compositions

Sign ● Neg ● Pos abs(Resid) ● 0 ● 1 ● 2 ● 3



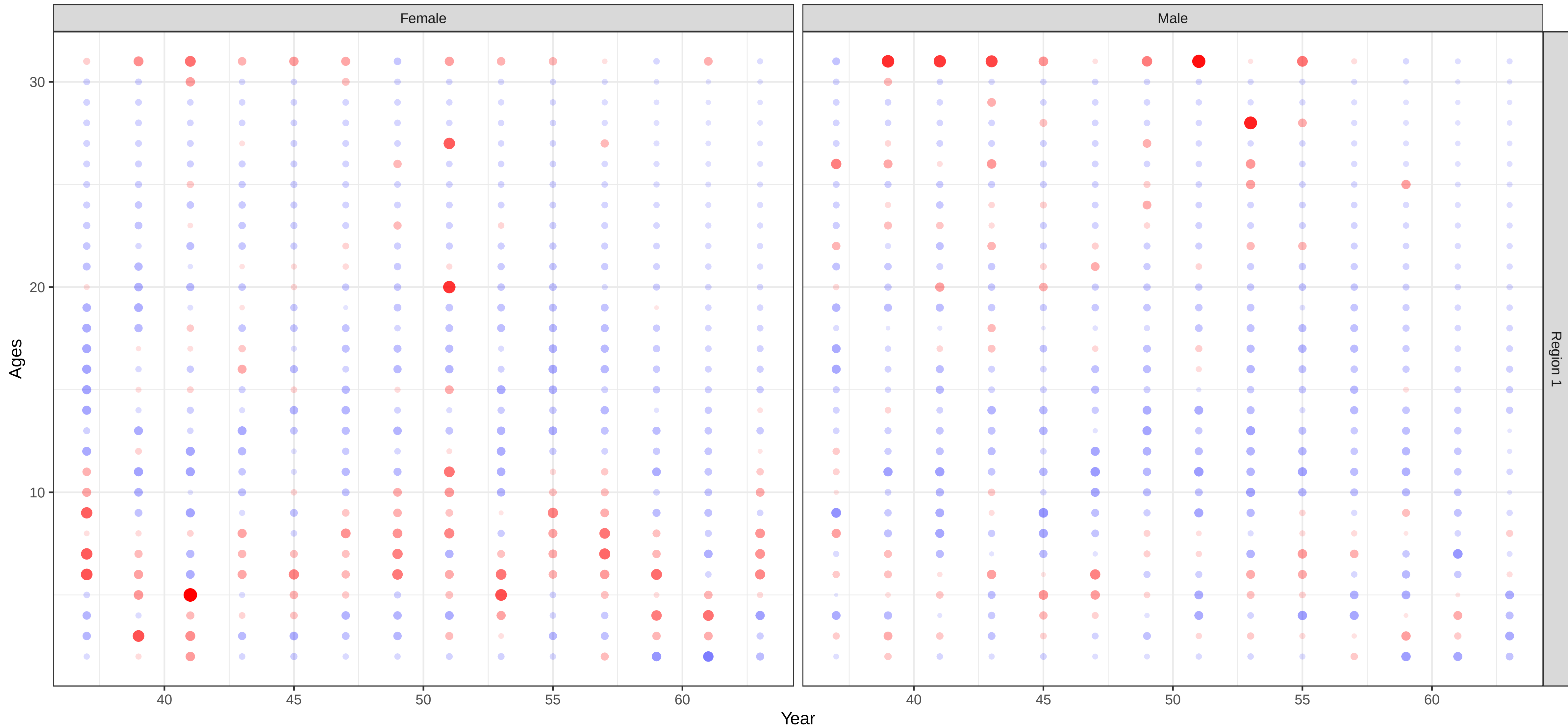
BS LLS Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 1 ● 2 ● 3 ● 4



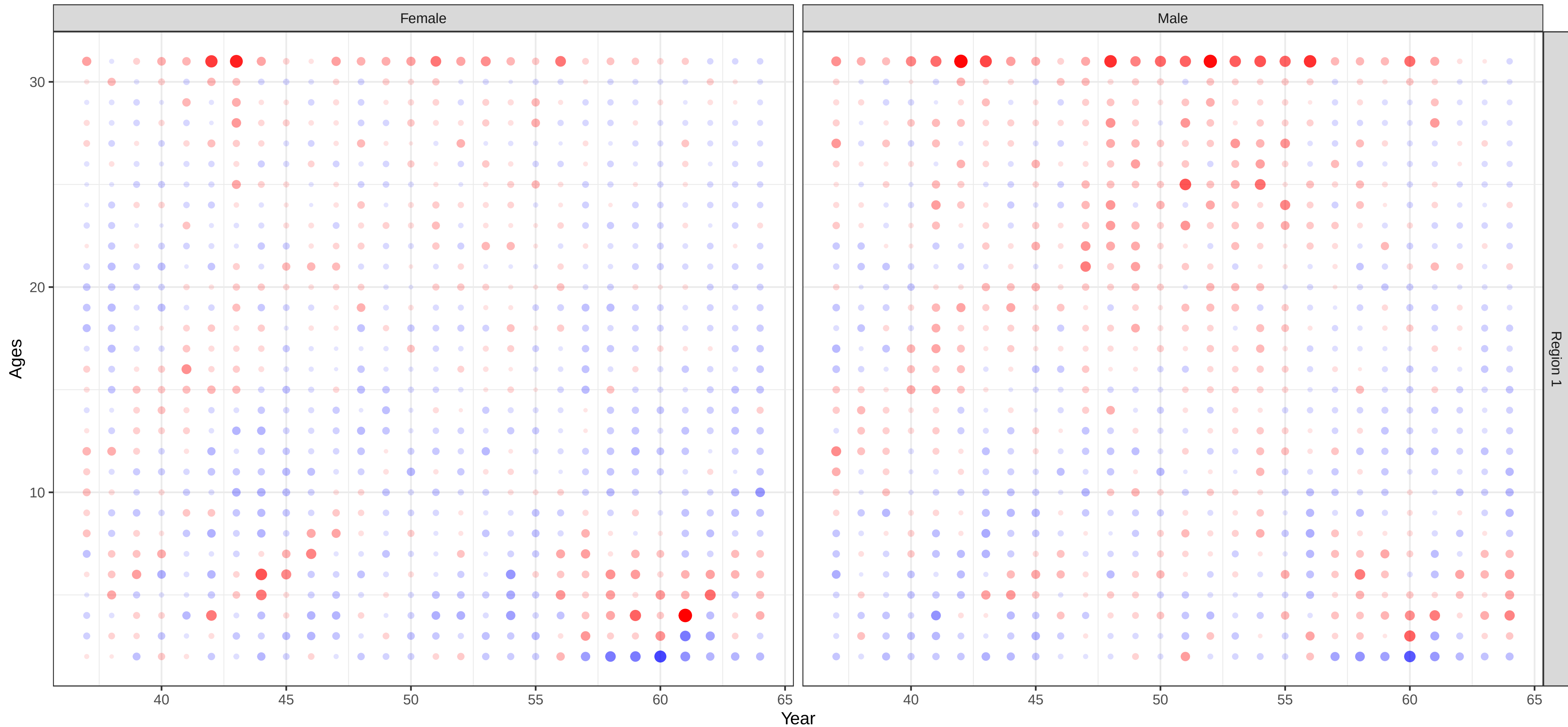
AI LLS Age Compositions

abs(Resid) 1 2 3 Sign ● Neg ● Pos



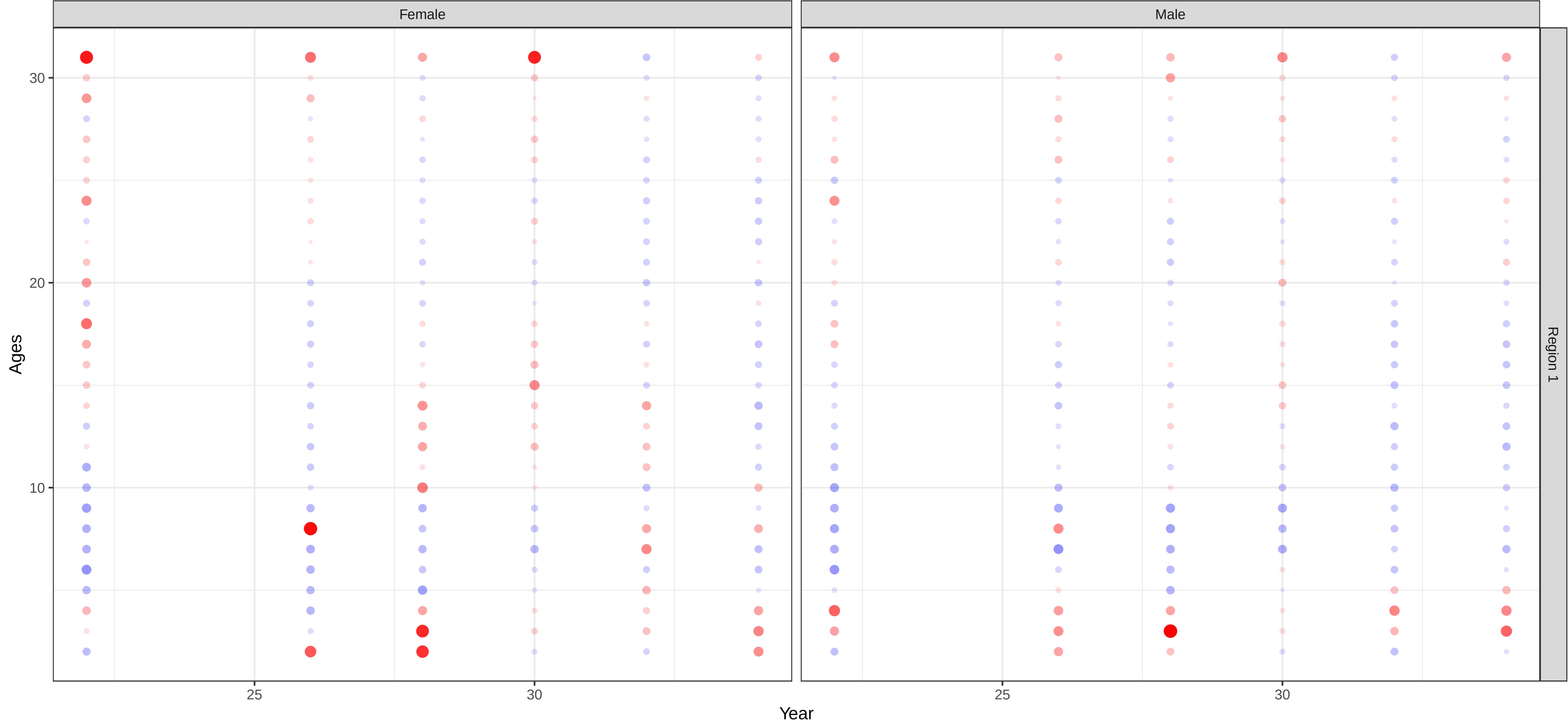
GOA LLS Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0 ● 2.5



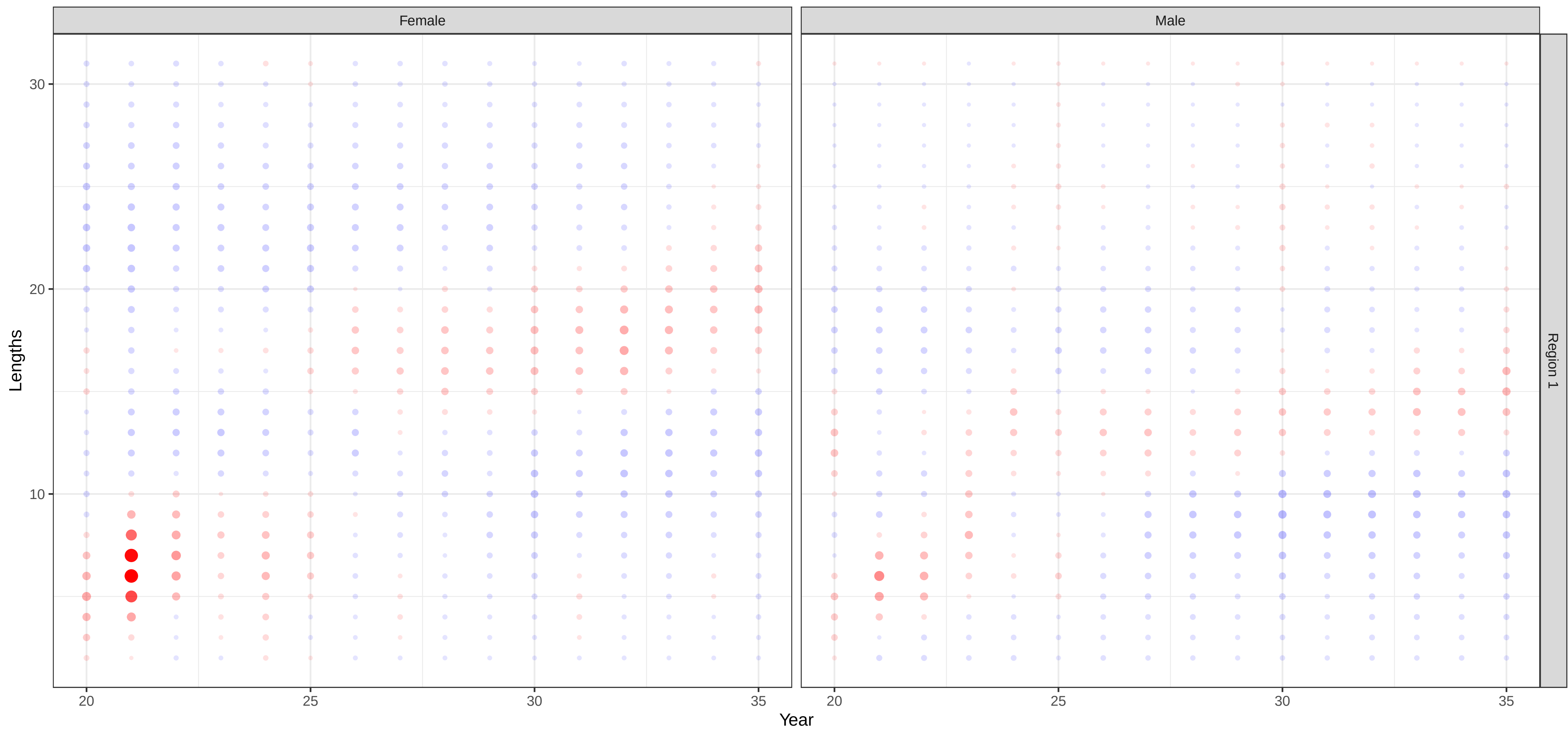
JPN LLS Age Compositions

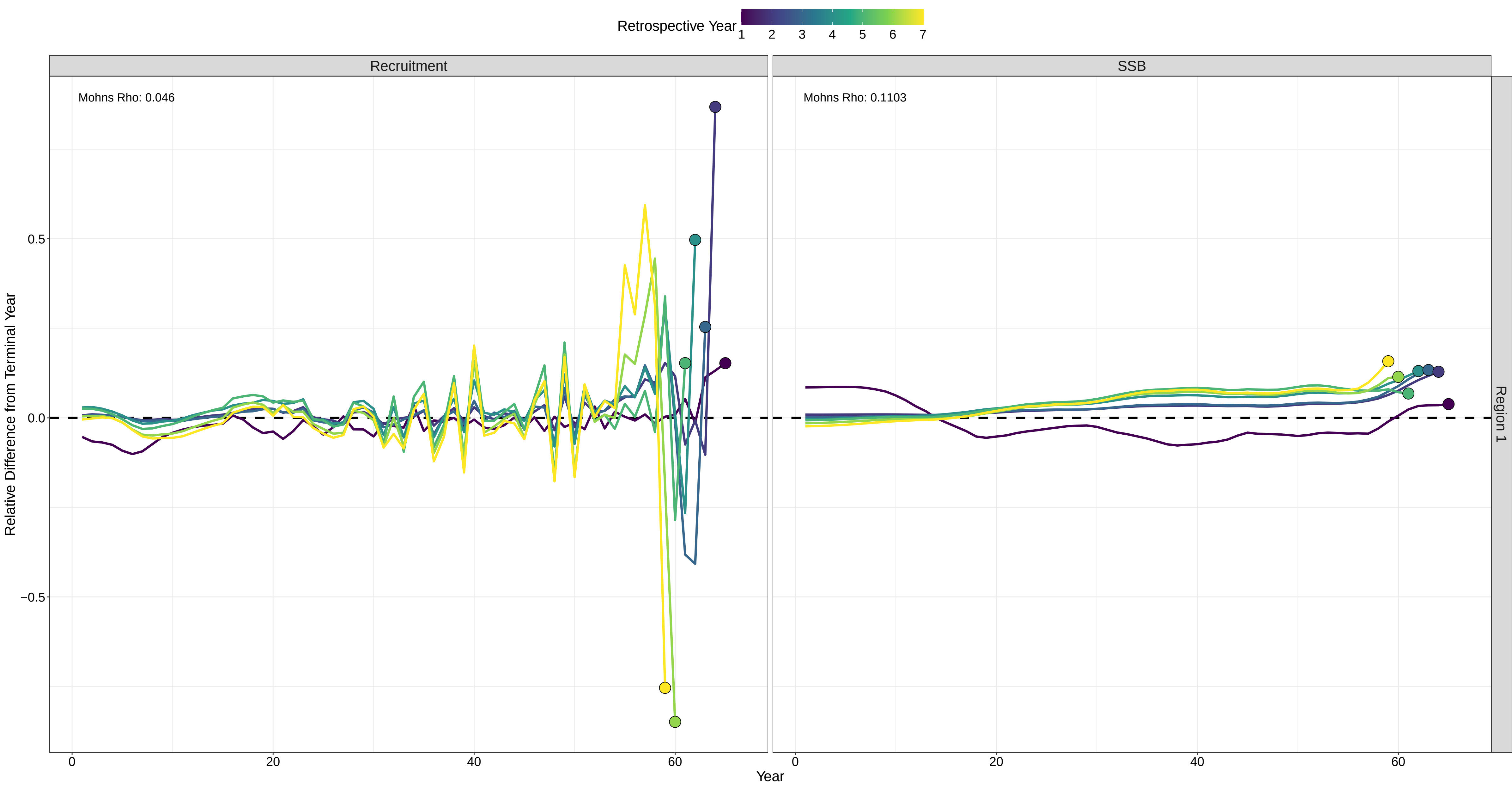
Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0

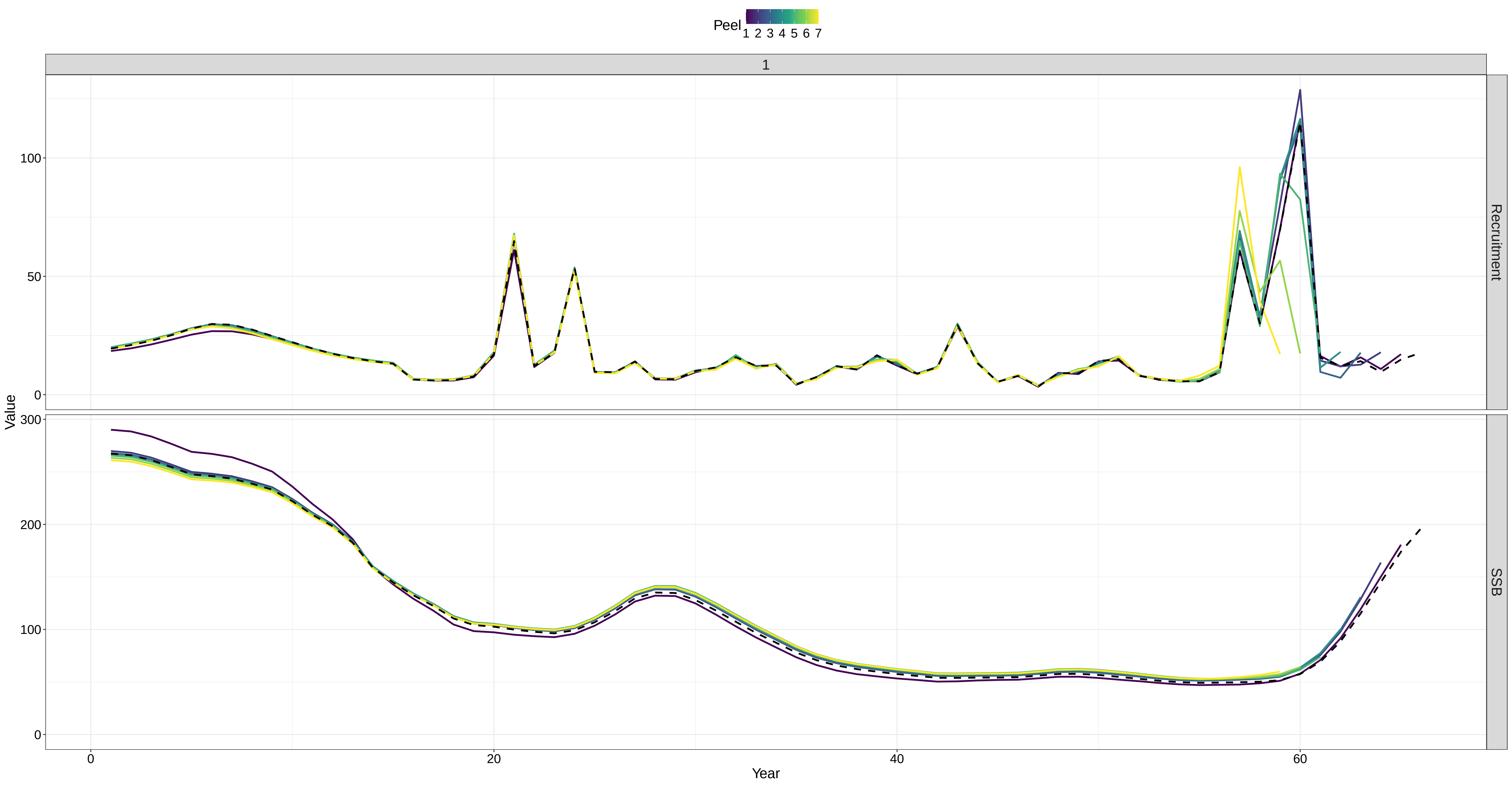


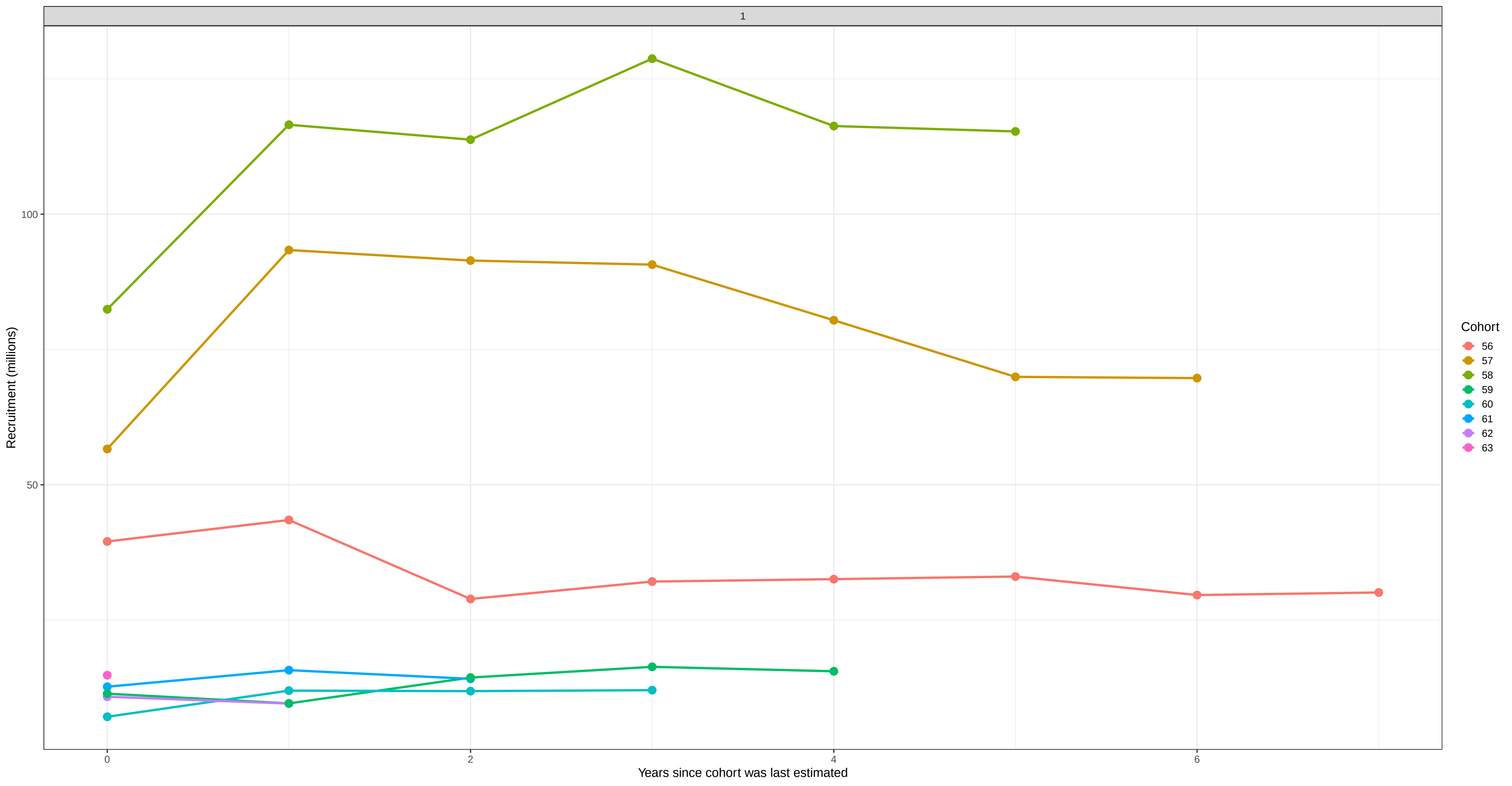
JPN LLS Length Compositions

Sign ● Neg ● Pos abs(Resid) ● 0 ● 1 ● 2

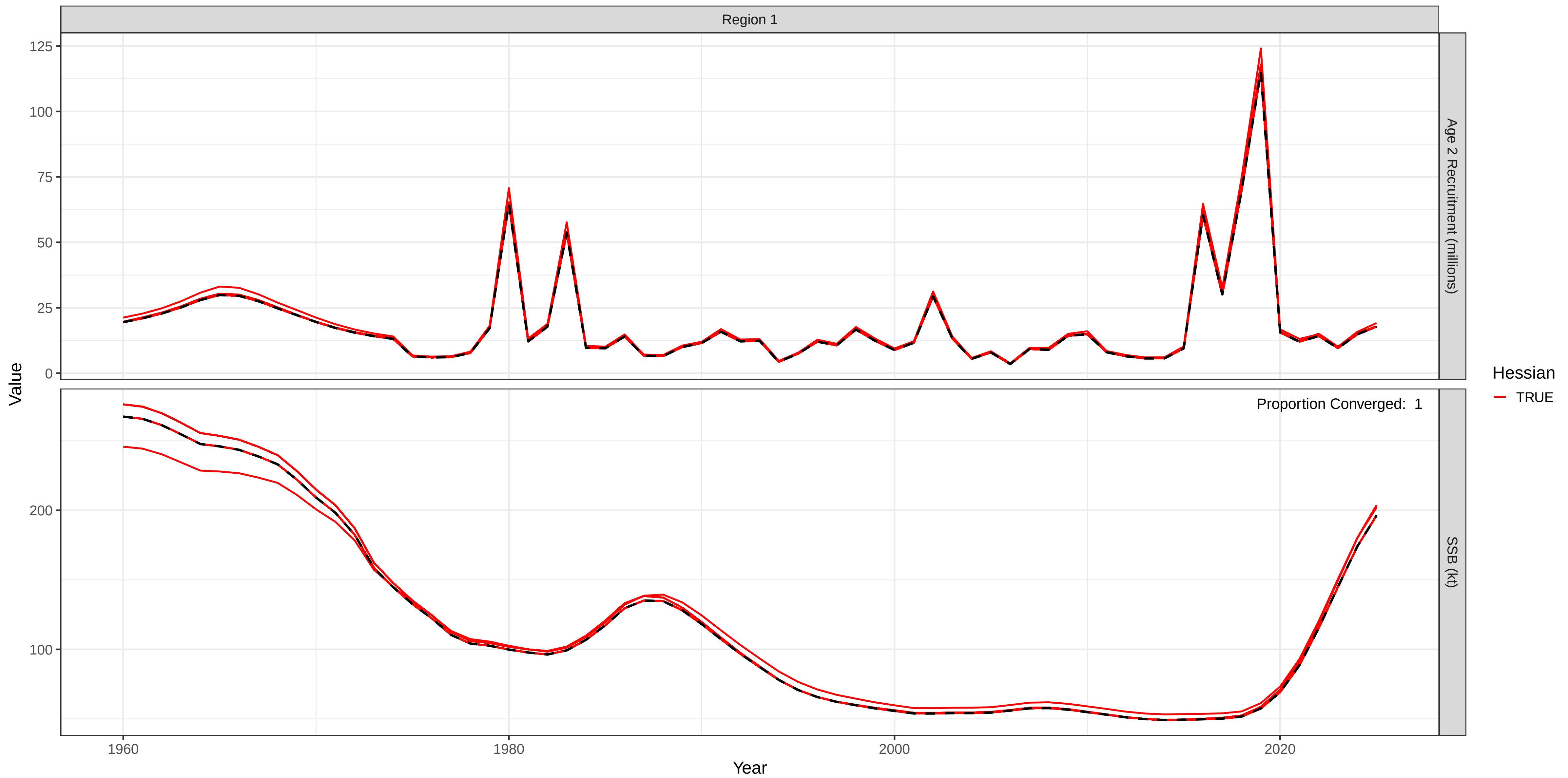




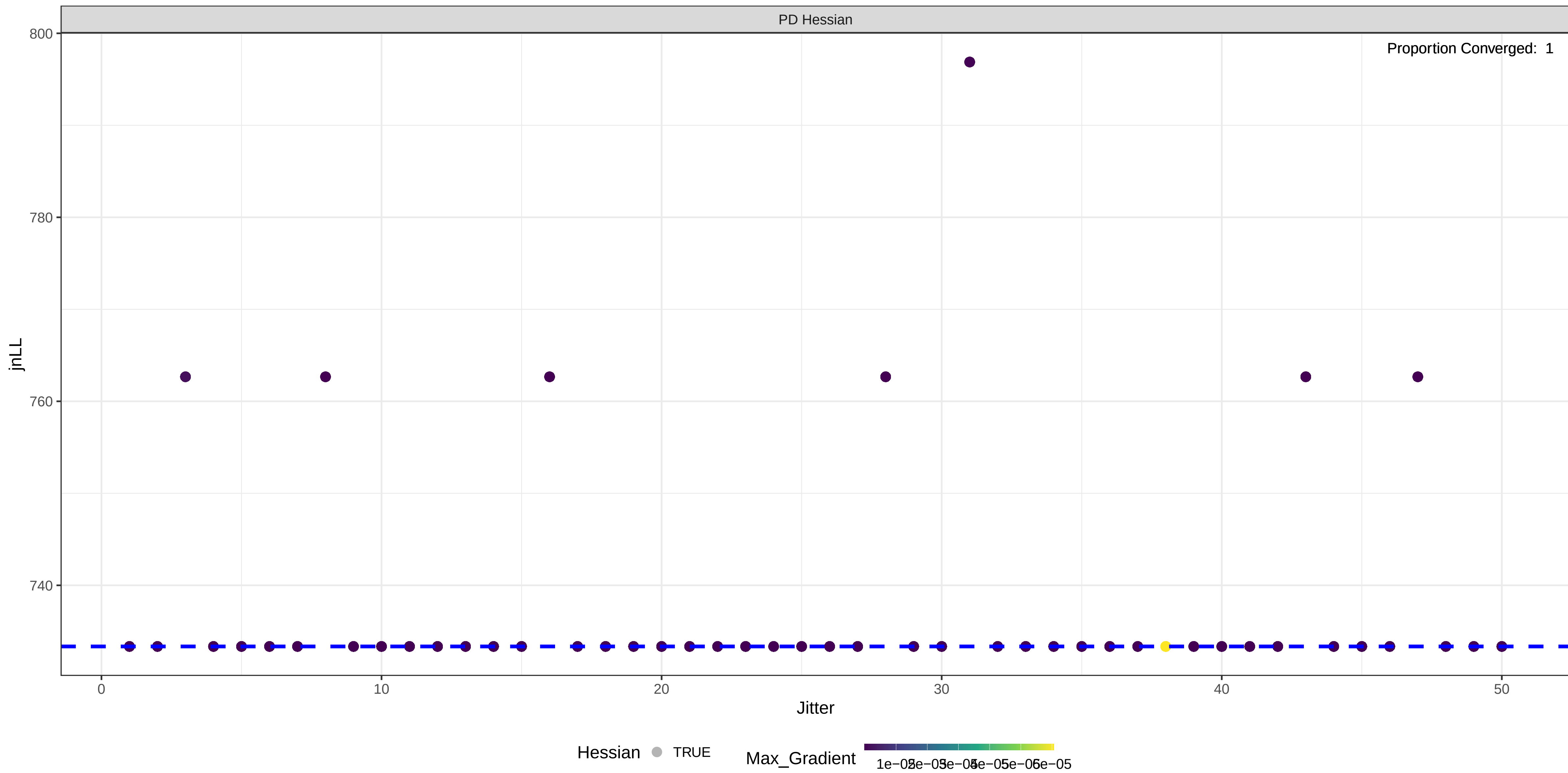




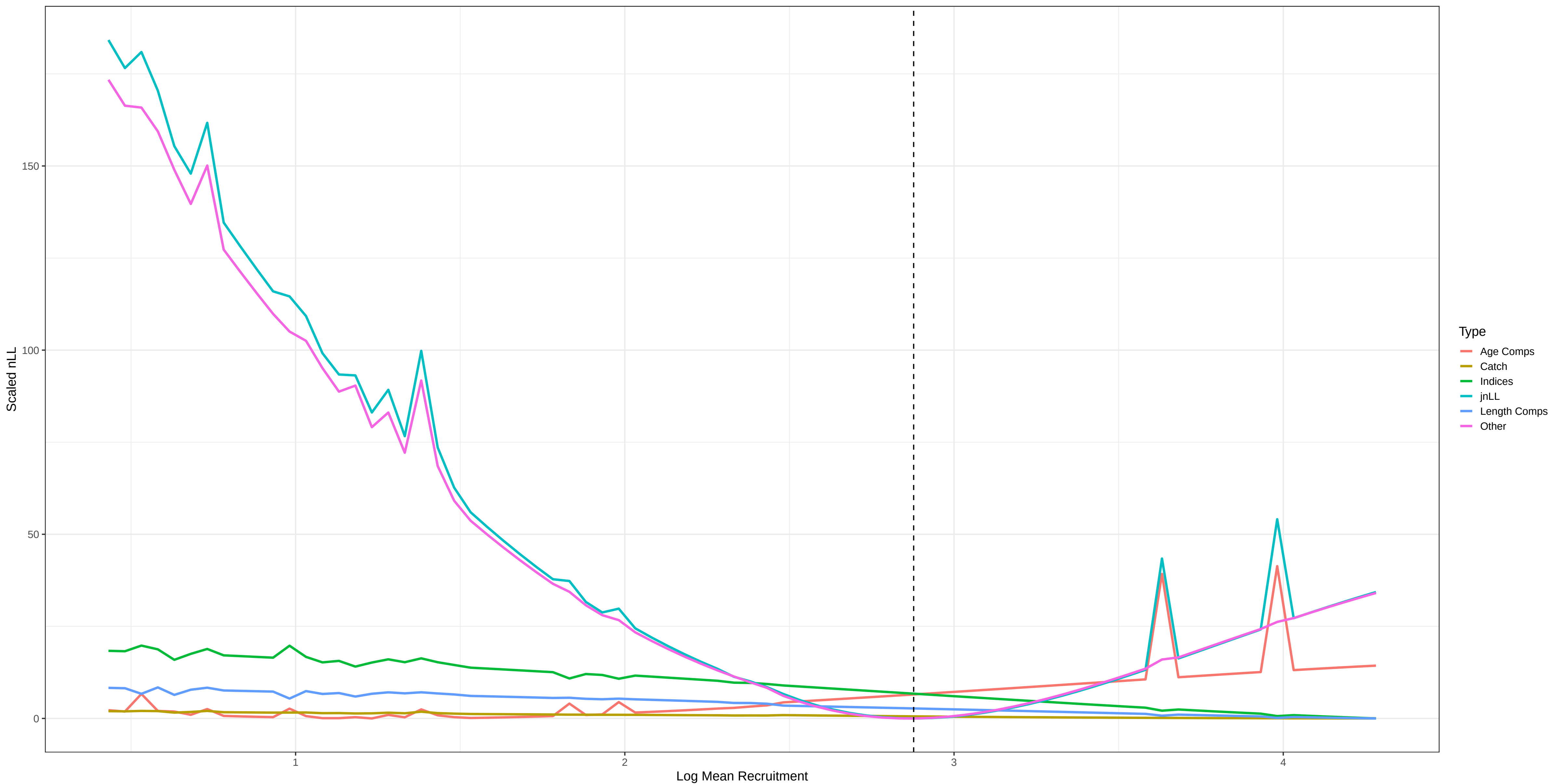
Jitter SSB and Recruitment



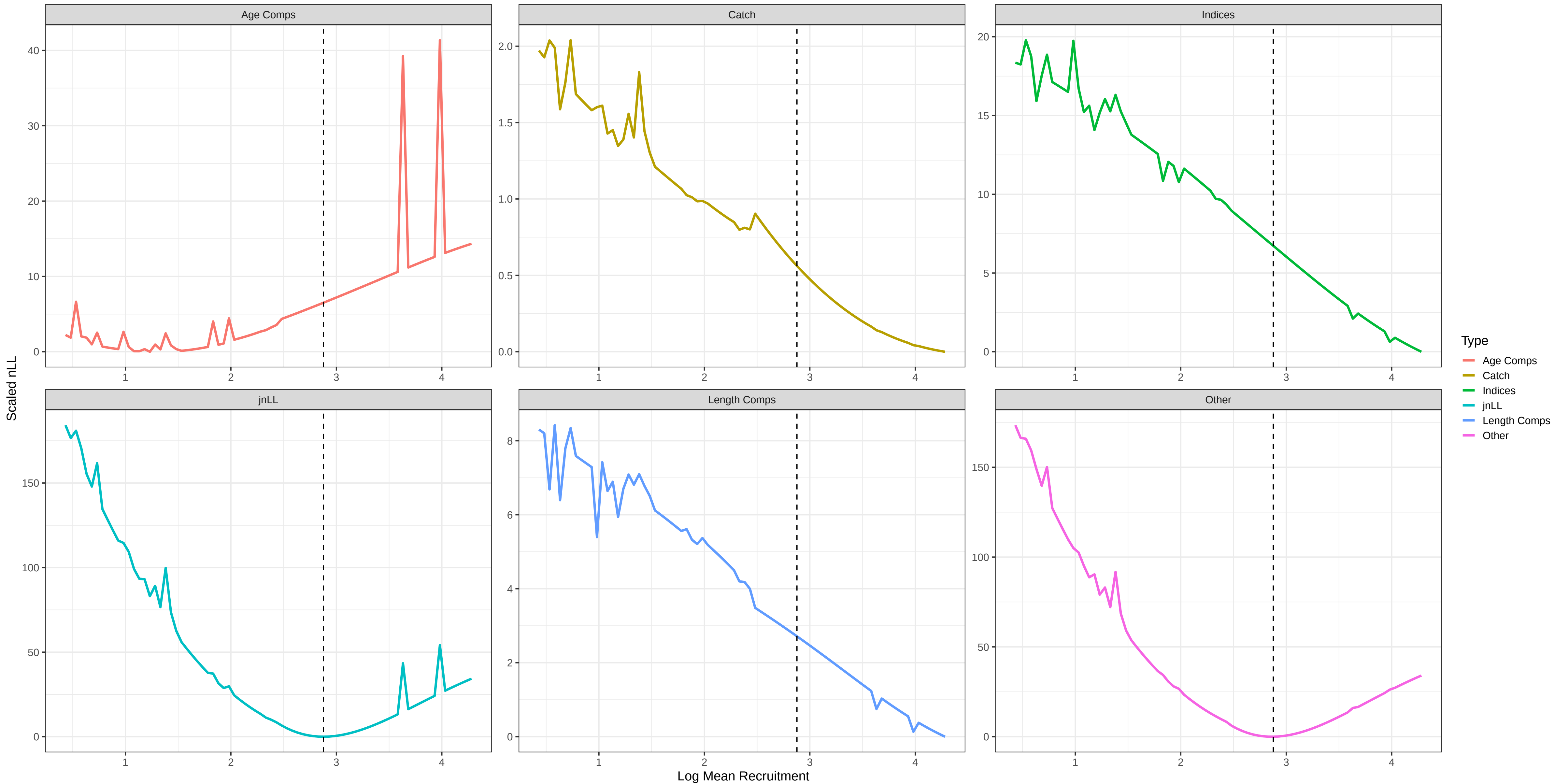
Jitter nLL and Convergence



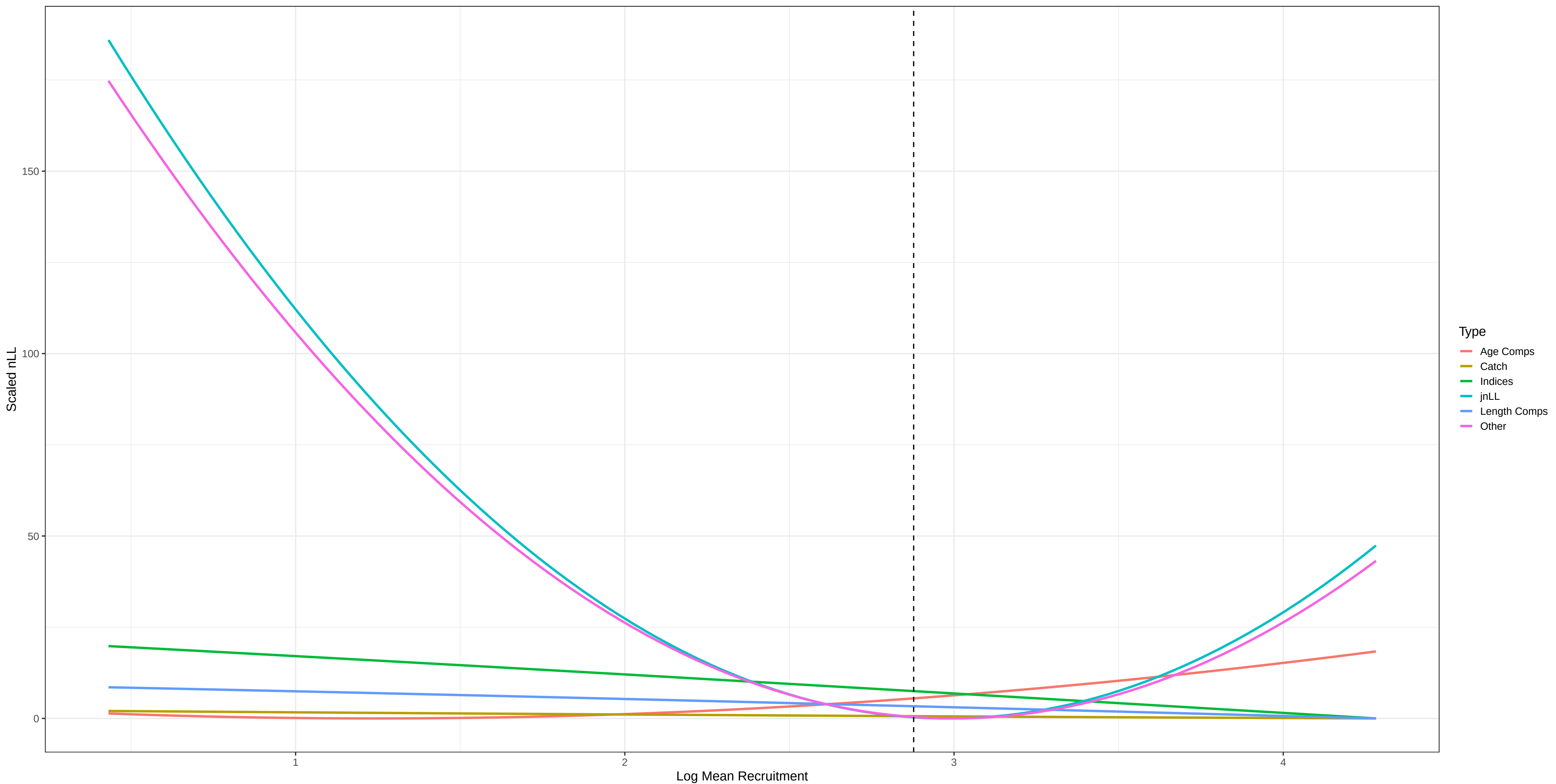
Profile Likelihood R0



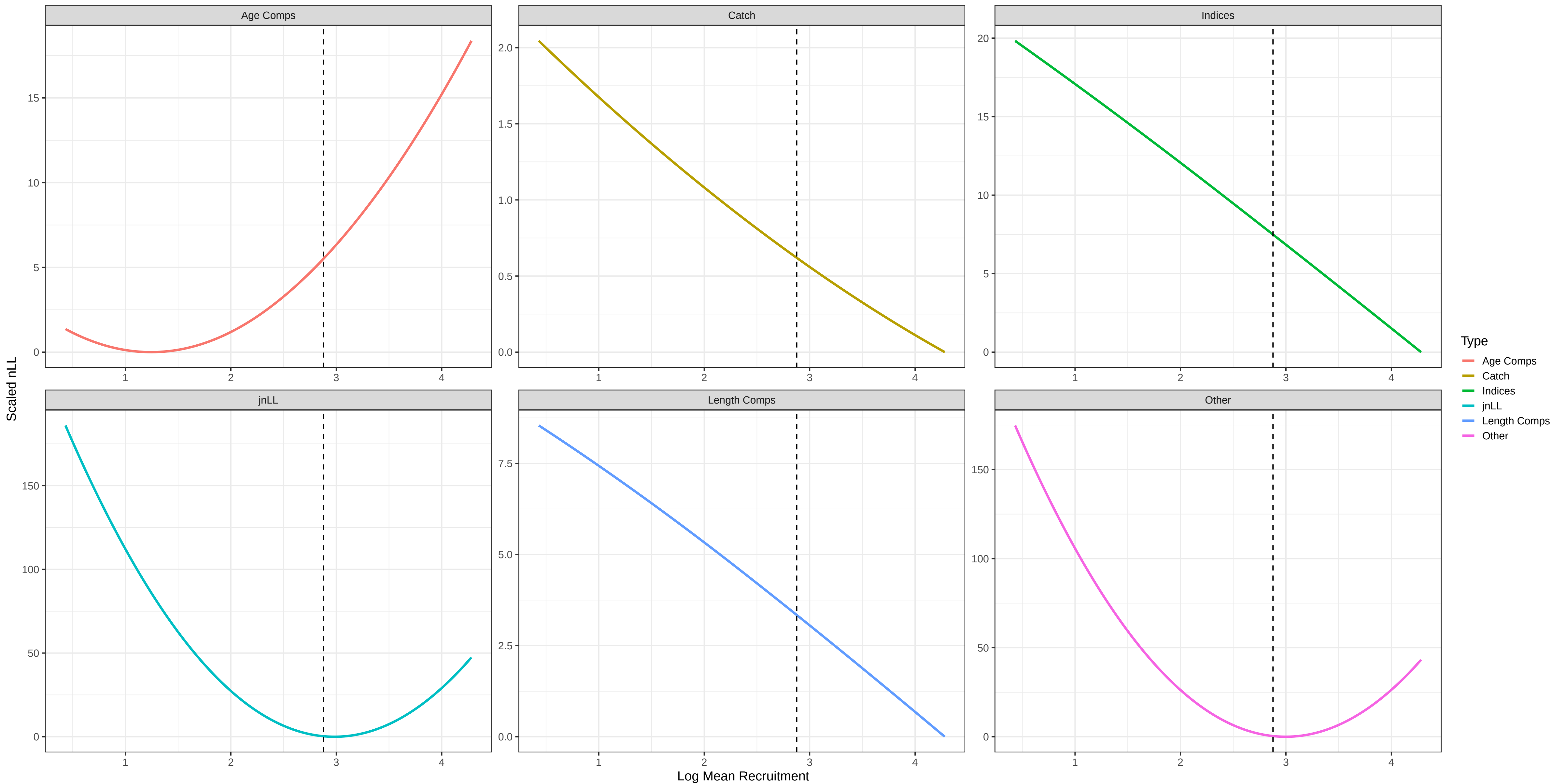
Profile Likelihood R0



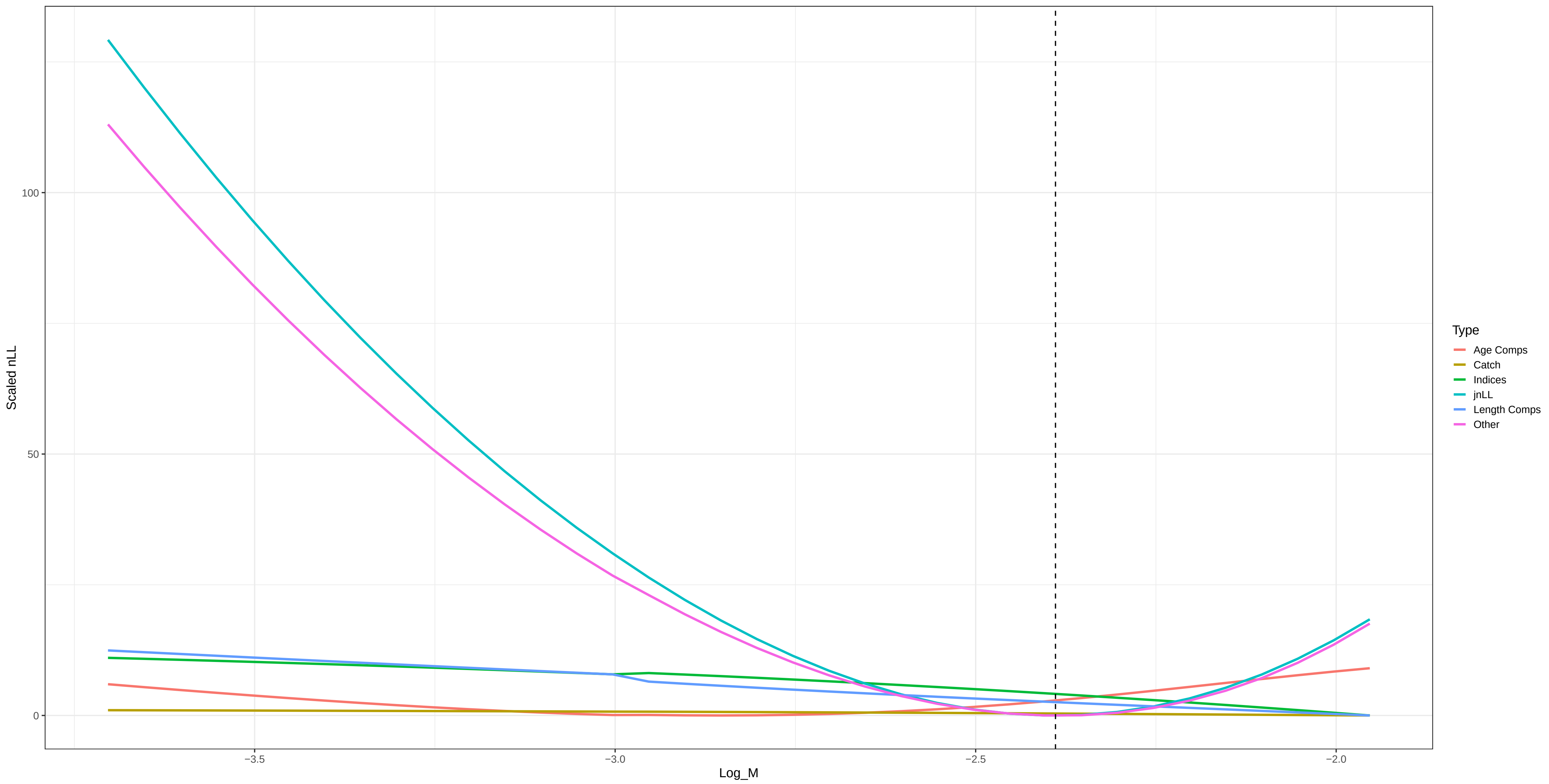
Profile Likelihood R0



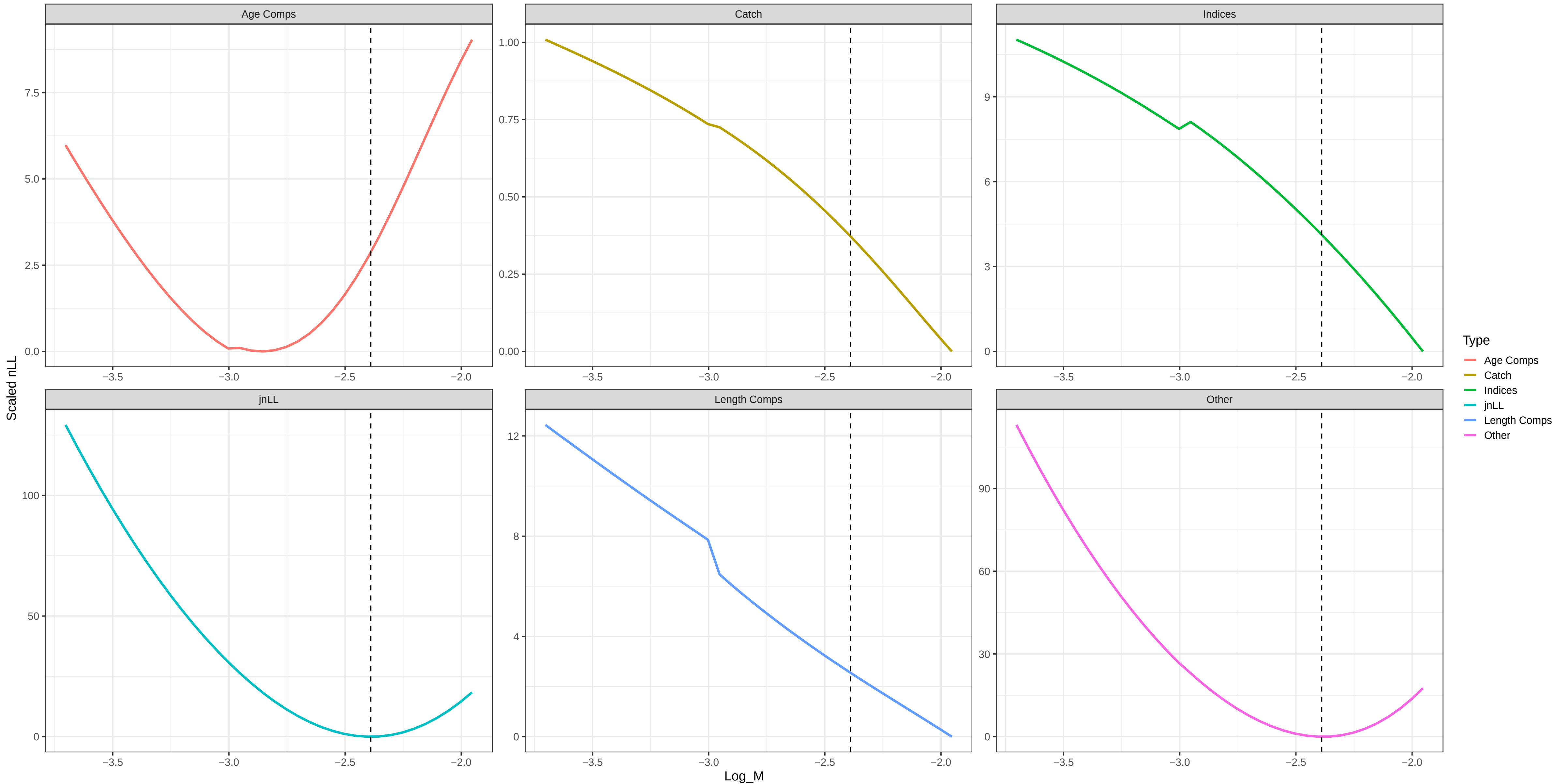
Profile Likelihood R0



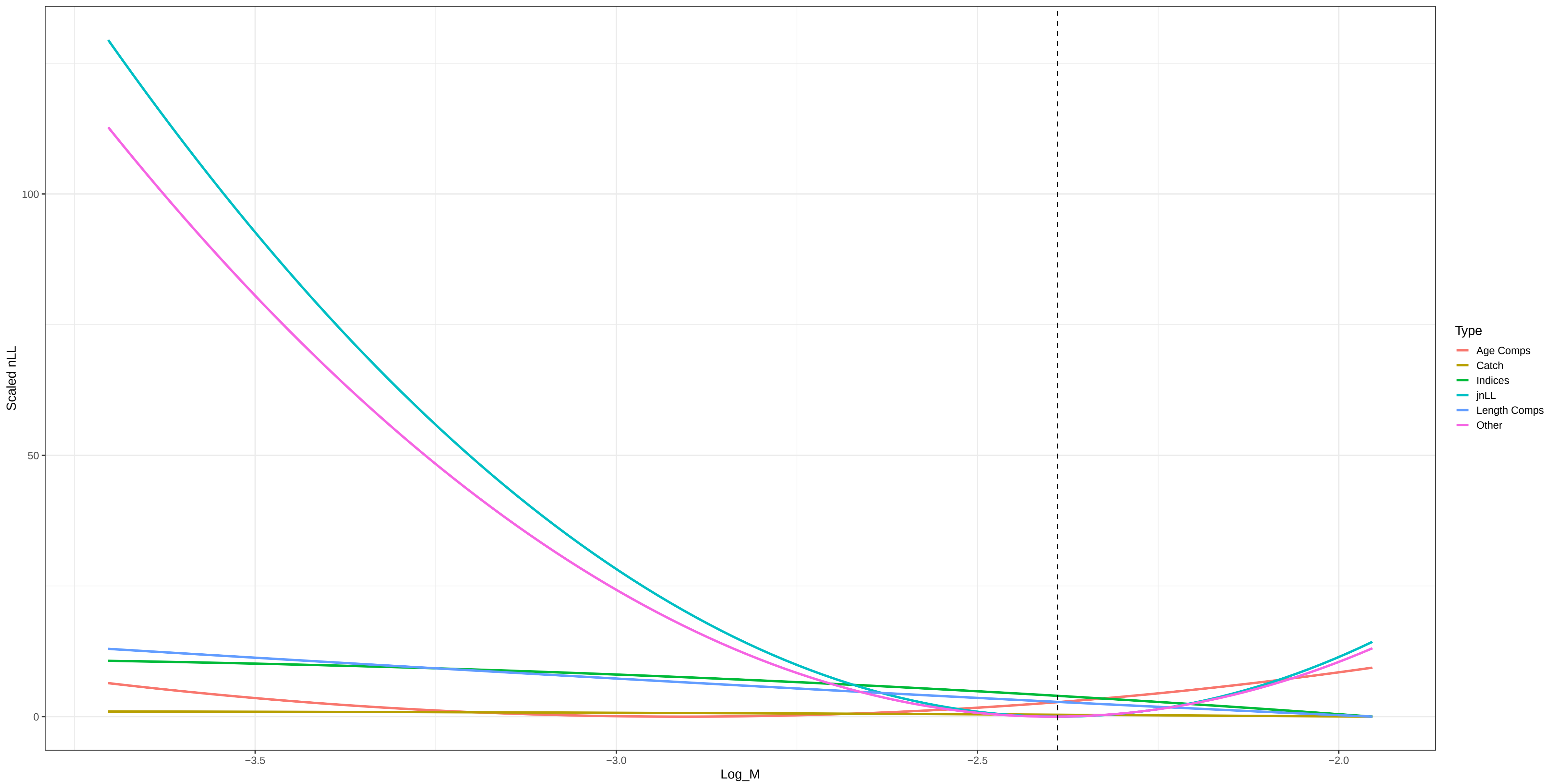
Profile Likelihood Female M



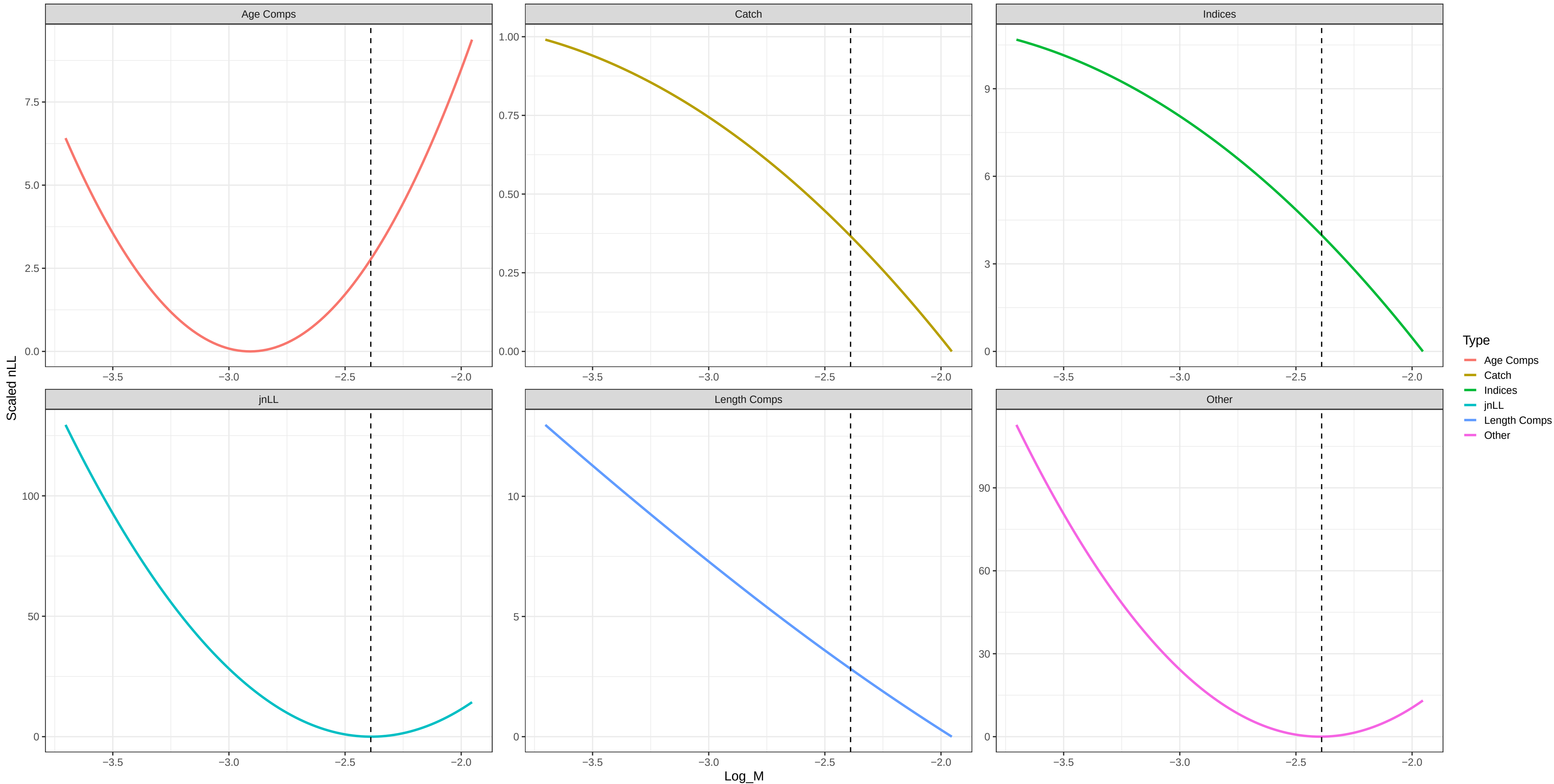
Profile Likelihood Female M



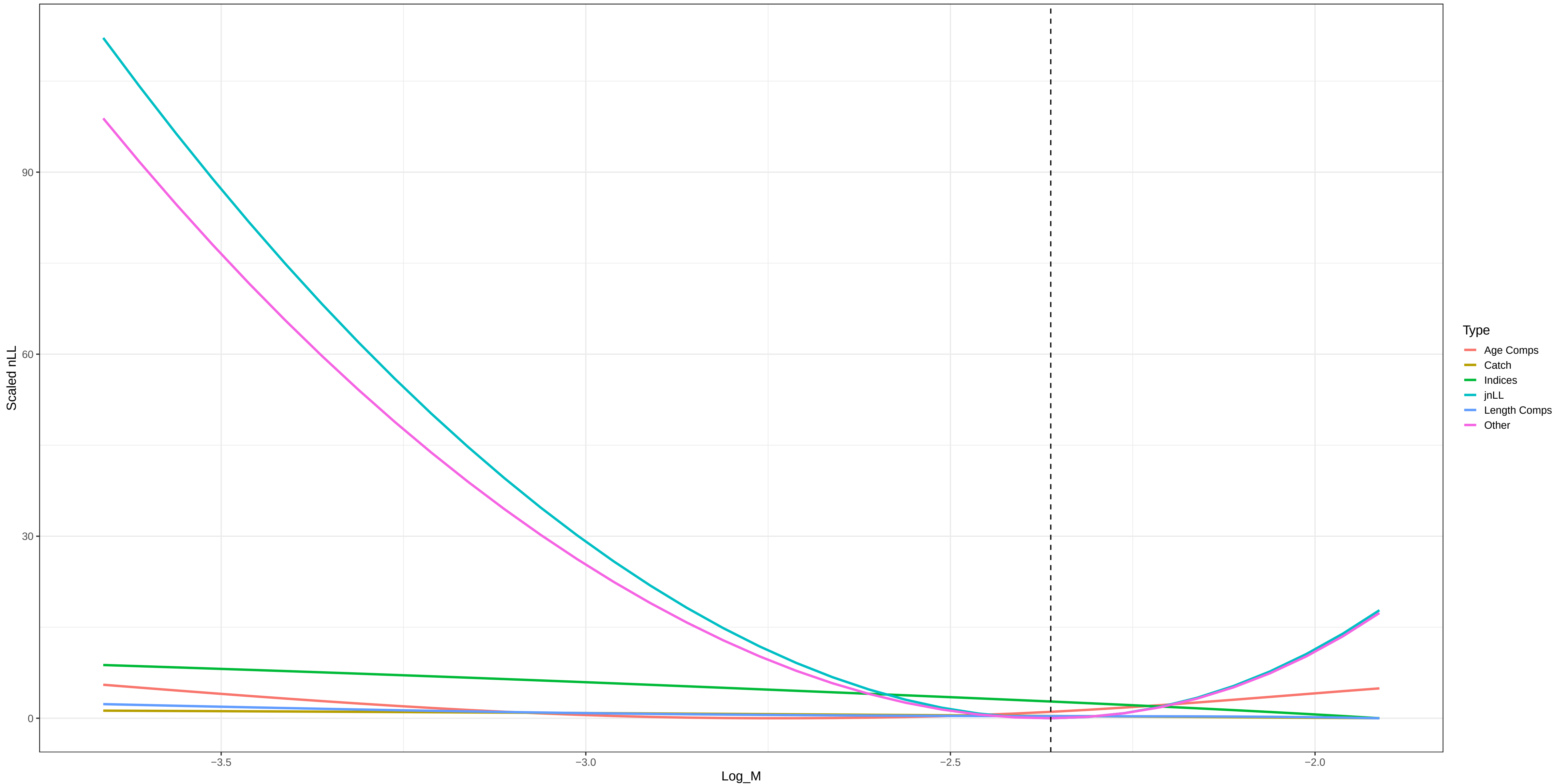
Profile Likelihood Female M



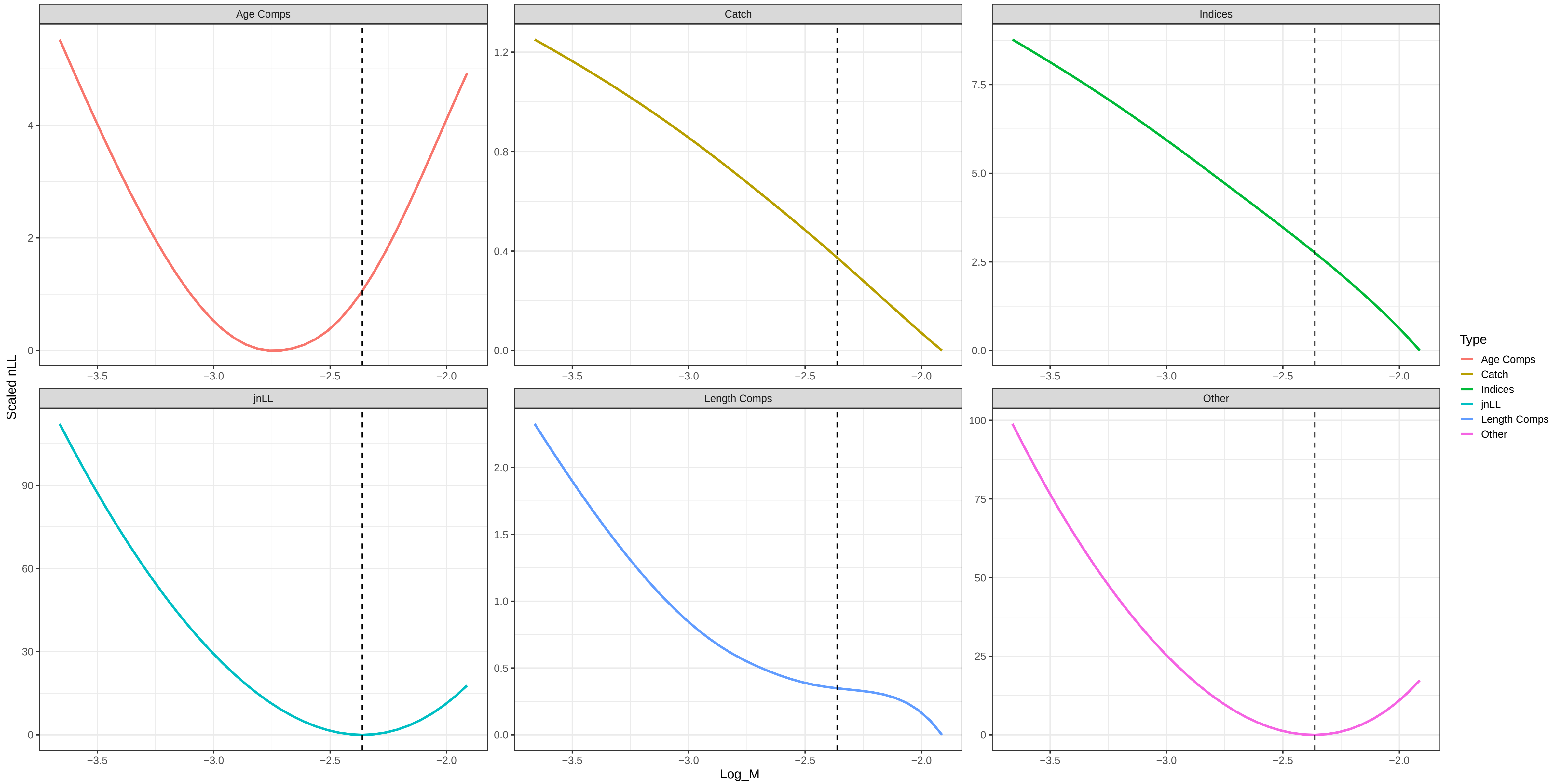
Profile Likelihood Female M



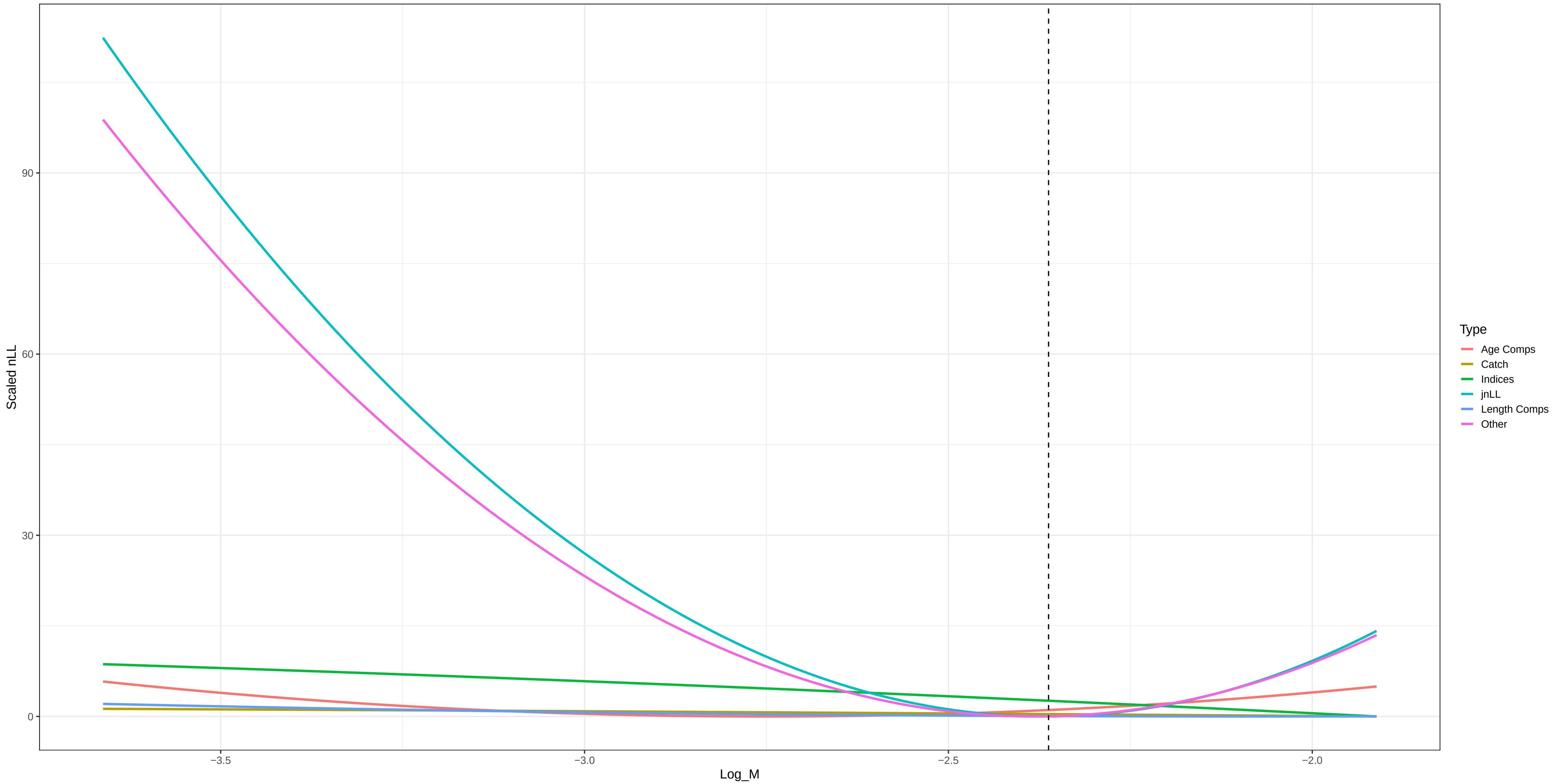
Profile Likelihood Male M



Profile Likelihood Male M



Profile Likelihood Male M



Profile Likelihood Male M

